



KESHAV MEMORIAL INSTITUTE OF TECHNOLOGY
(AN AUTONOMOUS INSTITUTION)



**Accredited by NBA & NAAC, Approved by AICTE, Affiliated to JNTUH,
Narayanguda, Hyderabad, Telangana – 500029**



**DEPARTMENT OF
COMPUTER SCIENCE AND ENGINEERING**

LAB RECORD

SOFTWARE ENGINEERING LAB

B. Tech. III YEAR I SEM (KR23)
ACADEMIC YEAR
2025-26



**KESHAV MEMORIAL INSTITUTE OF TECHNOLOGY
(AN AUTONOMOUS INSTITUTION)**



**Accredited by NBA & NAAC, Approved by AICTE, Affiliated to JNTUH, Hyderabad
Narayanguda, Hyderabad, Telangana – 500029**

Certificate

This is to certify that following is a Bonafide Record of the workbook task done by
_____ bearing Roll No _____ of _____
Branch of _____ year B. Tech. Course in the _____
Subject during the Academic year _____ & _____ under our supervision.

Number of week tasks completed: _____

Signature of Staff Member Incharge

Signature of Head of the Dept.

Signature of Internal Examiner

Signature of External Examiner

INDEX

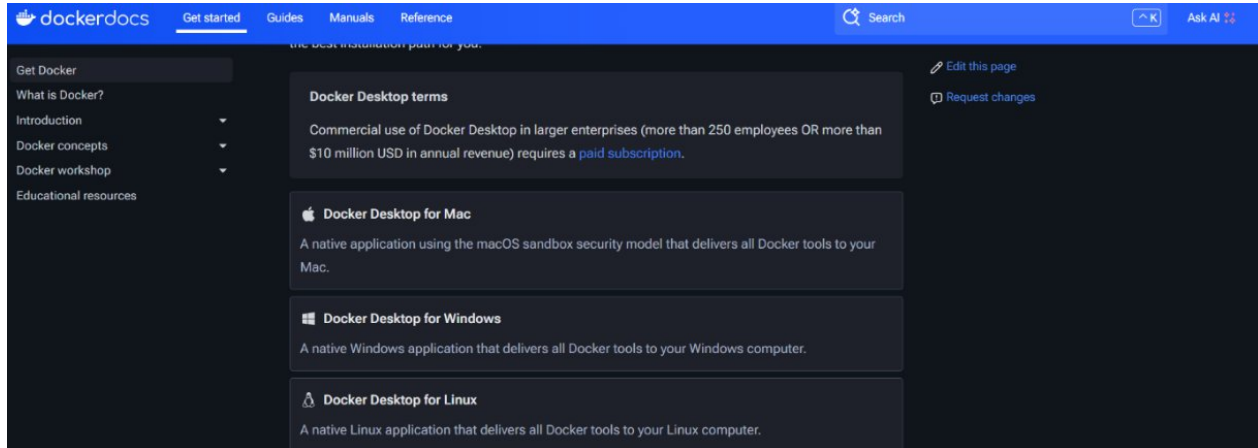
Sr. NO.	CONTENTS	PAGE NO.
1.	Software Installation & SRS Document - Abstract - Functional Requirements (FR) - Non-Functional Requirements (NFR) - User Identification - Workflow of Each User - Use Cases	
2.	<u>Exploring git local and remote commands on the multi-folder project</u> - Pushing multi-folder project into private repository (by student). - Students must explore all listed git commands on the multi-folder project in local and remote repository. - Students must explore all git commands on given scenario-based question	
3.	<u>Collaborative coding using git</u> - To work on collaborative coding by: - Creating Organization. - Coordinating with others through a shared repository - To resolve conflicts when collaborating on same part of code. - To create and apply patch.	
4.	<u>Build and package Java and Web applications using Maven</u> - Understand the structure and lifecycle of a Maven project. - Build and package Java and Web applications using Maven. - Add dependencies using pom.xml, compile and test using plugins. - Resolve errors and conflicts arising from dependency mismatches. - Work with parent and multi-module Maven projects. - Generate executable JARs and deployable WARs using Maven.	
5.	<u>Docker CLI commands</u> - Learn how to pull, run, stop, start, remove, and inspect containers and images. - Gain the ability to create, monitor, and troubleshoot running containers. - Configure and manage networks for container communication. - Create and manage persistent storage for containers. - Learn how to list, remove, and manage images efficiently.	
6.	<u>Docker</u> Learn how to define and run multiple interdependent services (e.g., web server, database) in a single configuration file.	

	b. Gain skills in writing and interpreting docker-compose.yml files for service setup. c. Deploy the same setup across different machines without manual configuration. d. Configure container networking and persistent storage within Compose. e. Reduce setup time and enable faster iteration during application development.	
7.	<u>Creating a Multi-Module Maven Project</u> a. Build and package Java and Web applications using Maven. b. Add dependencies using pom.xml, compile and test using plugins. c. Resolve errors and conflicts arising from dependency mismatches. d. Work with parent and multi-module Maven projects. e. Generate executable JARs and deployable WARs using Maven	
8.	<u>Jenkins Automation</u> a. Hands-on practice on manual creation of Jenkins pipeline using Maven projects from Github b. Create the job and build the pipeline for maven-java and maven-web project.	
9.	<u>Pipeline Creation using script</u> a. Evaluation of Jenkins pipeline. b. WORKING ON BUILD TRIGGERS FOR LAST JENKINS PIPELINE c. Hands-on practice on creation of scripted Jenkins pipeline. d. Take the screenshots for above task	
10.	<u>Working with minikube and Nagios</u> a. Hands-on practice of creating, running and scaling pods in minikube. b. Running Nginx server on specified port number by explaining the Nginx monitoring tool c. Running Nagios server and Understanding the Monitoring tool using Docker. d. AWS-free Trier account Creation steps	
11.	<u>Jenkins-CI/CD</u> a. CI-Continuous Integration using Webhooks. b. Sending E-mail Notification on Build Failure or success	
12.	<u>Creation of virtual machine for Ubuntu OS and Deploying the web application</u> 1. Creation of virtual machine 2. Deploying the web application 3. Accessing it publicly	

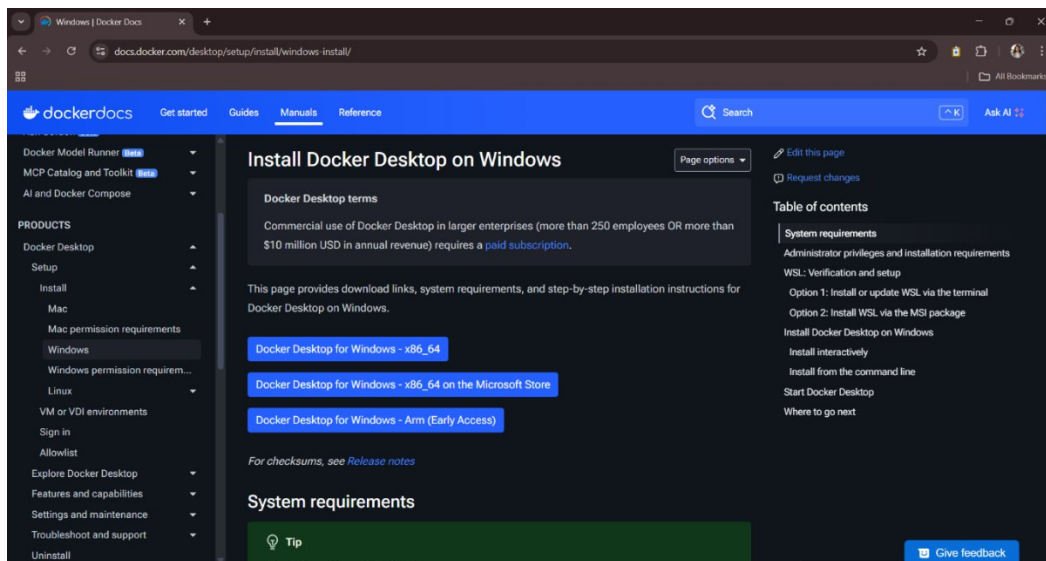
1. Software Installation & SRS Document:

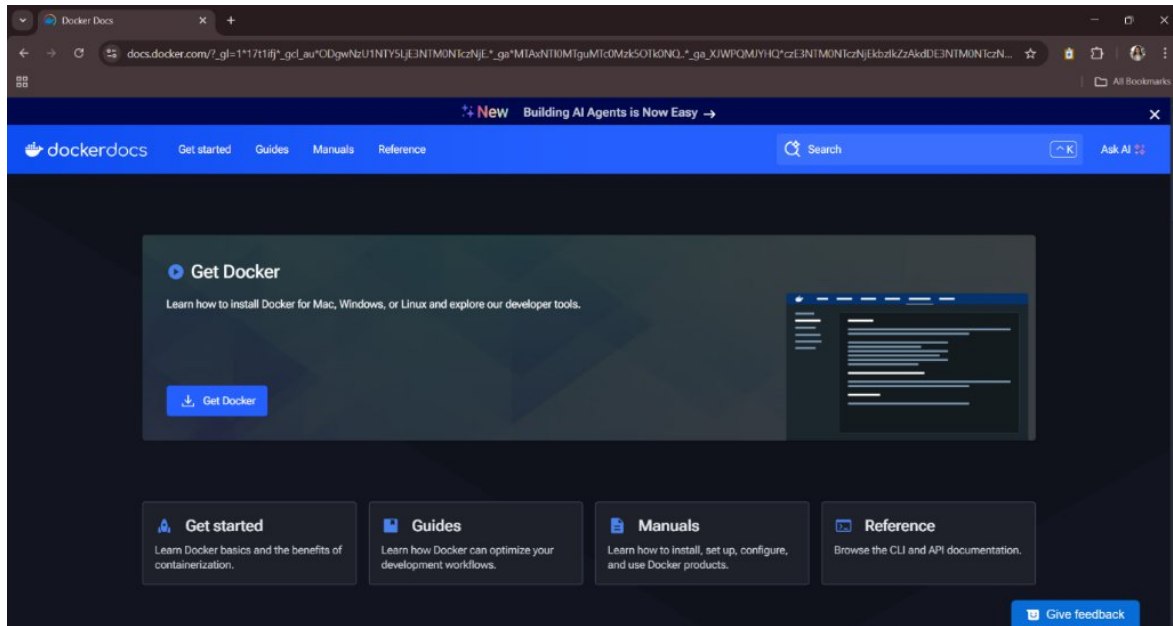
DOCKER- INSTALLATION

Step-1: Go to docker website

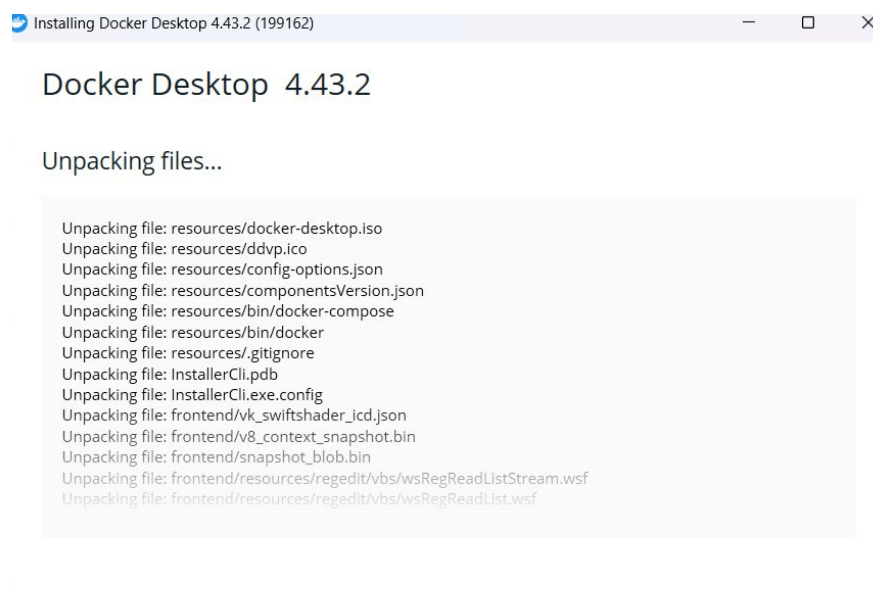


Step-2: Select the suitable one for your system

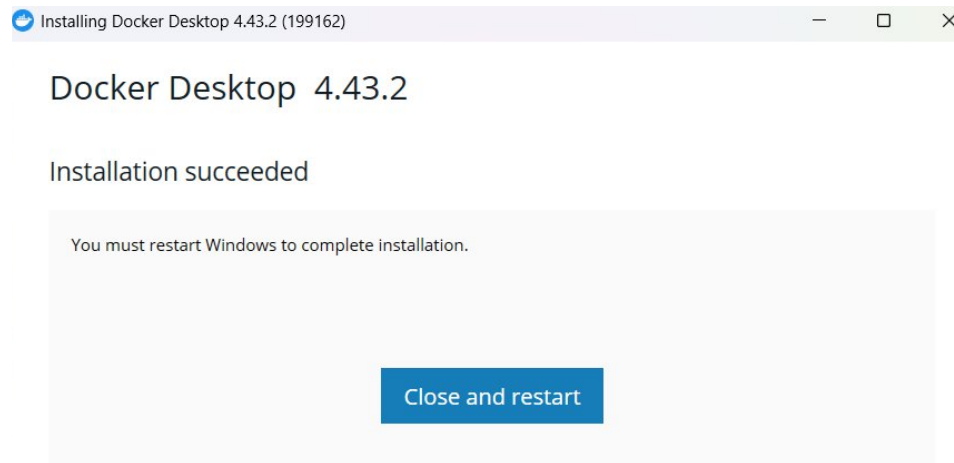




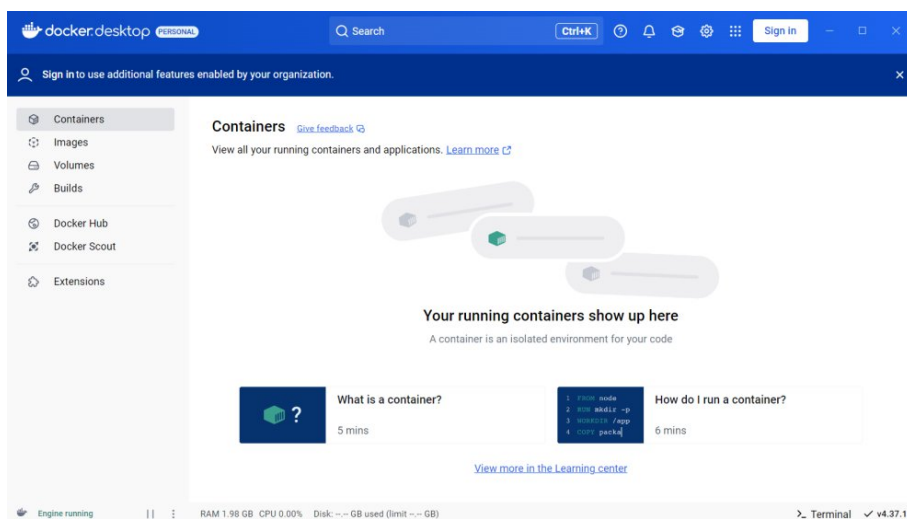
Step-3: After clicking on get docker it starts initializing



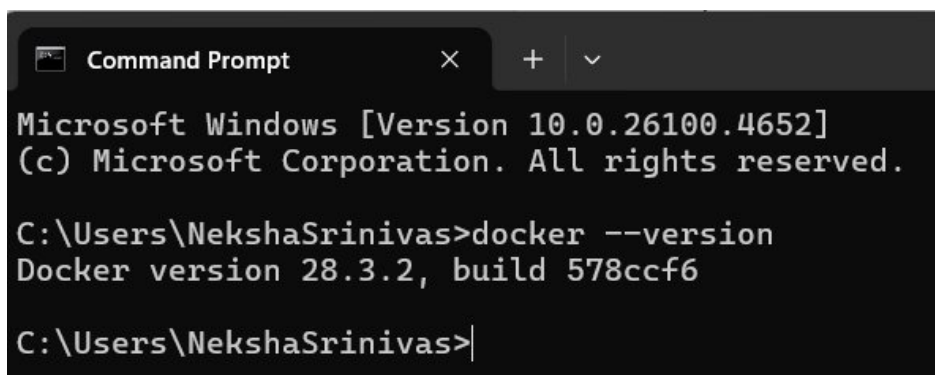
Step-4: Installation successful



Step-5: Docker interface

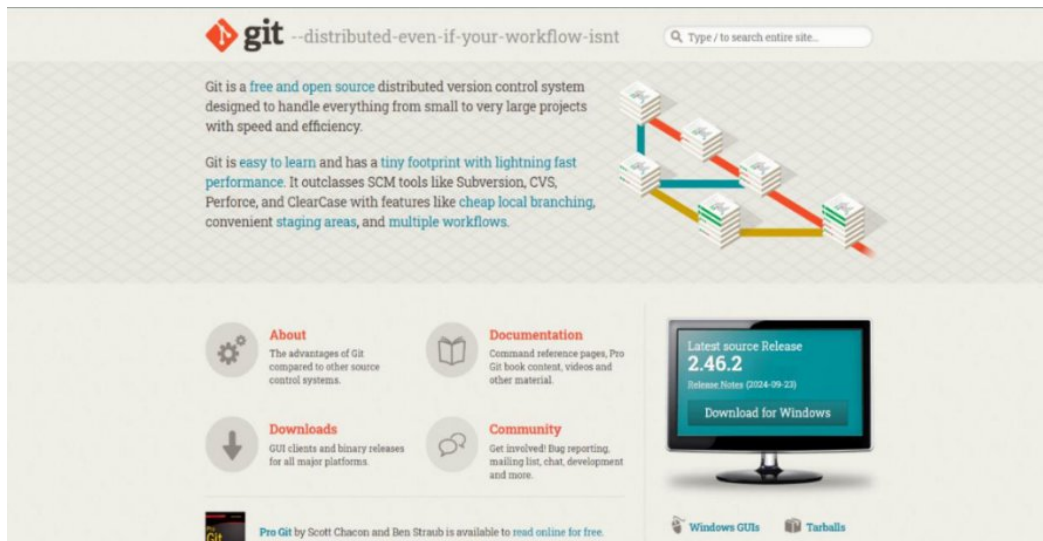


Step-6: docker version

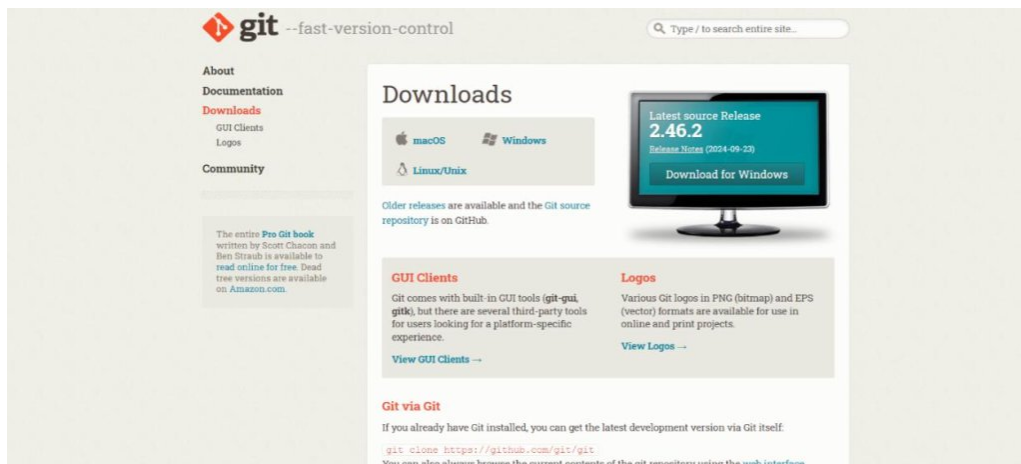


GIT - INSTALLATION:

Step-1: Go to Git website



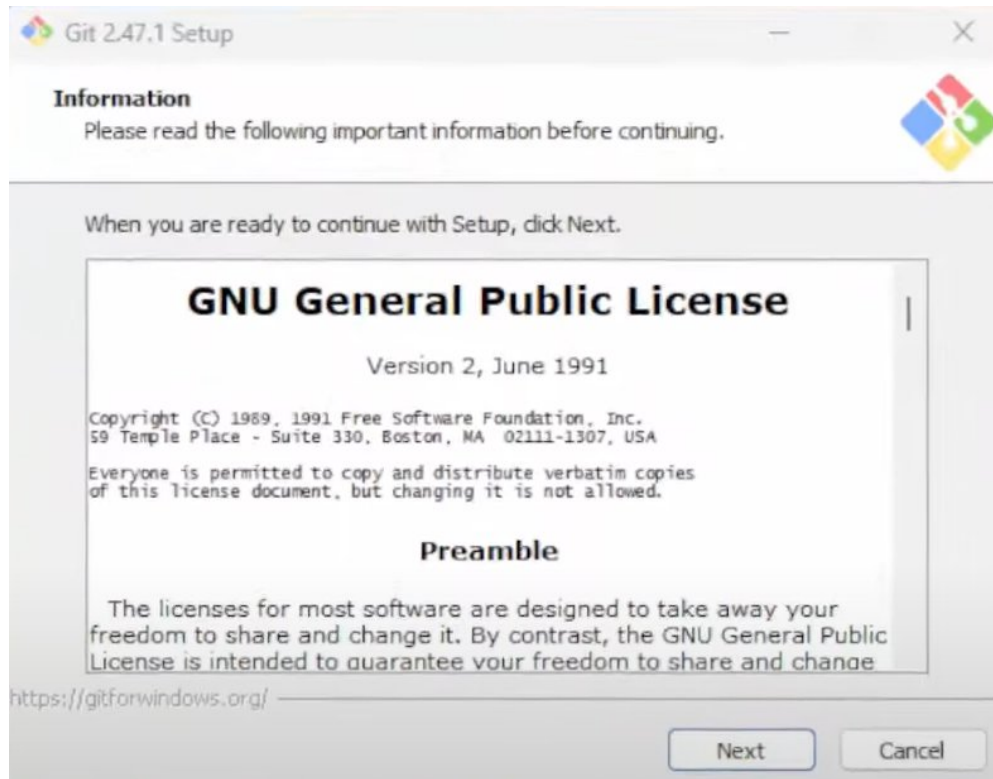
Step-2: click on downloads and options will be displayed



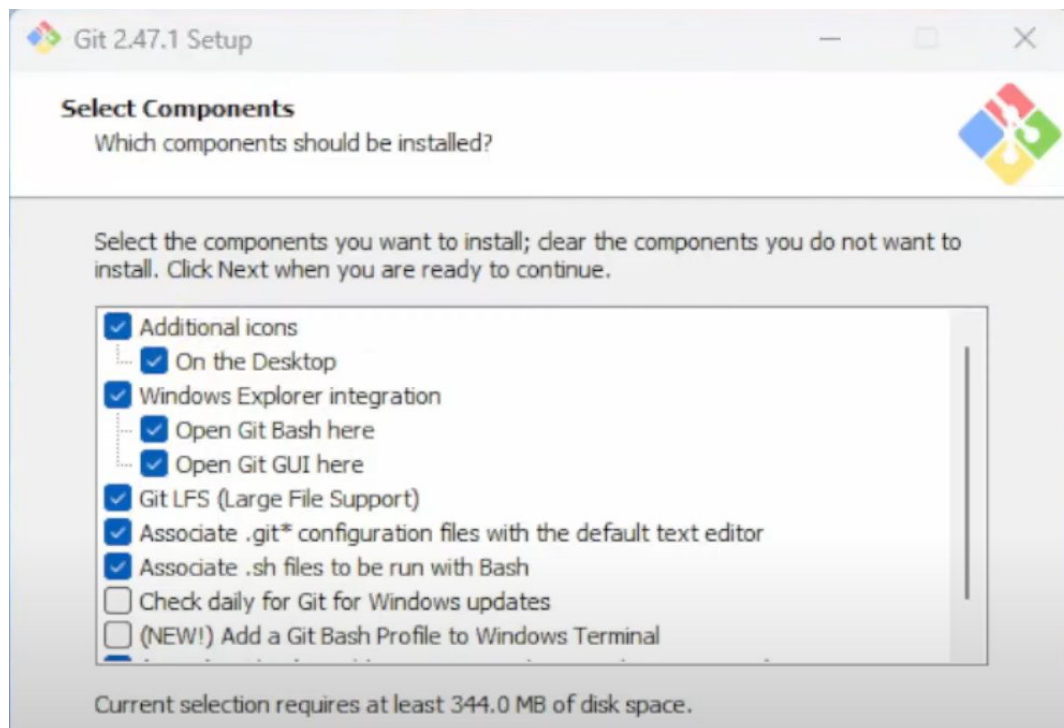
Step-3: Download for windows(suitable one for your system)



Step-4: License will be displayed click on next

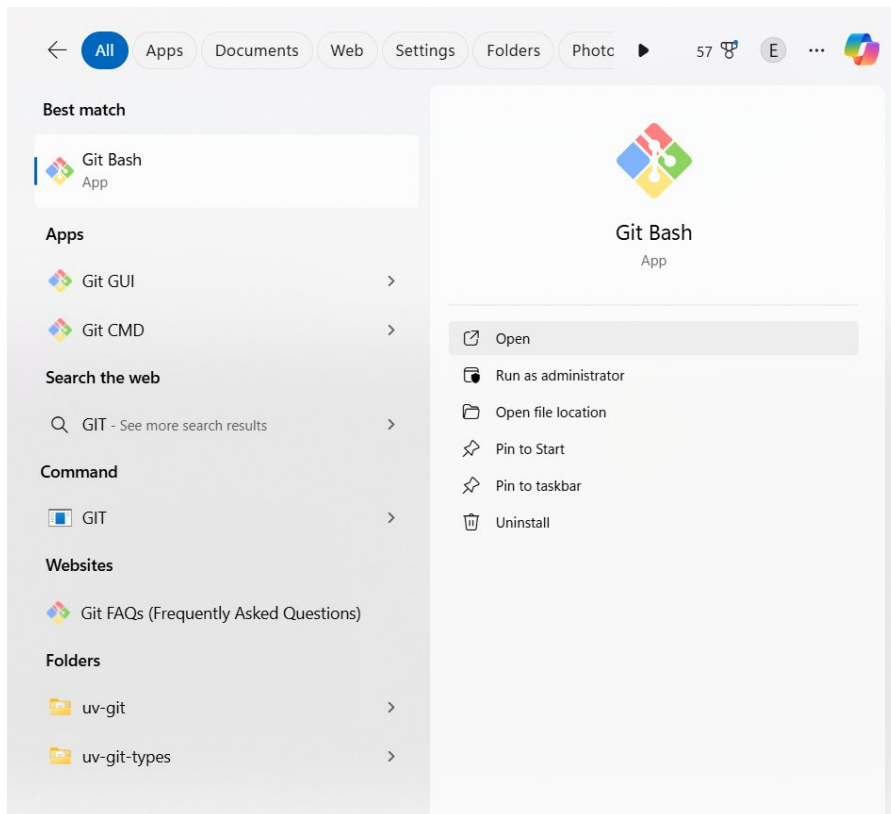


Step-5: Select the componets and click next

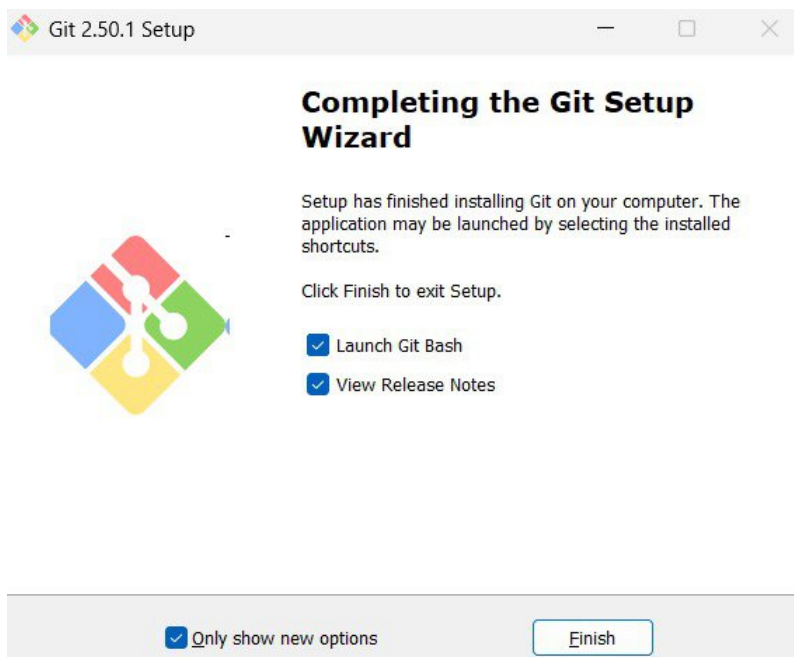


Git bash:

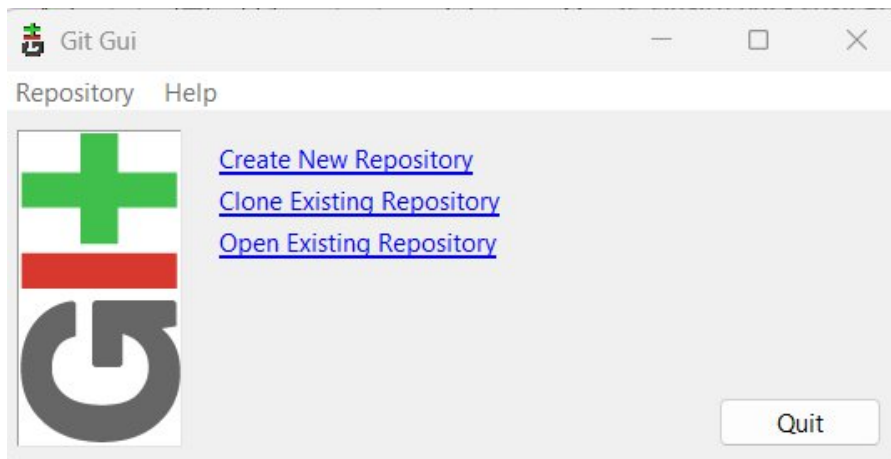
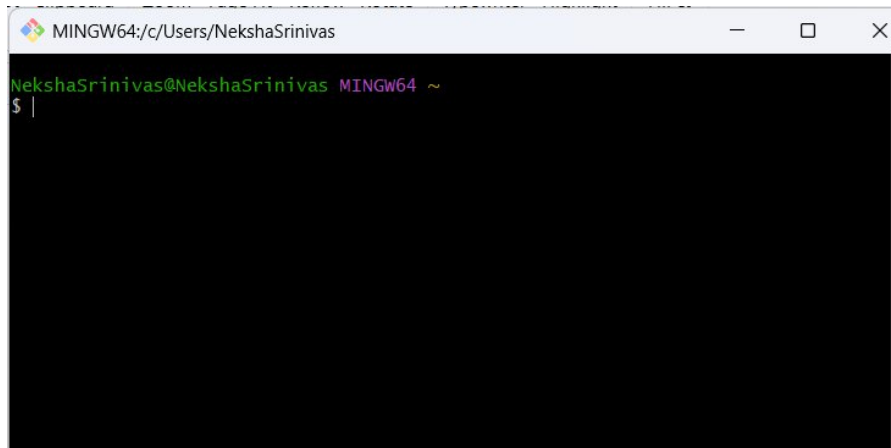
Step-1: Go to search bar and click git bash



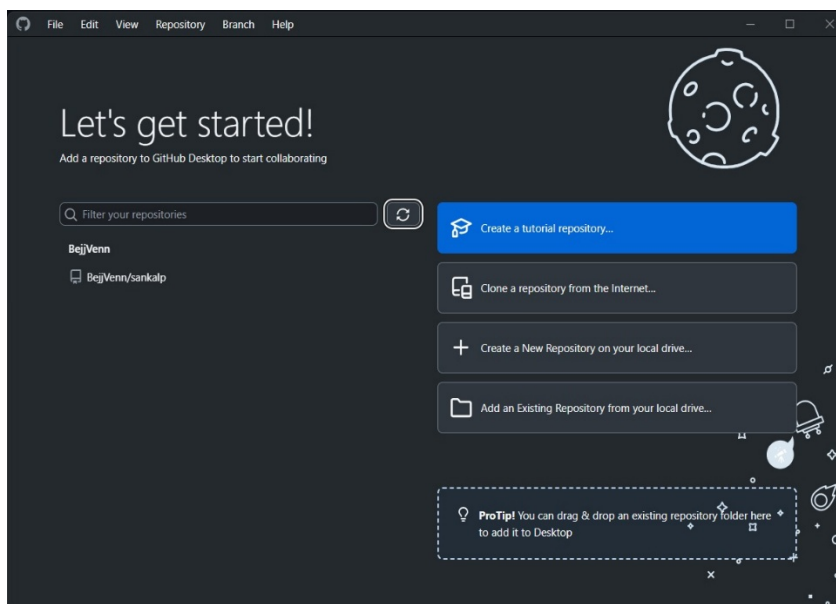
Step-2: Click on finish



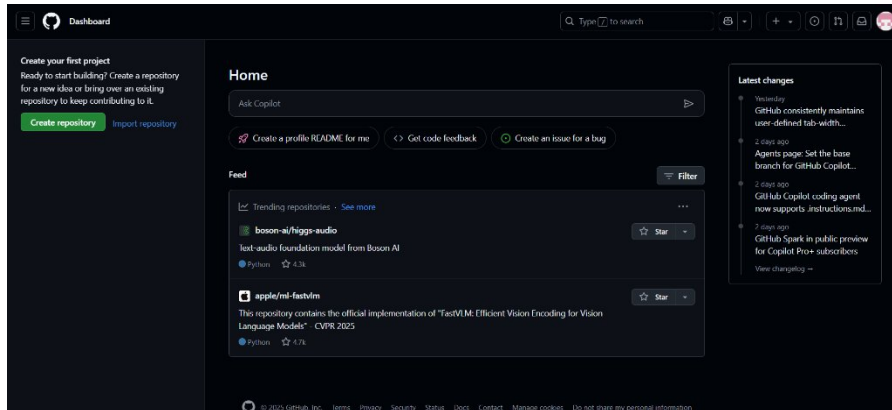
Step-3: git bash interface



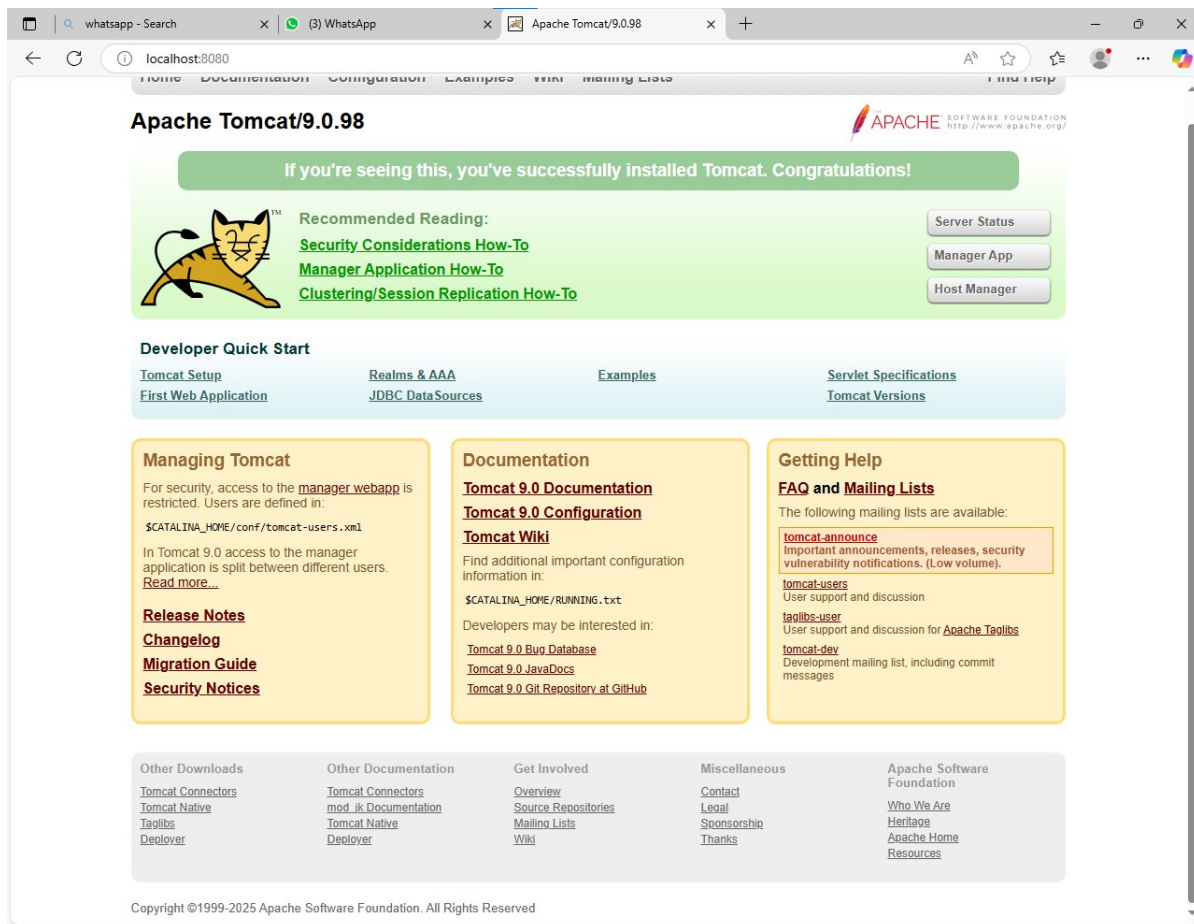
GIT-ACCOUNT



GIT-ACCOUNT



Tomcat



Java and maven versions

```
Command Prompt
Microsoft Windows [Version 10.0.19045.6093]
(c) Microsoft Corporation. All rights reserved.

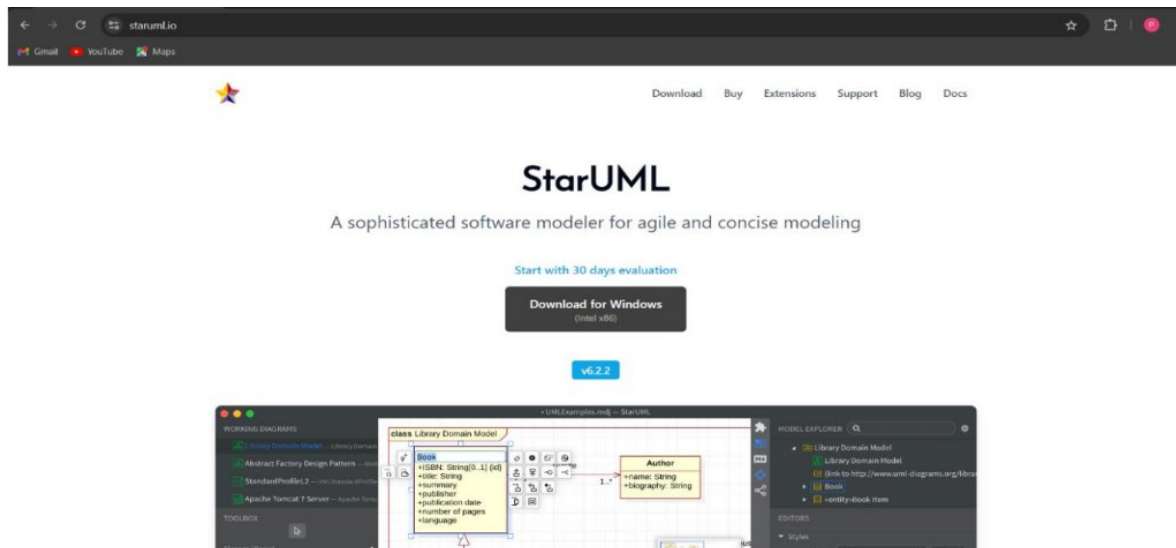
C:\Users\User>java --version
java 21.0.5 2024-10-15 LTS
Java(TM) SE Runtime Environment (build 21.0.5+9-LTS-239)
Java HotSpot(TM) 64-Bit Server VM (build 21.0.5+9-LTS-239, mixed mode, sharing)

C:\Users\User>mvn --version
Apache Maven 3.9.9 (8e8579a9e76f7d015ee5ec7bfc9cd97d260186937)
Maven home: C:\apache-maven-3.9.9
Java version: 21.0.5, vendor: Oracle Corporation, runtime: C:\Program Files\Java\jdk-21
Default locale: en_IN, platform encoding: UTF-8
OS name: "windows 10", version: "10.0", arch: "amd64", family: "windows"

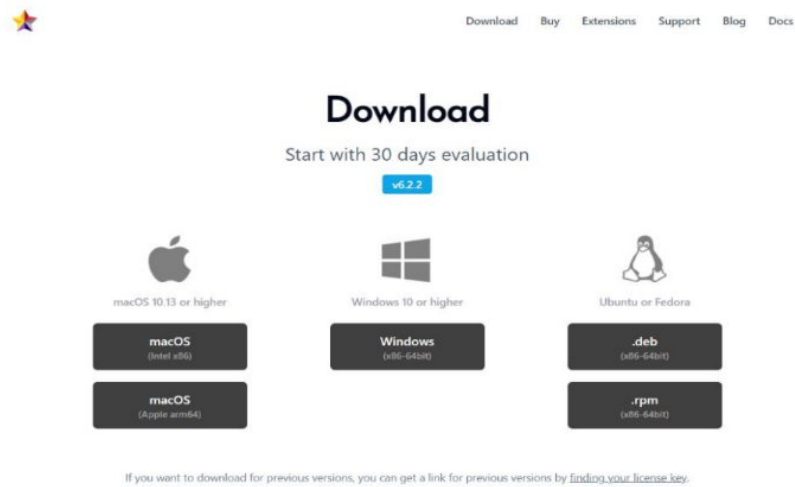
C:\Users\User>
```

StarUML INSTALLATION

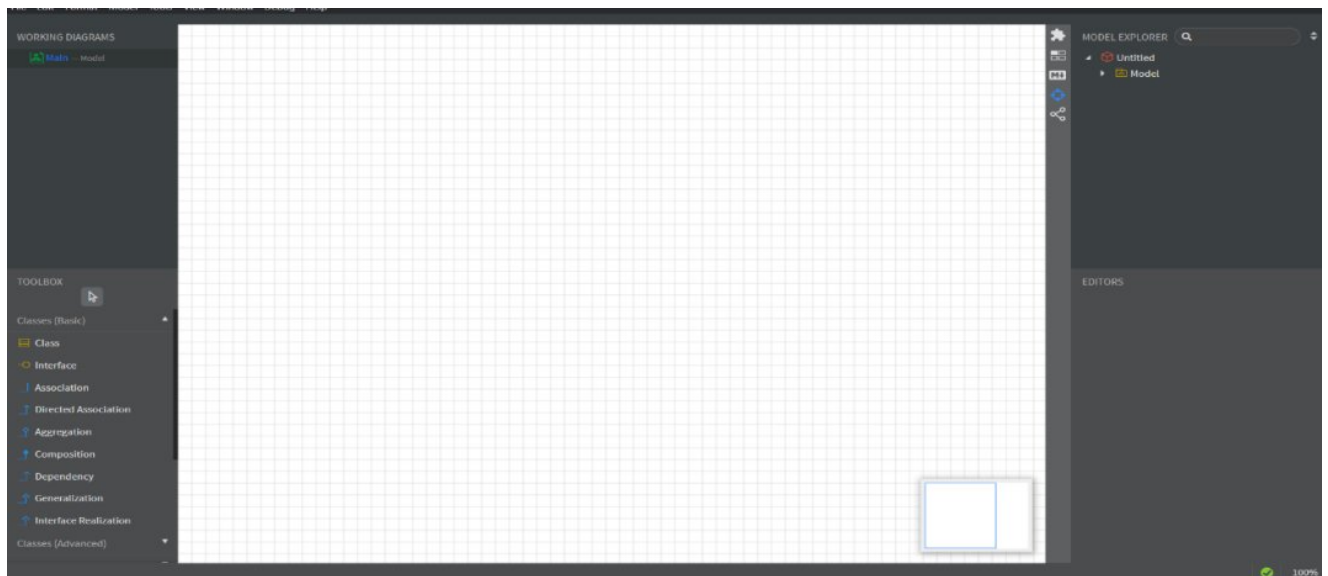
Step-1: Go to startuml website



Step-2: from the given options select the suitable one for your system



Step-3: Interface



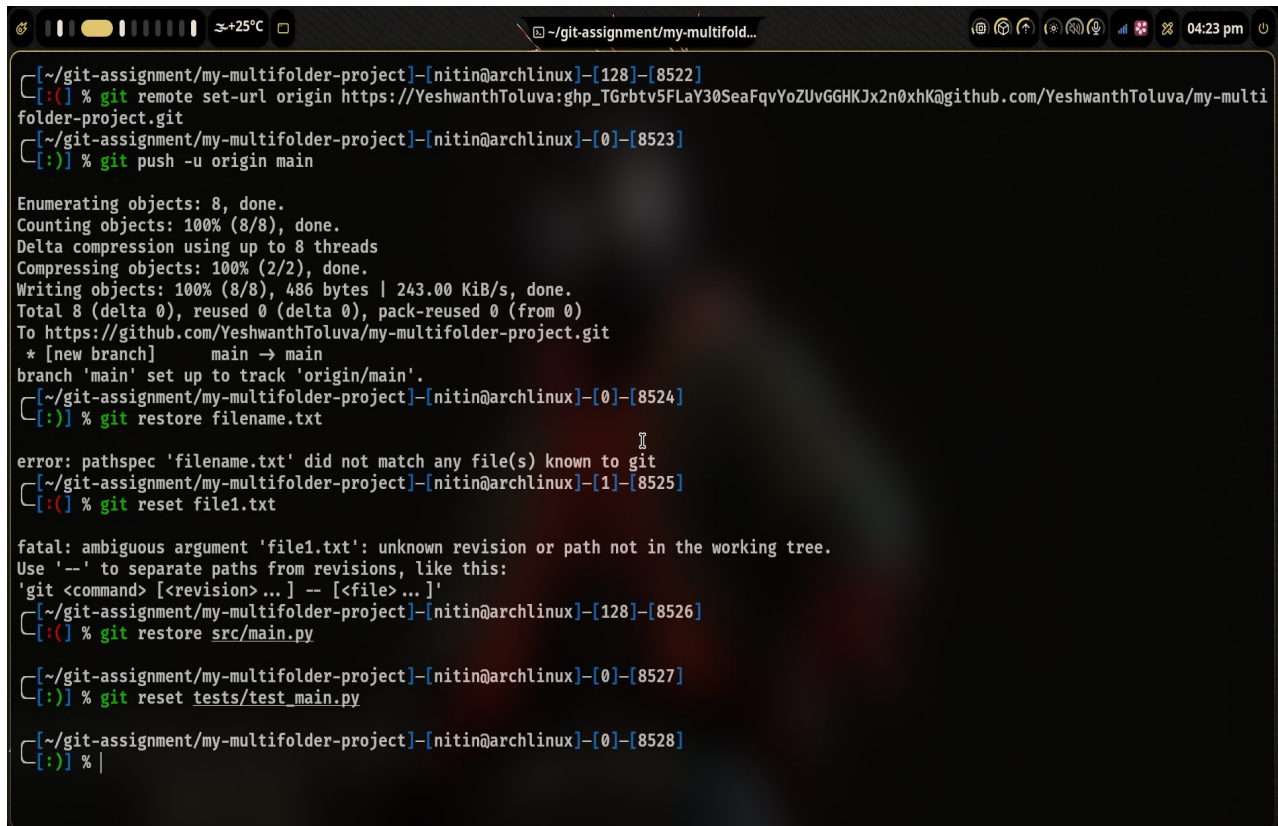
2. Exploring git local and remote commands on the multi-folder project

Github Global Configuration:

```
C:\Users\NekshaSrinivas>git config --global --list
core.editor="C:\Users\NekshaSrinivas\AppData\Local\Programs\Microsoft VS Code\bin\code" --wait
user.name=Edigirala-Neksha
user.email=edigiralaneksha@gmail.com

C:\Users\NekshaSrinivas>
```

Git Push to GitHub Public Repository with Remote Set



```
~/git-assignment/my-multifolder... 04:23 pm
[~/git-assignment/my-multifolder-project]-[nitin@archlinux]-[128]-[8522]
[.] % git remote set-url origin https://YeshwanthToluva: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@github.com/YeshwanthToluva/my-multi
folder-project.git
[~/git-assignment/my-multifolder-project]-[nitin@archlinux]-[0]-[8523]
[.] % git push -u origin main

Enumerating objects: 8, done.
Counting objects: 100% (8/8), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (8/8), 486 bytes | 243.00 KiB/s, done.
Total 8 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/YeshwanthToluva/my-multifolder-project.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
[~/git-assignment/my-multifolder-project]-[nitin@archlinux]-[0]-[8524]
[.] % git restore filename.txt

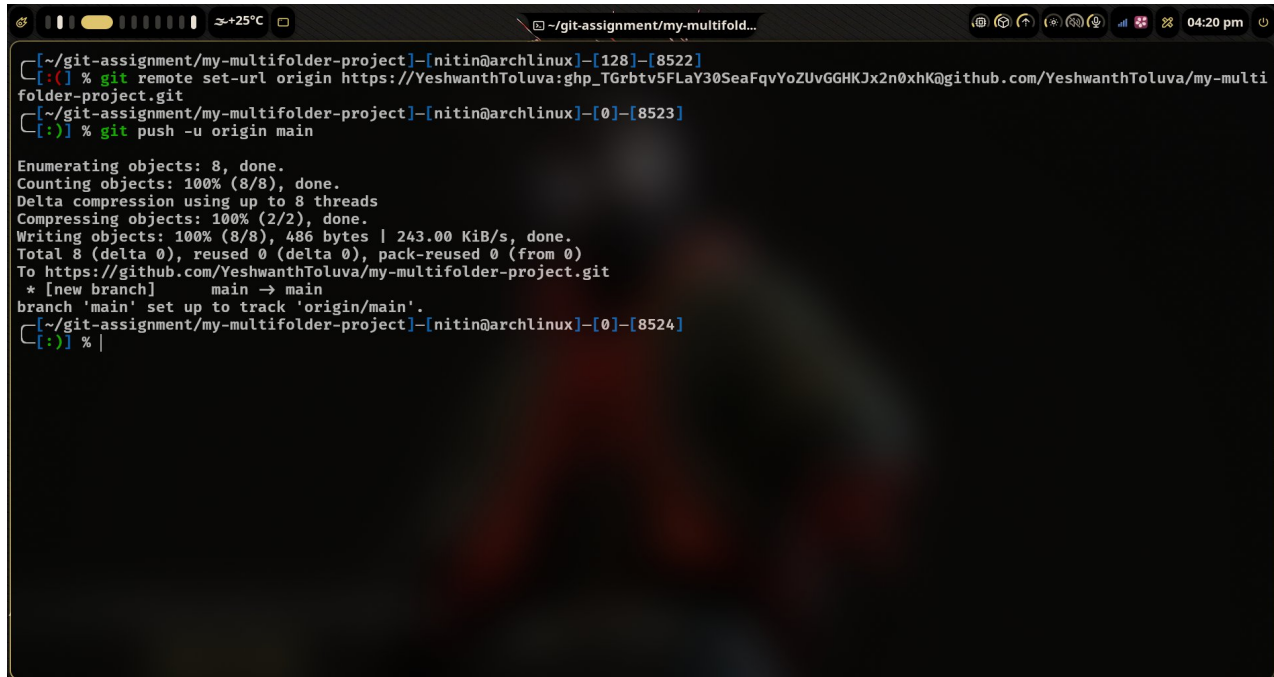
error: pathspec 'filename.txt' did not match any file(s) known to git
[~/git-assignment/my-multifolder-project]-[nitin@archlinux]-[1]-[8525]
[.] % git reset file1.txt

fatal: ambiguous argument 'file1.txt': unknown revision or path not in the working tree.
Use '-' to separate paths from revisions, like this:
'git <command> [<revision>...] -- [<file>...]'
[~/git-assignment/my-multifolder-project]-[nitin@archlinux]-[128]-[8526]
[.] % git restore src/main.py

[~/git-assignment/my-multifolder-project]-[nitin@archlinux]-[0]-[8527]
[.] % git reset tests/test_main.py

[~/git-assignment/my-multifolder-project]-[nitin@archlinux]-[0]-[8528]
[.] % |
```

Scenario-Based Git Commands: Discarding and Unstaging Changes

A terminal window with a dark background and light-colored text. The window title is '~/.git-assignment/my-multifolder...'. The prompt is '[~/git-assignment/my-multifolder-project]--[nitin@archlinux]--[128]--[8522]'. The user enters '% git remote set-url origin https://YeshwanthToluva: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@github.com/YeshwanthToluva/my-multifolder-project.git'. The prompt changes to '[~/git-assignment/my-multifolder-project]--[nitin@archlinux]--[0]--[8523]'. The user enters '% git push -u origin main'. The terminal shows the output of the push command: 'Enumerating objects: 8, done.', 'Counting objects: 100% (8/8), done.', 'Delta compression using up to 8 threads', 'Compressing objects: 100% (2/2), done.', 'Writing objects: 100% (8/8), 486 bytes | 243.00 KiB/s, done.', 'Total 8 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)', 'To https://github.com/YeshwanthToluva/my-multifolder-project.git', '* [new branch] main -> main', 'branch 'main' set up to track 'origin/main'.', and '[~/git-assignment/my-multifolder-project]--[nitin@archlinux]--[0]--[8524]'. The prompt is now '[~/git-assignment/my-multifolder-project]--[nitin@archlinux]--[0]--[8524] % |'.

1. You've cloned a repository and made some changes to a local branch. Now you want to push these changes to the remote repository, but you're getting an error saying "rejected - non-fast-forward." How would you resolve this?

This error occurs when the remote branch has changes that your local branch doesn't. To resolve it:

git pull --rebase origin <branch-name>

This rebases your local changes on top of the latest remote changes. After resolving any conflicts, push your changes:

2. You've been working on a feature branch, and now you need to push it to the remote repository. However, the remote repository already has a main branch. How do you push your feature branch without affecting the main branch?

You can push your feature branch independently:

git push origin feature/feat-1

This creates a new remote branch and does not affect the main branch.

3. You cloned a remote repository, but after a while, the repository's structure changed and new

branches were added. How would you keep your local repository updated with the latest changes from the remote repository?

Use the following commands:

git fetch origin

This updates your local copy with all branches and changes from the remote. You can then check out new branches using:

git checkout branch-name

4. A colleague has pushed some changes to the main branch, but you have local changes in the same branch. You want to pull their changes, but you want to avoid merge conflicts. What steps would you take?

Use rebase to integrate their changes on top of your work:

git stash # Temporarily store your changes

git pull --rebase origin main

git stash pop # Apply your changes on top

This reduces the chance of conflicts and keeps history clean.

5. You accidentally pushed a sensitive file (e.g., API keys) to the remote repository. How would you fix this situation?

Steps to remove the sensitive data:

Remove the file and commit:

git rm --cached path/to/file

git commit -m "Remove sensitive file"

git push origin main

If the secret is in history, use git filter-branch or BFG Repo-Cleaner to rewrite history:

**git filter-branch --force --index-filter **

**"git rm --cached --ignore-unmatch path/to/file" **

--prune-empty --tag-name-filter cat -- --all

Force push and rotate the secret.

6. You're working on a feature branch, and your manager requests that you integrate the latest changes from main into your feature branch. What steps would you take?

Use rebase or merge:

Rebase:

git checkout feature/your-feature

git fetch origin

git rebase origin/main

7. You cloned a remote repository, but later you find that you need to push your changes to a different remote repository. How do you configure your local repository to push to this new remote?

Then push your changes:

git push origin branch-name

8. After running git pull, you notice that your local branch is behind the remote branch. How would you proceed to bring your local branch up to date without losing your local changes?

Use stash or rebase:

git stash

git pull --rebase origin branch-name

git stash pop

This ensures a clean rebase and retains your changes.

9. You're working on a project with multiple collaborators, and you notice that your local changes conflict with changes that have been pushed by others. How would you resolve the conflicts?

Pull the latest changes:

git pull origin branch-name

Git will highlight conflicts. Open the files, manually resolve the <<<<<<, =====, and >>>>>> markers.

Mark as resolved and commit:

git add .

git commit

10. You've pushed a feature branch to a remote repository, but now you need to delete the branch from the remote. How would you do that?

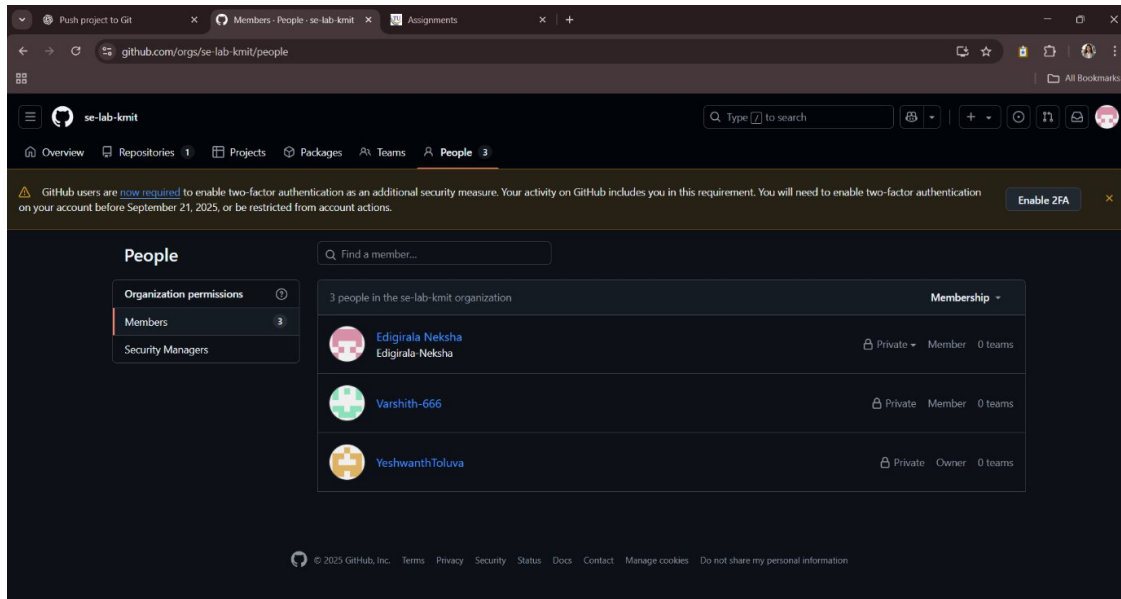
Use the following command:

git push origin --delete feature/branch-name

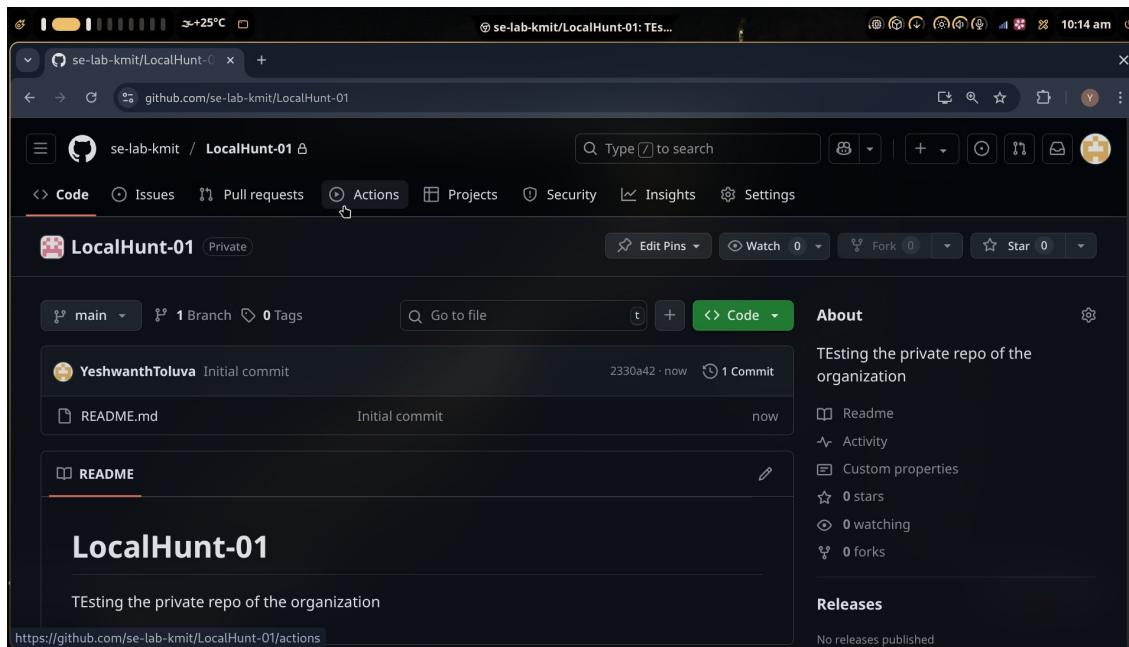
This will remove the branch from the remote repository.

3. Collaborative coding using git

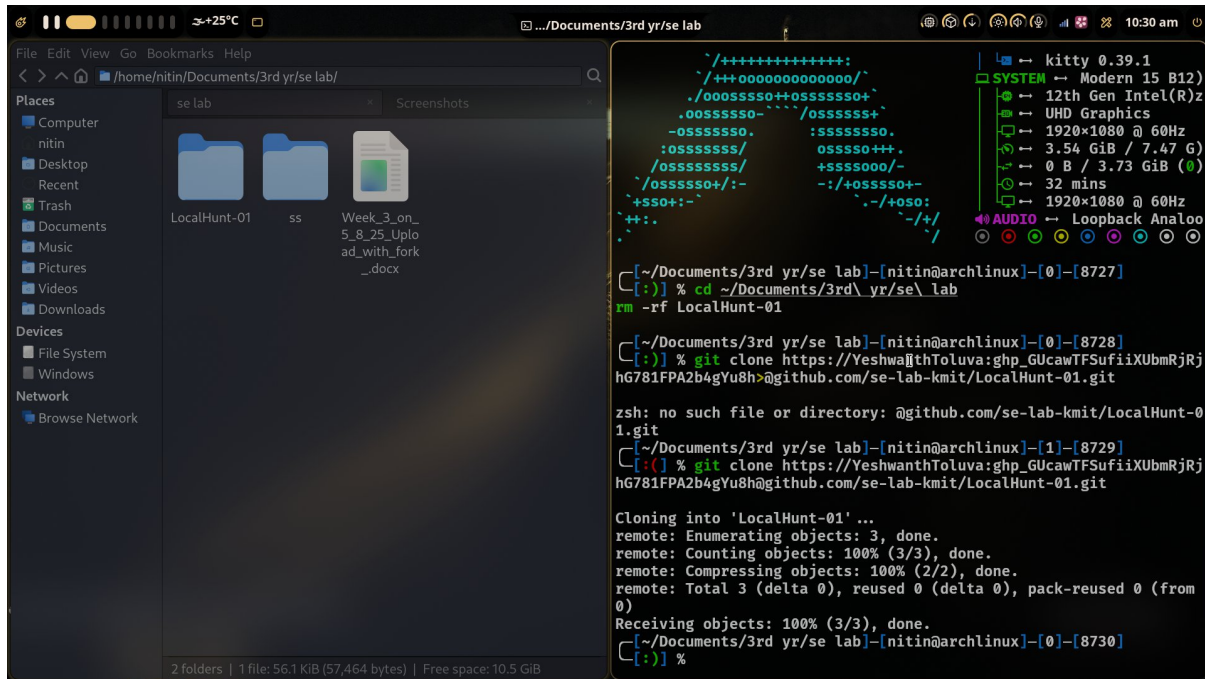
GitHub Organization Members Page - se-lab-kmit Team Overview



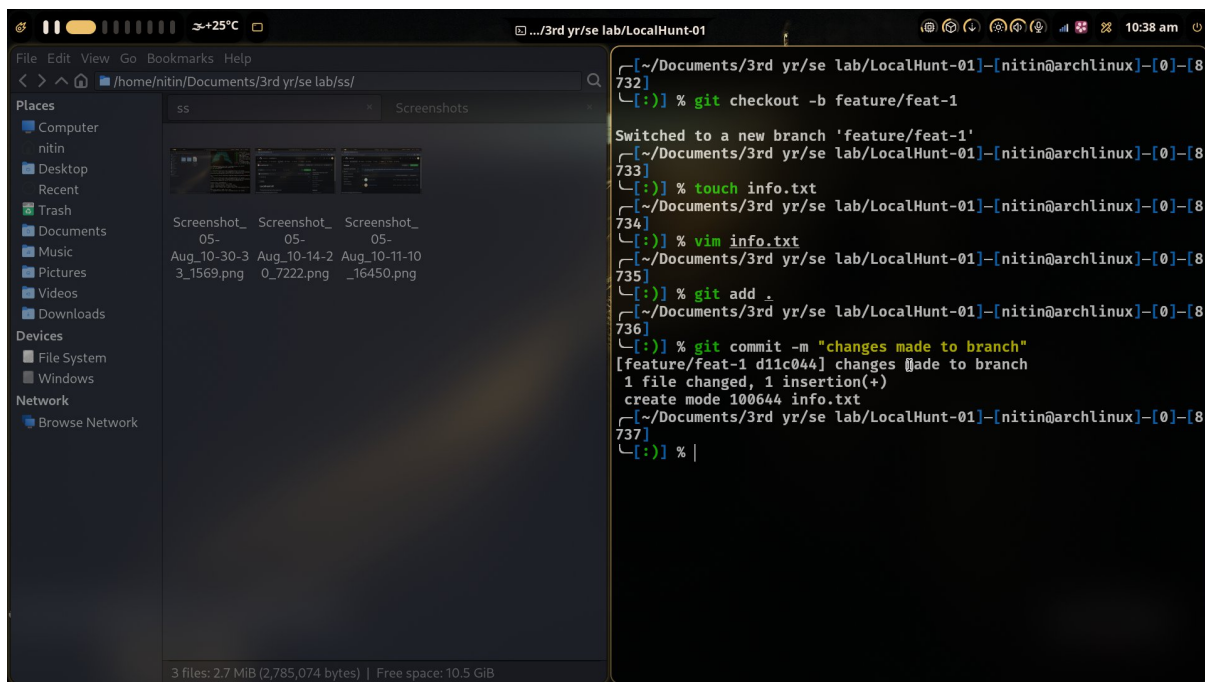
GitHub Repository Overview - LocalHunt-01 Private Repository



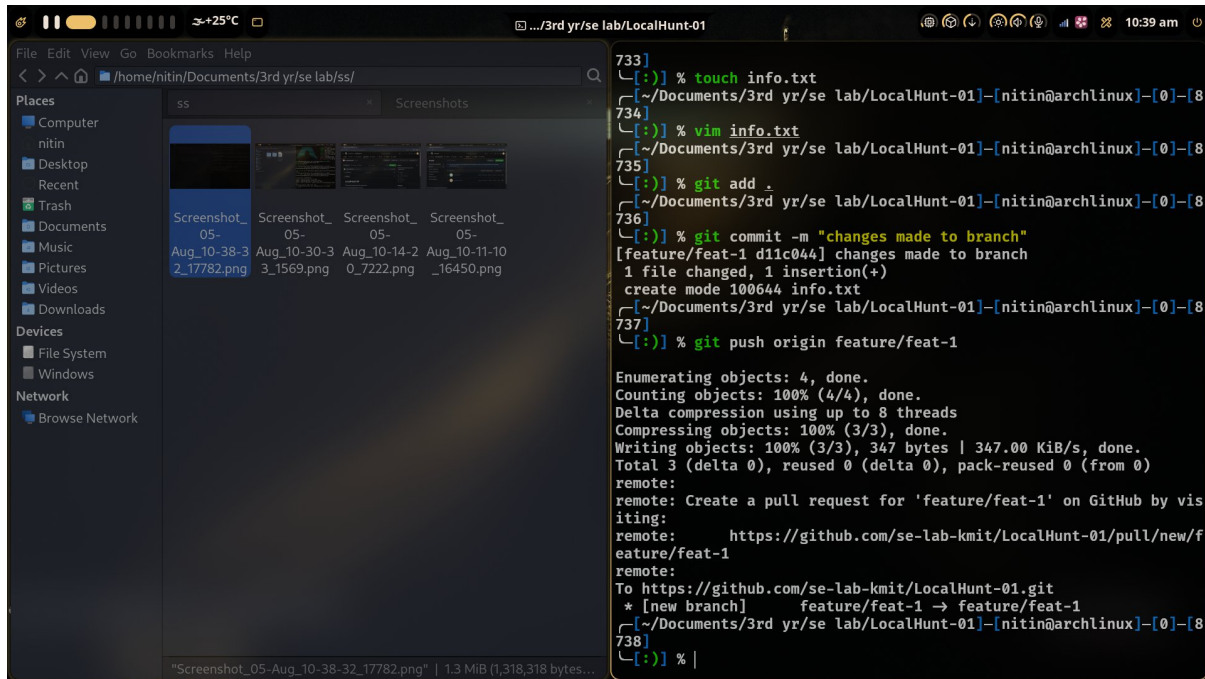
Terminal Git Clone Operations - LocalHunt-01 Repository Setup



Git Branch Operations - Feature Branch Creation and File Management



Git Push and Pull Request Creation - Feature Branch Workflow

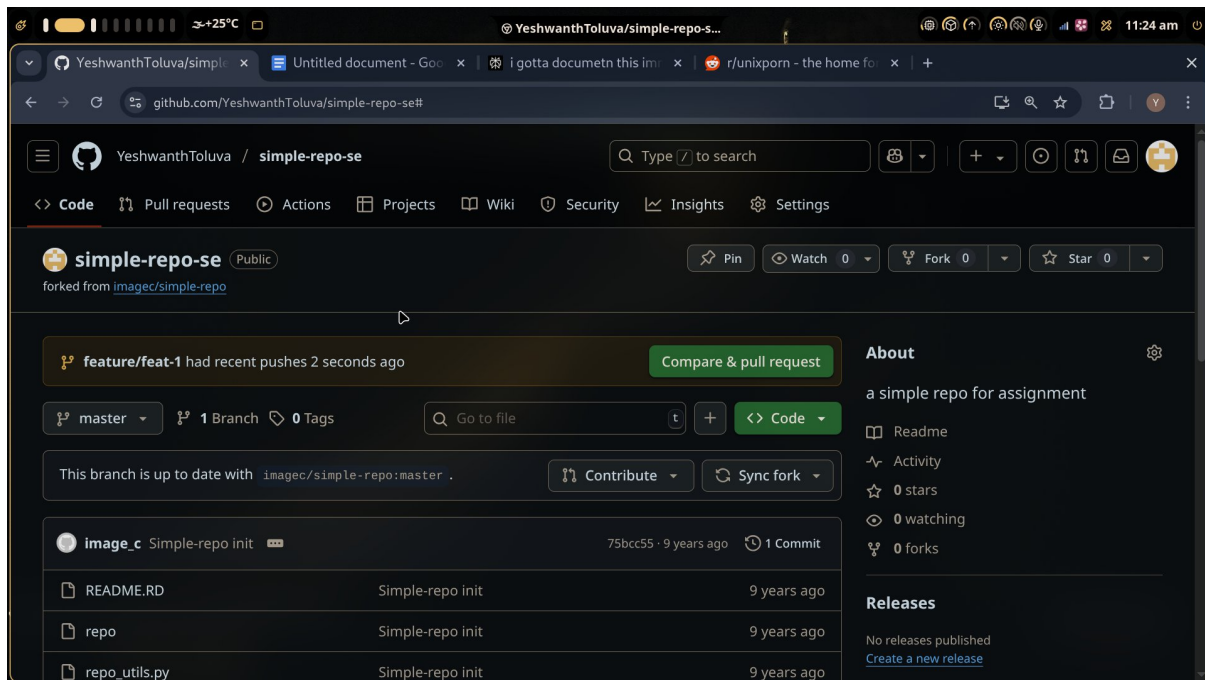


A terminal window on a Linux system showing a series of commands and their outputs. The user is in a directory `~/Documents/3rd yr/se lab/LocalHunt-01`. The commands executed are:

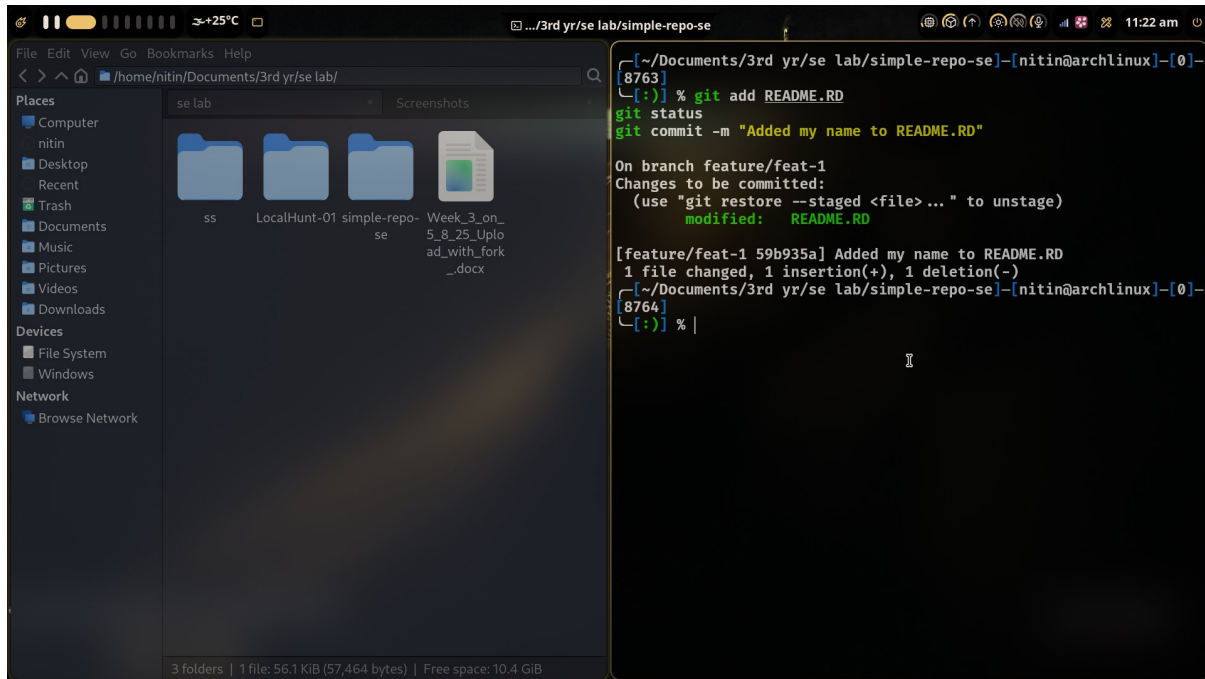
- `touch info.txt`
- `vim info.txt`
- `git add .`
- `git commit -m "changes made to branch"`
- `git push origin feature/feat-1`

The output of the push command shows the objects being enumerated, counted, and compressed, and then a pull request is created on GitHub for the `feature/feat-1` branch.

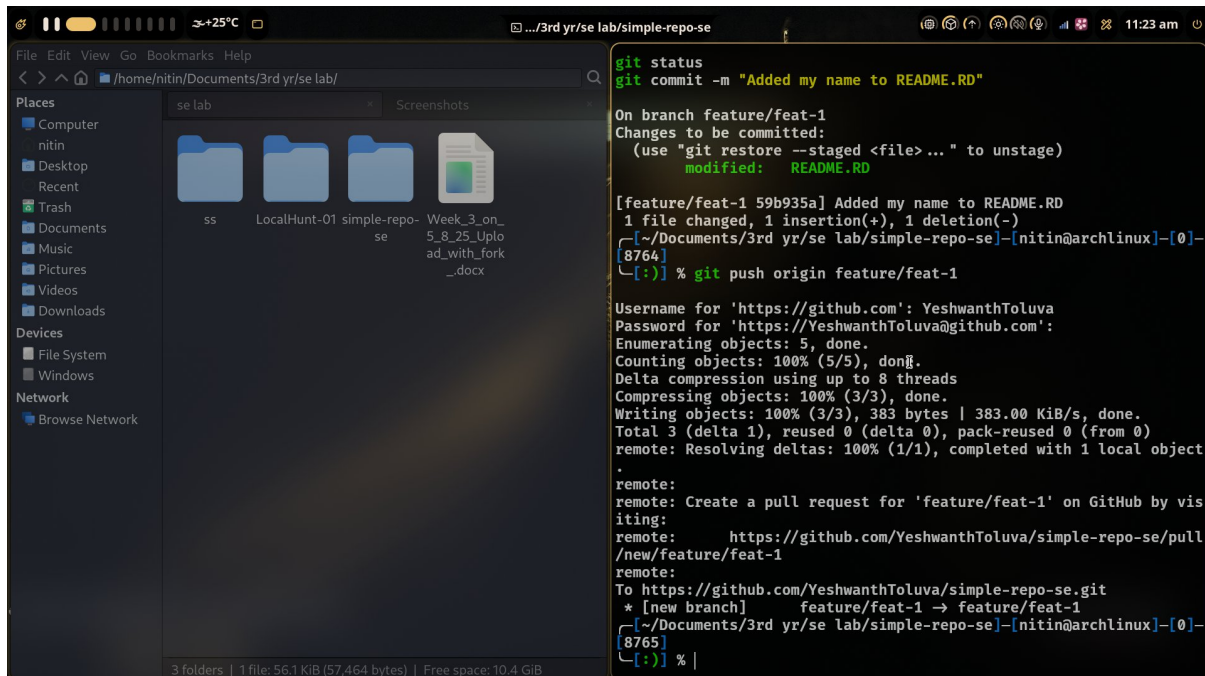
GitHub Repository Fork - simple-repo-se Overview and Setup



Git Commit and Status - README.RD File Modifications in Feature Branch

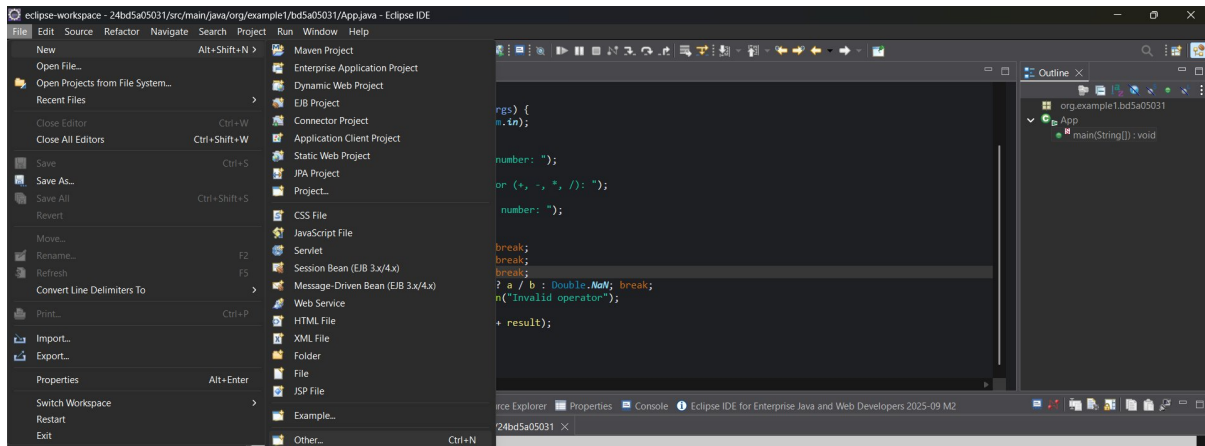


Git Push to Forked Repository - Feature Branch Upload and Pull Request Creation

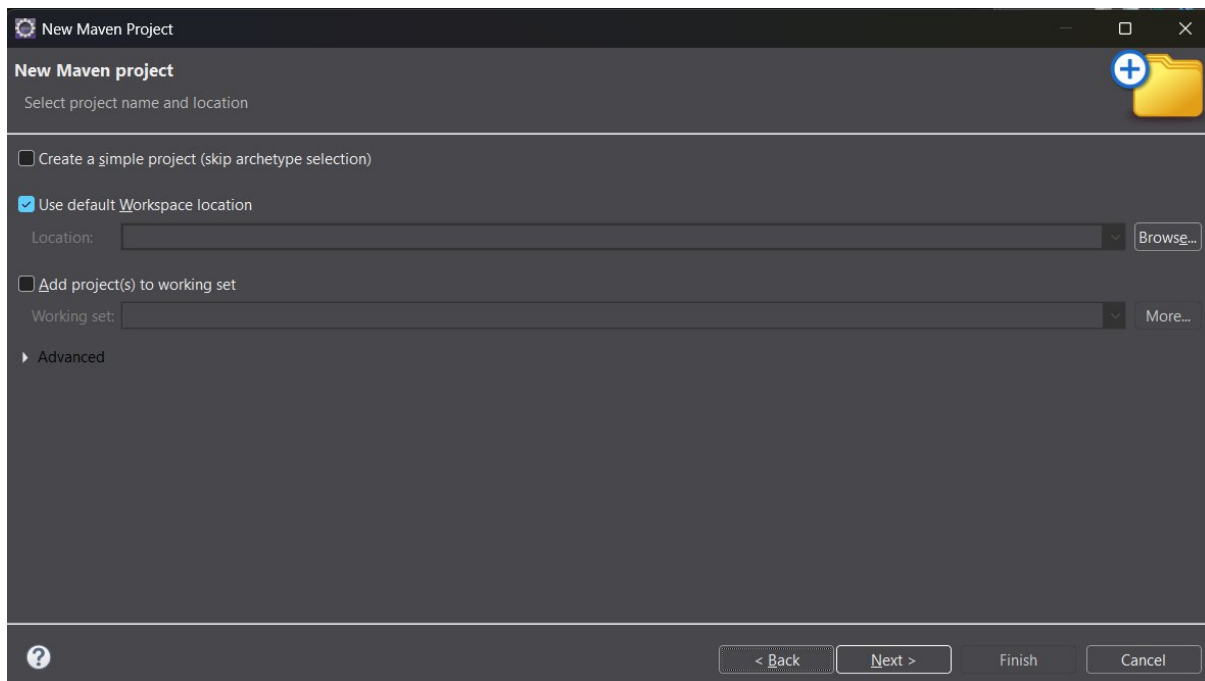


4. Build and package Java and Web applications using Maven

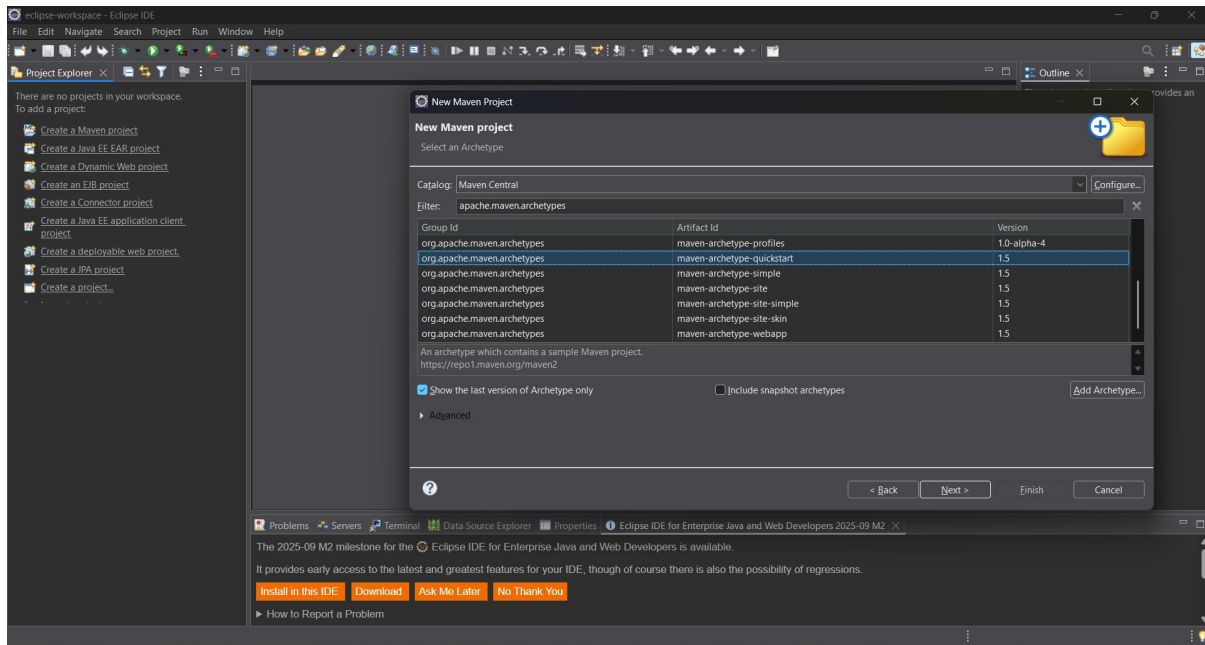
Step-1: Open the eclipse and click on file>new>Maven project



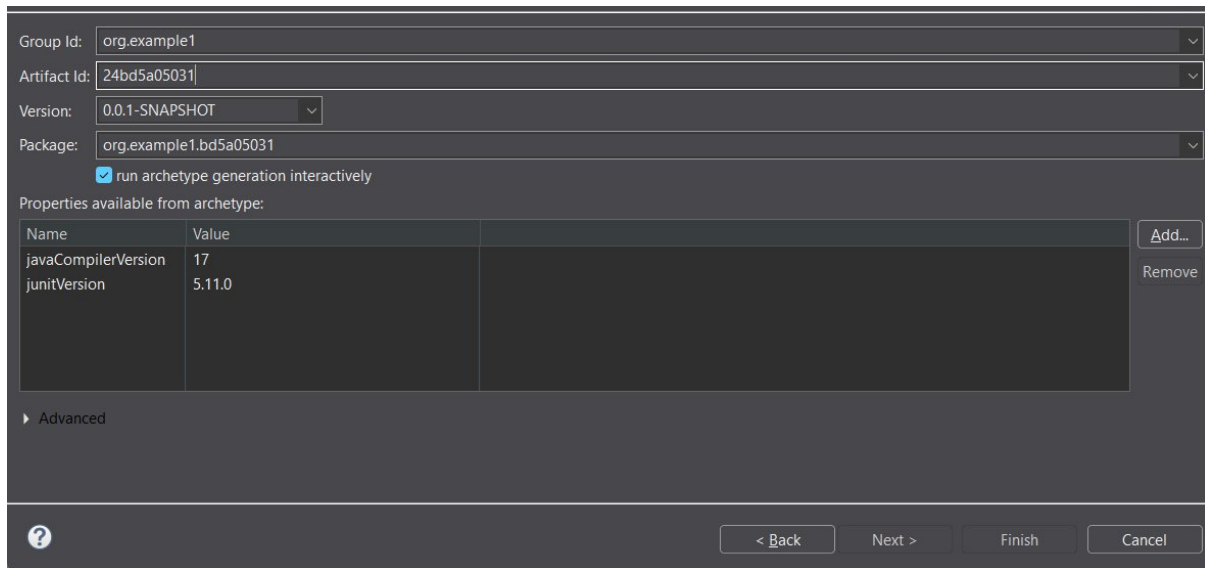
Step-2: select the default workspace and click on next



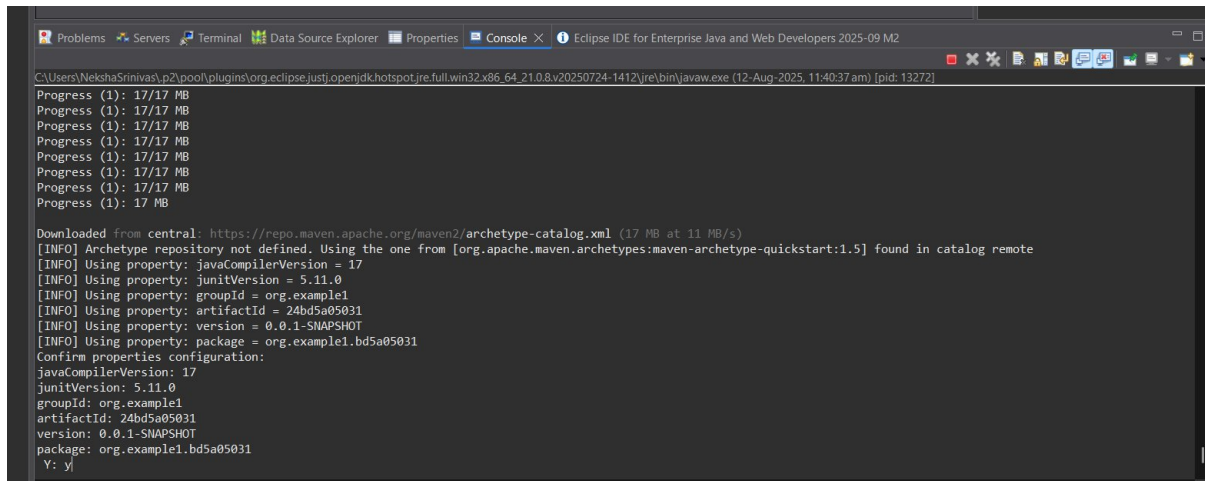
Step-3: in the filter option select the one maven-archetype-quickstart



Step-4: give the Group Id and Artifact Id and click on next



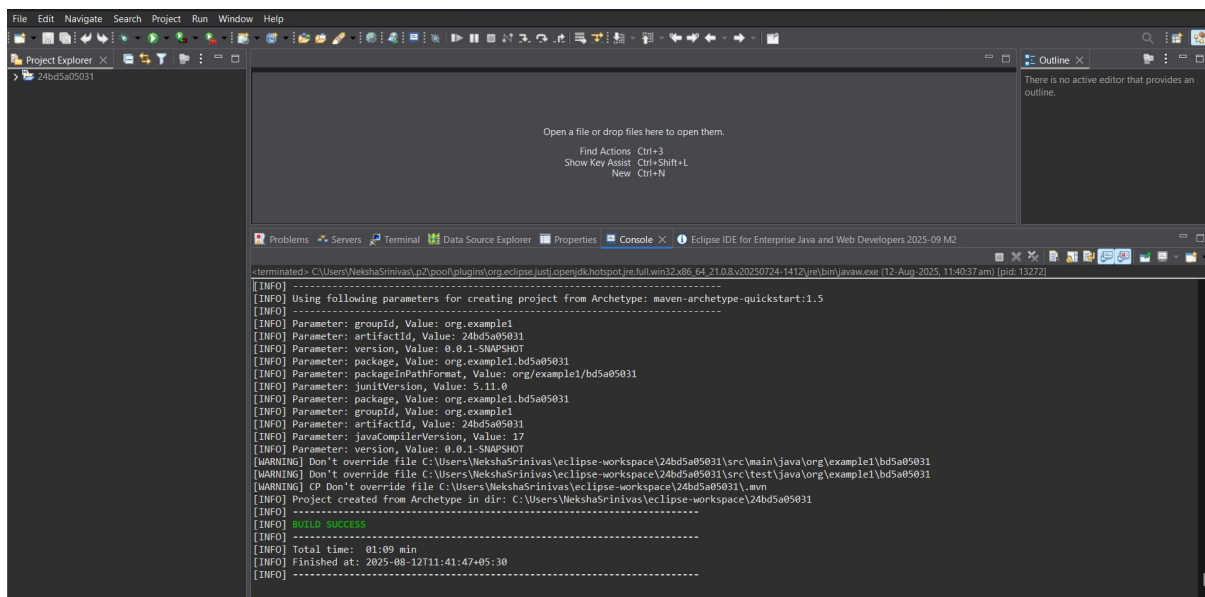
Step-5: In the console the progress will be showed type y (refers to yes) and press enter



The screenshot shows the Eclipse IDE console with the following output:

```
C:\Users\NekshaSrinivas\p2\pool\plugins\org.eclipse.justi.openjdk.hotspot.jre.full.win32.x86_64.21.0.8.v20250724-1412\jre\bin\javaw.exe (12-Aug-2025, 11:40:37 am) [pid: 13272]
Progress (1): 17/17 MB
Progress (1): 17/17 MB
Progress (1): 17/17 MB
Progress (1): 17/17 MB
Progress (1): 17/17 MB
Progress (1): 17/17 MB
Progress (1): 17 MB
Downloaded from central: https://repo.maven.apache.org/maven2/archetype-catalog.xml (17 MB at 11 MB/s)
[INFO] Archetype repository not defined, Using the one from [org.apache.maven.archetypes:maven-archetype-quickstart:1.5] found in catalog remote
[INFO] Using property: javaCompilerVersion = 17
[INFO] Using property: junitVersion = 5.11.0
[INFO] Using property: groupId = org.example1
[INFO] Using property: artifactId = 24bd5a05031
[INFO] Using property: version = 0.0.1-SNAPSHOT
[INFO] Using property: package = org.example1.bd5a05031
Confirm properties configuration:
javaCompilerVersion: 17
junitVersion: 5.11.0
groupId: org.example1
artifactId: 24bd5a05031
version: 0.0.1-SNAPSHOT
package: org.example1.bd5a05031
Y: y
```

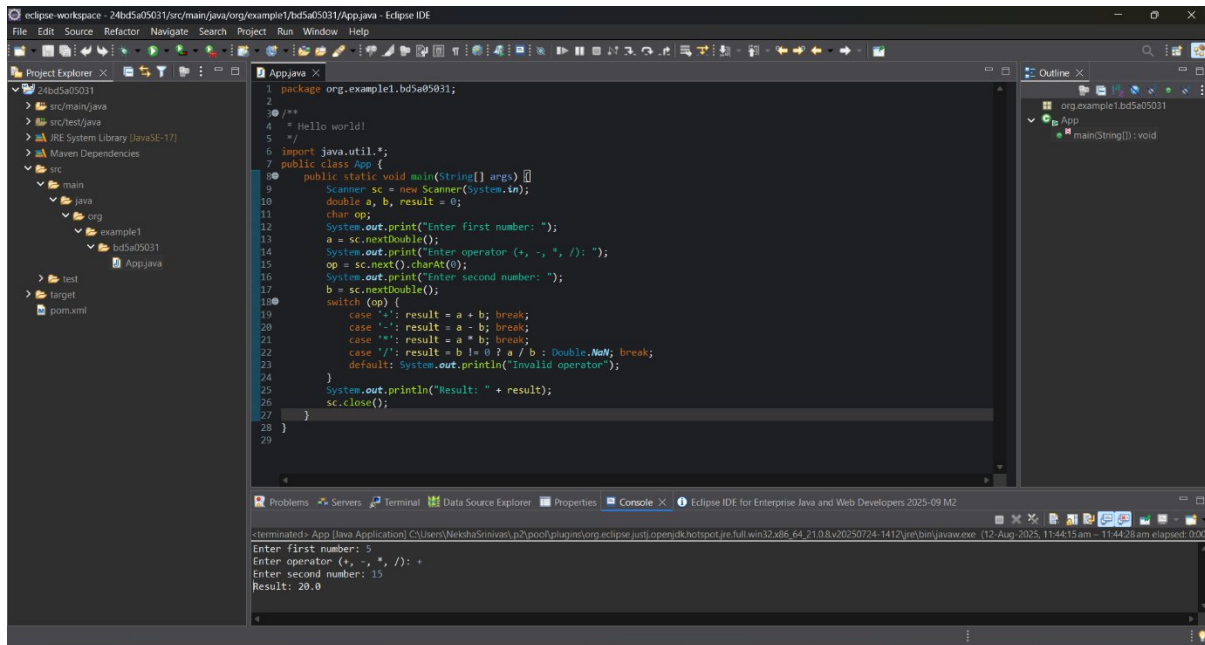
Step-6: BUILD SUCCESS will be shown



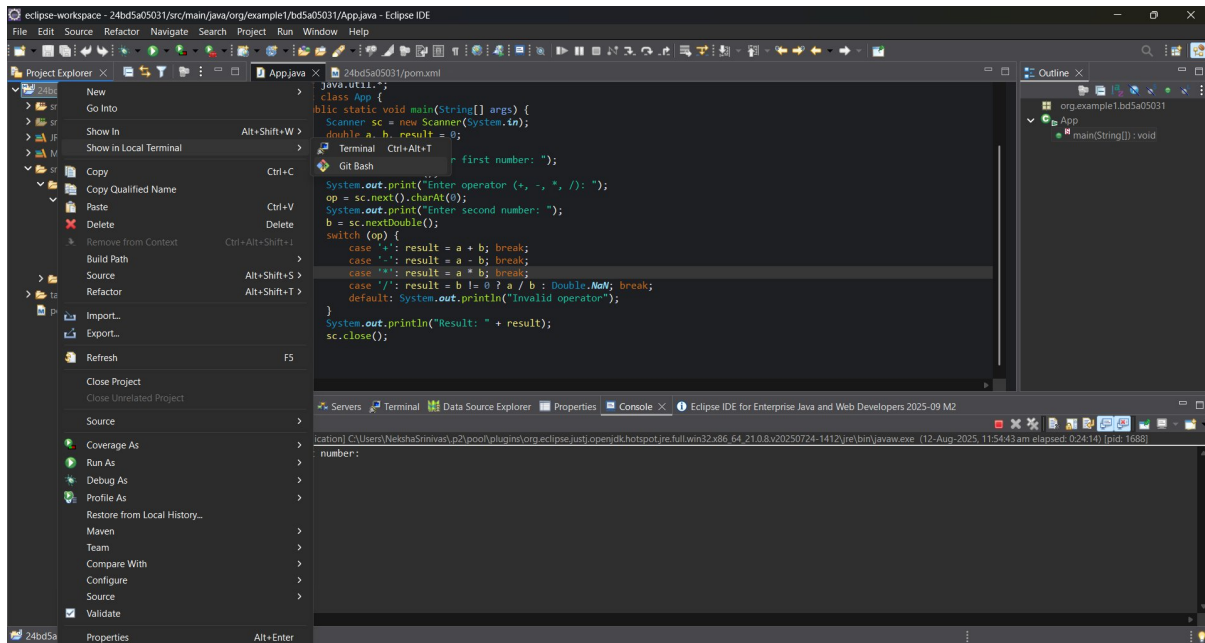
The screenshot shows the Eclipse IDE interface with the console output at the bottom. The Project Explorer on the left shows a project named '24bd5a05031'. The console output is as follows:

```
terminated C:\Users\NekshaSrinivas\p2\pool\plugins\org.eclipse.justi.openjdk.hotspot.jre.full.win32.x86_64.21.0.8.v20250724-1412\jre\bin\javaw.exe (12-Aug-2025, 11:40:37 am) [pid: 13272]
[INFO] -----
[INFO] Using following parameters for creating project from Archetype: maven-archetype-quickstart:1.5
[INFO] -----
[INFO] Parameter: groupId, Value: org.example1
[INFO] Parameter: artifactId, Value: 24bd5a05031
[INFO] Parameter: version, Value: 0.0.1-SNAPSHOT
[INFO] Parameter: package, Value: org.example1.bd5a05031
[INFO] Parameter: packageInPathFormat, Value: org/example1/bd5a05031
[INFO] Parameter: junitVersion, Value: 5.11.0
[INFO] Parameter: package, Value: org.example1.bd5a05031
[INFO] Parameter: groupId, Value: org.example1
[INFO] Parameter: artifactId, Value: 24bd5a05031
[INFO] Parameter: javaCompilerVersion, Value: 17
[INFO] Parameter: version, Value: 0.0.1-SNAPSHOT
[WARNING] Don't override file C:\Users\NekshaSrinivas\workspace\24bd5a05031\src\main\java\org\example1\bd5a05031
[WARNING] CP Don't override file C:\Users\NekshaSrinivas\workspace\24bd5a05031\src\test\java\org\example1\bd5a05031
[INFO] Project created from Archetype in dir: C:\Users\NekshaSrinivas\workspace\24bd5a05031
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 01:09 min
[INFO] Finished at: 2025-08-12T11:41:47+05:30
[INFO] -----
```

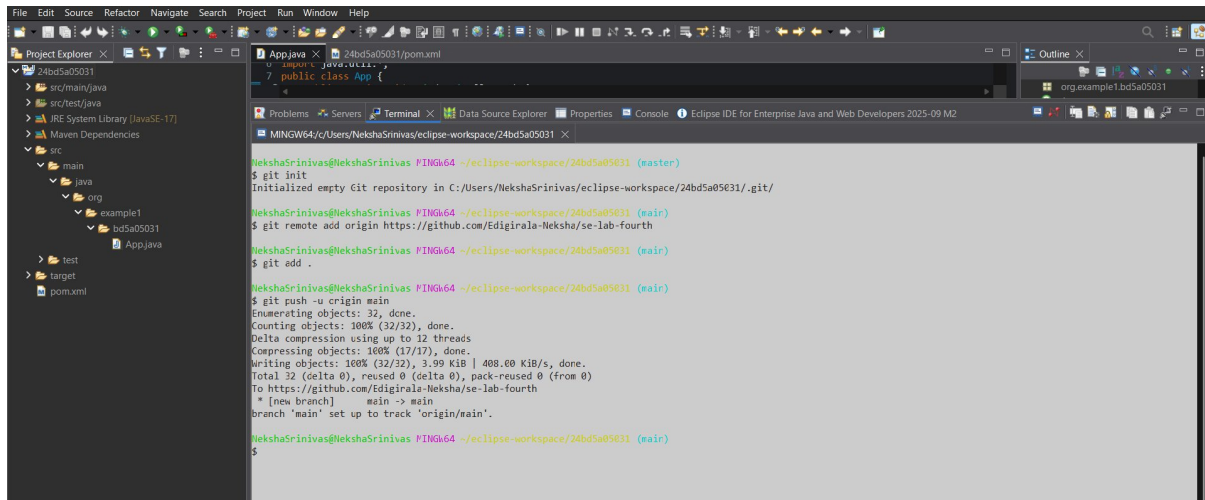
Step-6: write the code in the App.java file



Step-7: right click on the root folder and select show in git bash



Step-8: push to the git repo



The screenshot shows the Eclipse IDE interface. On the left is the Project Explorer showing a project named '24bd5a05031' with a file 'App.java'. The main editor displays the 'pom.xml' file. The bottom panel shows a terminal window with the following commands and output:

```
NekshaSrinivas@NekshaSrinivas ~/eclipse-workspace/24bd5a05031 (master)
$ git init
Initialized empty Git repository in C:/Users/NekshaSrinivas/eclipse-workspace/24bd5a05031/.git/

NekshaSrinivas@NekshaSrinivas ~/eclipse-workspace/24bd5a05031 (main)
$ git remote add origin https://github.com/Edigirala-Neksha/se-lab-fourth

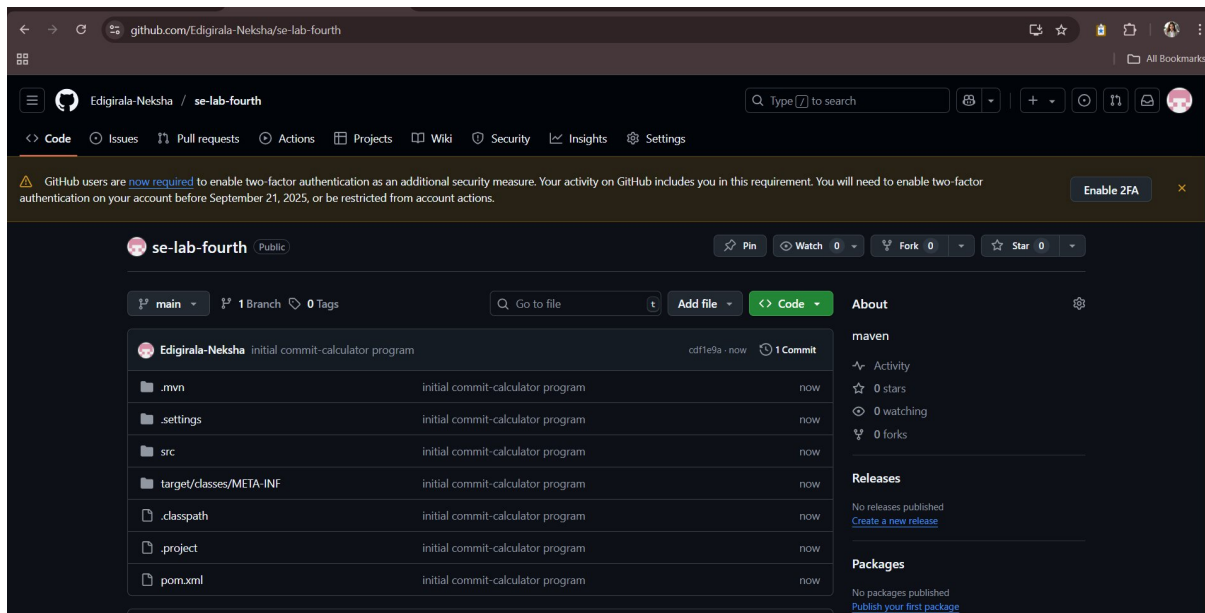
NekshaSrinivas@NekshaSrinivas ~/eclipse-workspace/24bd5a05031 (main)
$ git add .

NekshaSrinivas@NekshaSrinivas ~/eclipse-workspace/24bd5a05031 (main)
$ git push -u origin main
Enumerating objects: 32, done.
Counting objects: 100% (32/32), done.
Delta compression using up to 12 threads
Compressing objects: 100% (17/17), done.
Writing objects: 100% (32/32), 3.99 KiB | 408.00 KiB/s, done.
Total 32 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/Edigirala-Neksha/se-lab-fourth
 * [new branch] main -> main
branch 'main' set up to track 'origin/main'.

NekshaSrinivas@NekshaSrinivas ~/eclipse-workspace/24bd5a05031 (main)
$
```

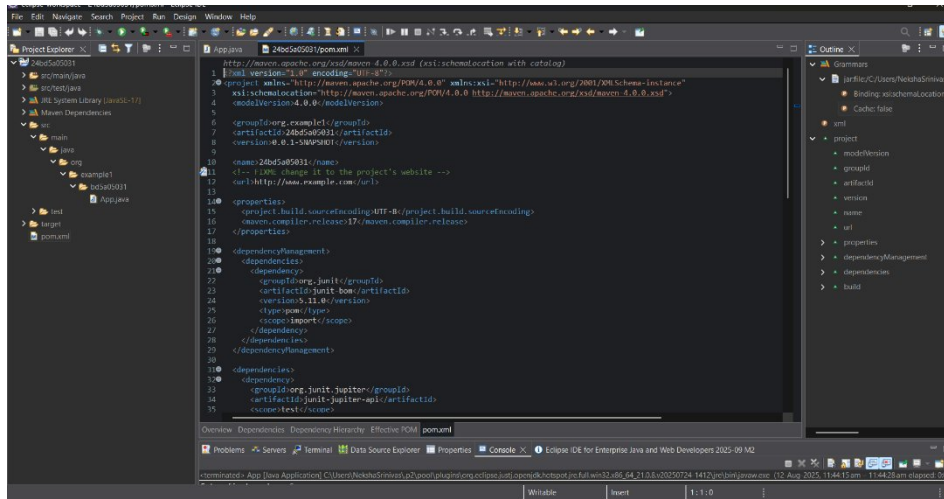
Git repo:

Git repo link: <https://github.com/Edigirala-Neksha/se-lab-fourth>



pom.xml file:

Shows the structure-



Creating maven-web project:

Step 1: Create a new maven project and give the details

New Maven Project

New Maven project

Specify Archetype parameters

Group Id:web-app-22

Artifact Id:24bd5a0503-webapp-22

Version:0.0.1-SNAPSHOT

Package:web_app_22.bd5a0503_webapp_22

☒ run archetype generation interactively

Properties available from archetype:

Name	Value	
		Add... Remove

Advanced

?

< Back

Next >

Finish

Cancel

Step 2: Click y to continue the creation of project

```
C:\Users\NekshaSrinivas\p2\pool\plugins\org.eclipse.justi.openjdk.hotspot.jre.full.win32.x86_64.21.0.8.v20250724-1412\jre\bin\javaw.exe (02-Sept-2025, 7:19:56 pm) [pid: 13772]
Progress (1): 17/17 MB
Progress (1): 17/17 MB
Progress (1): 17/17 MB
Progress (1): 17/17 MB
Progress (1): 17/17 MB
Progress (1): 17 MB

Downloaded from central: https://repo.maven.apache.org/maven2/archetype-catalog.xml (17 MB at 9.1 MB/s)
[INFO] Archetype repository not defined. Using the one from [org.apache.maven.archetypes:maven-archetype-webapp:1.5] found in catalog remote
[INFO] Using property: groupId = web-app-22
[INFO] Using property: artifactId = 24bd5a0503-webapp-22
[INFO] Using property: version = 0.0.1-SNAPSHOT
[INFO] Using property: package = web_app_22.bd5a0503_webapp_22
Confirm properties configuration:
groupId: web-app-22
artifactId: 24bd5a0503-webapp-22
version: 0.0.1-SNAPSHOT
package: web_app_22.bd5a0503_webapp_22
Y: y
```

Step 3: If the build is success it will show the message

```
package: web_app_22.bd5a0503_webapp_22
Y: y
[INFO] -----
[INFO] Using following parameters for creating project from Old (1.x) Archetype: maven-archetype-webapp:1.0
[INFO] -----
[INFO] Parameter: basedir, Value: C:\Users\NekshaSrinivas\eclipse-workspace
[INFO] Parameter: package, Value: web_app_22.bd5a0503_webapp_22
[INFO] Parameter: groupId, Value: web-app-22
[INFO] Parameter: artifactId, Value: 24bd5a0503-webapp-22
[INFO] Parameter: packageName, Value: web_app_22.bd5a0503_webapp_22
[INFO] Parameter: version, Value: 0.0.1-SNAPSHOT
[INFO] project created from Old (1.x) Archetype in dir: C:\Users\NekshaSrinivas\eclipse-workspace\24bd5a0503-webapp-22
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 43.500 s
[INFO] Finished at: 2025-09-02T19:20:41+05:30
[INFO] -----
```

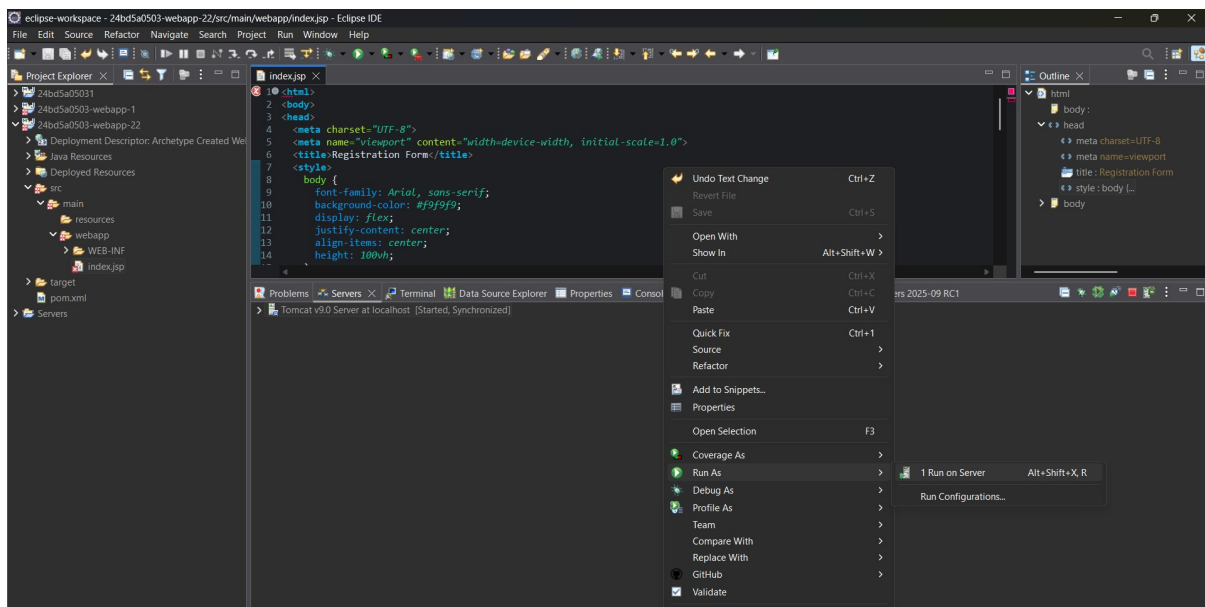
Step 4: write the html code for the web page:

```
index.jsp
1 <html>
2 <body>
3 <head>
4 <meta charset="UTF-8">
5 <meta name="viewport" content="width=device-width, initial-scale=1.0">
6 <title>Registration Form</title>
7 <style>
8   body {
9     font-family: Arial, sans-serif;
10    background-color: #f9f9f9;
11    display: flex;
12    justify-content: center;
13    align-items: center;
14    height: 100vh;
15  }
16  .form-container {
17    background: #fff;
18    padding: 20px 30px;
19    border-radius: 10px;
20    box-shadow: 0 4px 10px rgba(0,0,0,0.1);
21    width: 300px;
22  }
23  .form-container h2 {
24    text-align: center;
25    margin-bottom: 20px;
26  }
27  .form-container input {
28    width: 100%;
29    padding: 10px;
30    margin: 8px 0;
31    border: 1px solid #ccc;
32    border-radius: 5px;
33  }
34  .form-container button {
35    width: 100%;
36    padding: 10px;
37    background: #4CAF50;
```

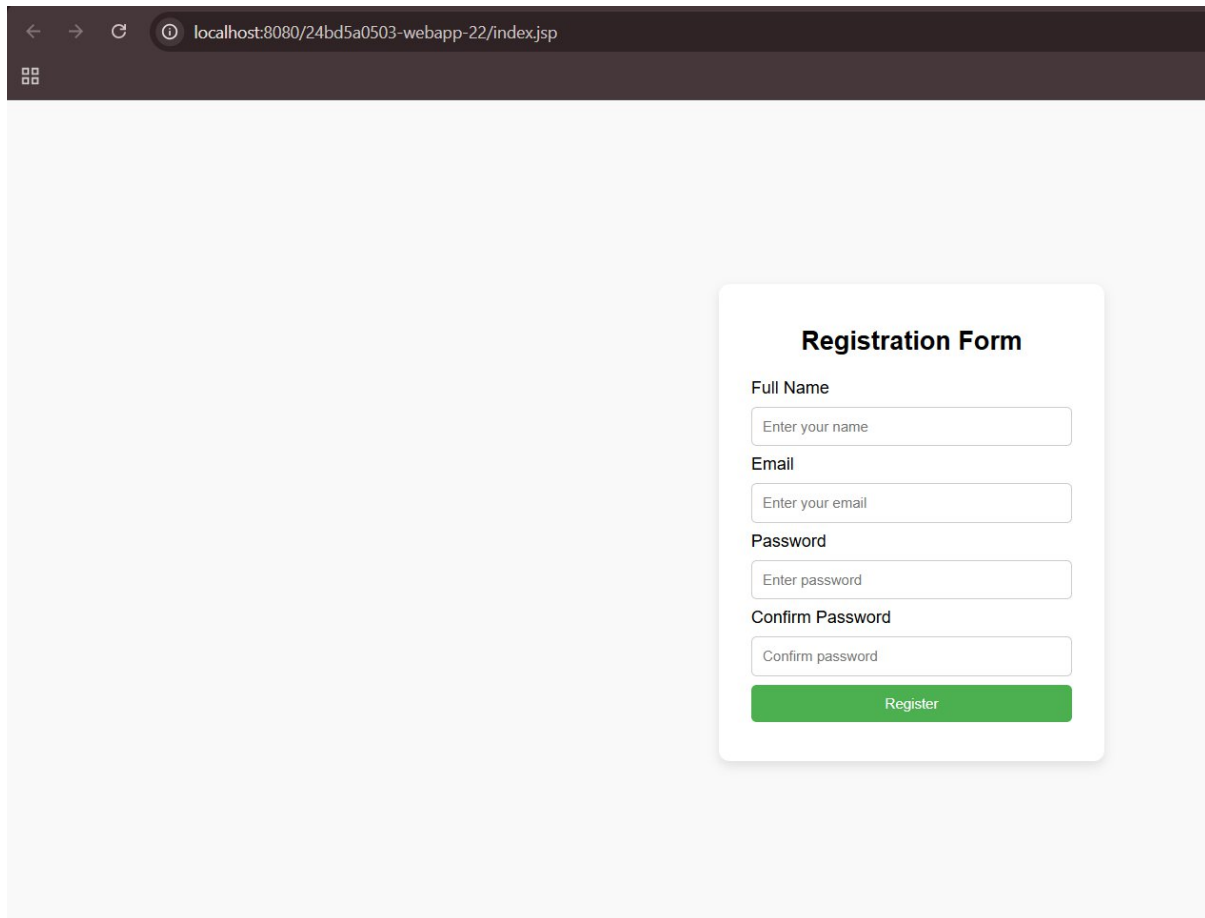
Web-page:

```
index.jsp X
32 border-radius: 5px;
33 }
34 .form-container button {
35 width: 100%;
36 padding: 10px;
37 background: #4CAF50;
38 color: white;
39 border: none;
40 border-radius: 5px;
41 cursor: pointer;
42 }
43 .form-container button:hover {
44 background: #45a049;
45 }
46 </style>
47 </head>
48 <body>
49 <div class="form-container">
50 <h2>Registration Form</h2>
51 <form action="#" method="post">
52 <label for="fullName">Full Name</label>
53 <input type="text" id="fullName" name="fullName" placeholder="Enter your name" required>
54
55 <label for="email">Email</label>
56 <input type="email" id="email" name="email" placeholder="Enter your email" required>
57
58 <label for="password">Password</label>
59 <input type="password" id="password" name="password" placeholder="Enter password" required>
60
61 <label for="confirm">Confirm Password</label>
62 <input type="password" id="confirm" name="confirm" placeholder="Confirm password" required>
63
64 <button type="submit">Register</button>
65 </form>
66 </div>
67 </body>
68 </html>
69
```

Step 5: Select run on server



Step 6: It will show the following output:



The screenshot shows a web browser window with the address bar displaying "localhost:8080/24bd5a0503-webapp-22/index.jsp". The browser's address bar has back, forward, and refresh buttons. Below the address bar, there is a grid icon. The main content area of the browser displays a registration form titled "Registration Form". The form contains four input fields: "Full Name" (placeholder: "Enter your name"), "Email" (placeholder: "Enter your email"), "Password" (placeholder: "Enter password"), and "Confirm Password" (placeholder: "Confirm password"). At the bottom of the form is a green button labeled "Register".

localhost:8080/24bd5a0503-webapp-22/index.jsp

Registration Form

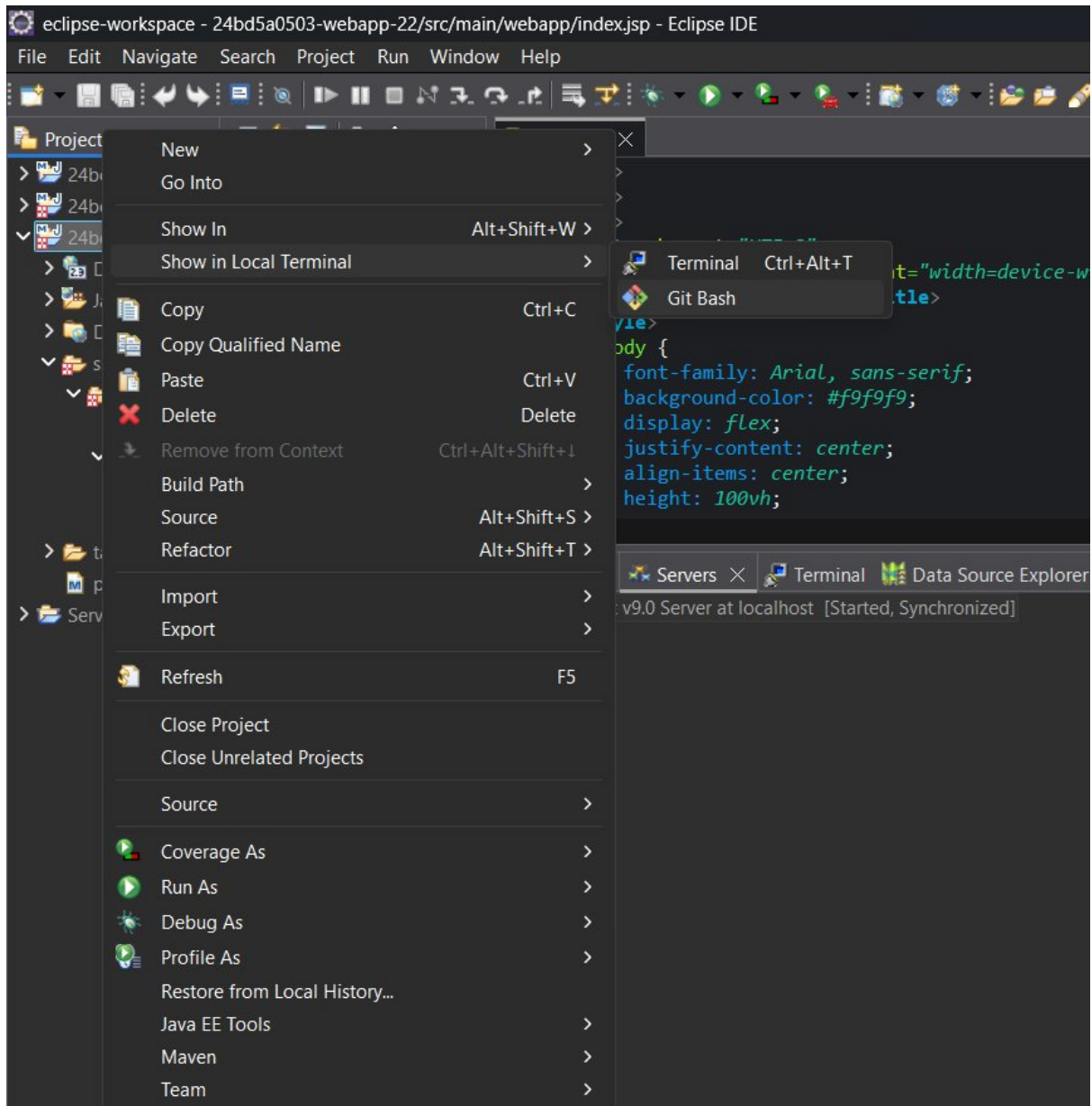
Full Name

Email

Password

Confirm Password

Step 7: To push it into git, select git bash from show in local terminal



Step 8: use the command of git to push the maven web project

```
MINGW64/c/Users/NekshaSrinivas/eclipse-workspace/24bd5a05... MINGW64/c/Users/NekshaSrinivas/eclipse-workspace/24bd5a05... MINGW64/c/Users/NekshaSrinivas/eclipse-workspace/24bd5a05...
NekshaSrinivas@NekshaSrinivas MINGW64 ~/eclipse-workspace/24bd5a0503-webapp-22 (master)
$ git init
Initialized empty Git repository in C:/Users/NekshaSrinivas/eclipse-workspace/24bd5a0503-webapp-22/.git/

NekshaSrinivas@NekshaSrinivas MINGW64 ~/eclipse-workspace/24bd5a0503-webapp-22 (main)
$ git add .

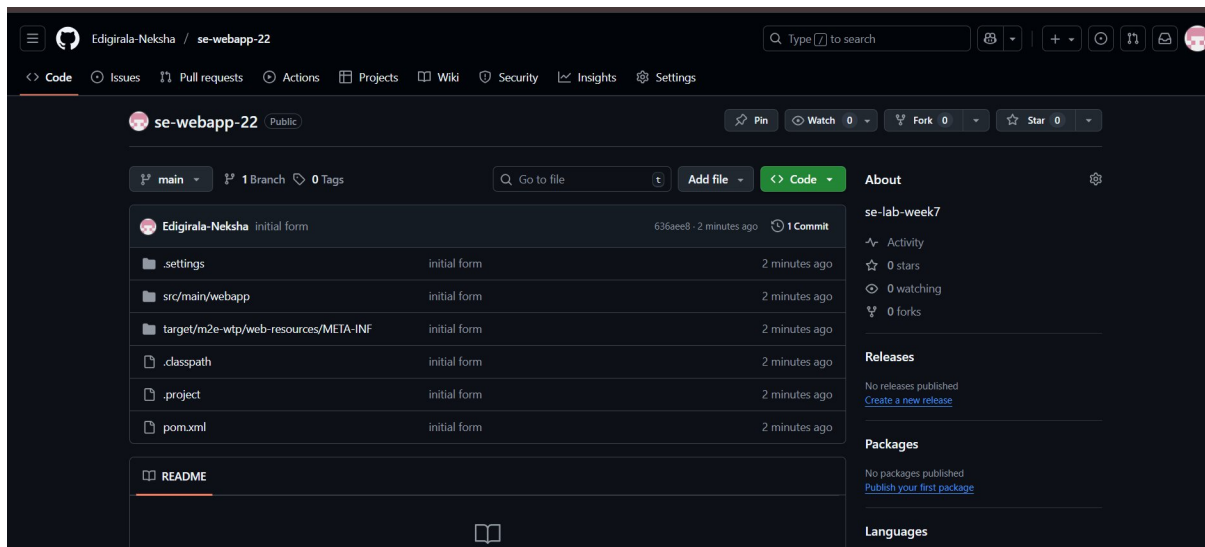
NekshaSrinivas@NekshaSrinivas MINGW64 ~/eclipse-workspace/24bd5a0503-webapp-22 (main)
$ git commit -m "initial form"
[main (root-commit) 636aee8] initial form
16 files changed, 254 insertions(+)
create mode 100644 .classpath
create mode 100644 .project
create mode 100644 .settings/.jsdt.scope
create mode 100644 .settings/crg.eclipse.jdt.core.preferences
create mode 100644 .settings/crg.eclipse.m2e.core.preferences
create mode 100644 .settings/crg.eclipse.wst.common.component
create mode 100644 .settings/crg.eclipse.wst.common.project.facet.core.xml
create mode 100644 .settings/crg.eclipse.wst.jsdt.ui.superType.container
create mode 100644 .settings/crg.eclipse.wst.jsdt.ui.superType.name
create mode 100644 .settings/crg.eclipse.wst.validation.preferences
create mode 100644 pom.xml
create mode 100644 src/main/webapp/WEB-INF/web.xml

NekshaSrinivas@NekshaSrinivas MINGW64 ~/eclipse-workspace/24bd5a0503-webapp-22 (main)
$ git branch
* main

NekshaSrinivas@NekshaSrinivas MINGW64 ~/eclipse-workspace/24bd5a0503-webapp-22 (main)
$ git push origin main
Enumerating objects: 29, done.
Counting objects: 100% (29/29), done.
Delta compression using up to 12 threads
Compressing objects: 100% (18/18), done.
Writing objects: 100% (29/29), 4.43 KiB | 283.00 KiB/s, done.
Total 29 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), done.
To https://github.com/Edigirala-Neksha/se-webapp-22.git
 * [new branch]      main -> main

NekshaSrinivas@NekshaSrinivas MINGW64 ~/eclipse-workspace/24bd5a0503-webapp-22 (main)
$
```

Step 9: verify the repo in git hub



5. Docker CLI commands

Installing Docker and Setting up Nginx

Introduction

Docker is a platform that allows us to run applications inside lightweight containers. Containers are isolated environments that include everything needed to run an application. This makes it easy to set up and deploy software without worrying about dependencies or configurations on the host system.

In this task, we used Docker to run an **Ubuntu container**, install **nginx** inside it, and serve a customized homepage

Step 1: Pulling the Ubuntu Image

First, we pulled the latest Ubuntu image from Docker Hub.

```
PS C:\Users\NekshaSrinivas> docker --version
Docker version 28.3.2, build 578ccf6
PS C:\Users\NekshaSrinivas> cd SE-1
PS C:\Users\NekshaSrinivas\SE-1> docker --version
Docker version 28.3.2, build 578ccf6
PS C:\Users\NekshaSrinivas\SE-1> docker pull ubuntu:latest
latest: Pulling from library/ubuntu
b71466b94f26: Pull complete
Digest: sha256:7c06e91f61fa88c08cc74f7e1b7c69ae24910d745357e0dfe1d2c0322aaf20f9
Status: Downloaded newer image for ubuntu:latest
docker.io/library/ubuntu:latest
```

Step 2: Running the Container

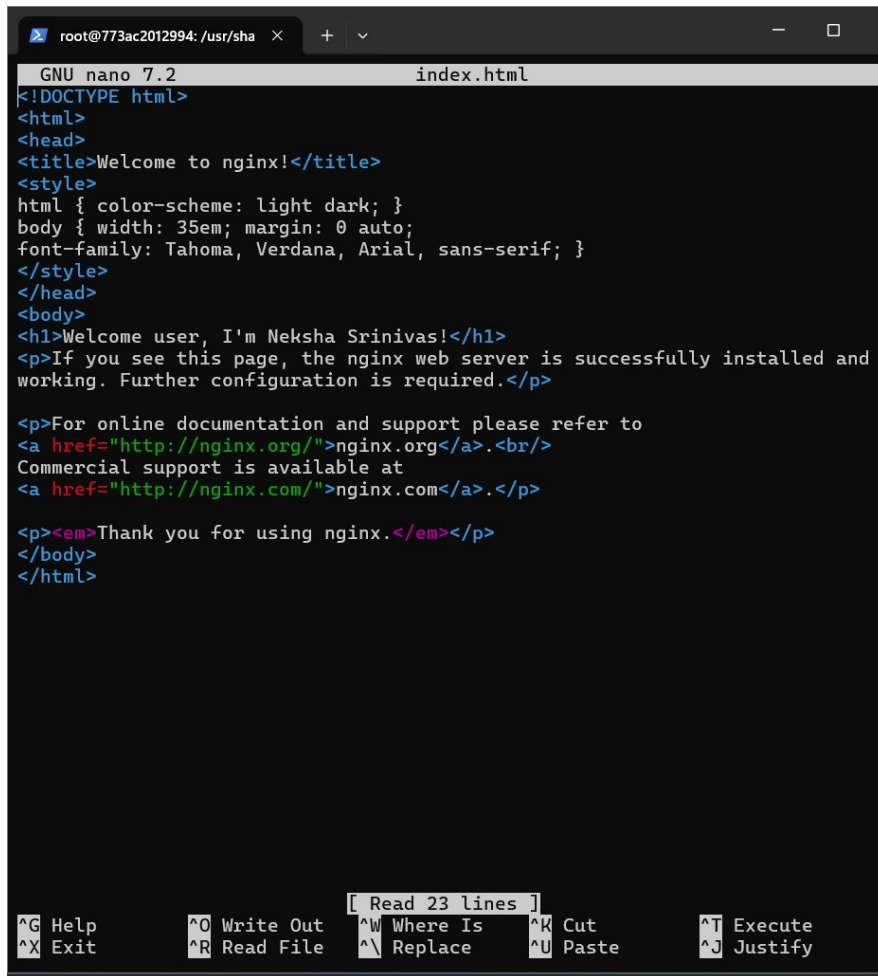
We created and started a new container named **myubuntu**, mapping port **3000** on the host to port **80** inside the container.

```
PS C:\Users\NekshaSrinivas\SE-1> docker run -it -p 9090:80 --name myubuntu1 ubuntu:latest
root@773ac2012994:/# apt update
Get:1 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble InRelease [256 kB]
Get:3 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [1135 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:5 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1355 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:7 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [2047 kB]
Get:8 http://archive.ubuntu.com/ubuntu noble/main amd64 Packages [1808 kB]
Get:9 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Packages [23.0 kB]
Get:10 http://archive.ubuntu.com/ubuntu noble/multiverse amd64 Packages [331 kB]
```

Step 3: Installing Nginx and redirecting to index.html page to edit the content

```
Processing triggers for libc-bin (2.39-0ubuntu8.5) ...
root@773ac2012994:/# ls
bin      dev      lib      mnt      root    sbin    usr-is-merged  tmp
bin.usr-is-merged  etc     lib64    opt      run     srv      usr
boot     home    media    proc     sbin    sys      var
root@773ac2012994:/# cd usr
root@773ac2012994:/usr# ls
bin  games  include  lib  lib64  libexec  local  sbin  share  src
root@773ac2012994:/usr# cd share
root@773ac2012994:/usr/share# ls
apport          gcc             pam
base-files      gdb             pam-configs
base-passwd     info           perl5
bash-completion info.dir        pixmaps
bug             keyrings        polkit-1
common-licenses lib64-bin        profile
debconf         libgcrypt20     profile.md5sums
debianutils     lintian          sensible-utils
dict            locale          staff-group-for-usr-local
doc             man             tabset
doc-base       menu            terminfo
dot.bashrc      misc            util-linux
dot.profile     motd            vim
dot.profile.md5sums  networks
dpkg            nginx
root@773ac2012994:/usr/share# cd nginx
root@773ac2012994:/usr/share/nginx# ls
html  modules
root@773ac2012994:/usr/share/nginx# cd html
root@773ac2012994:/usr/share/nginx/html# ls
index.html
root@773ac2012994:/usr/share/nginx/html# nano index.html
bash: nano: command not found
root@773ac2012994:/usr/share/nginx/html# apt install nano
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Suggested packages:
hunspell
The following NEW packages will be installed:
nano
```

Step 4: navigate to index.html using command – “nano index.html” Changed the content of h1 tag

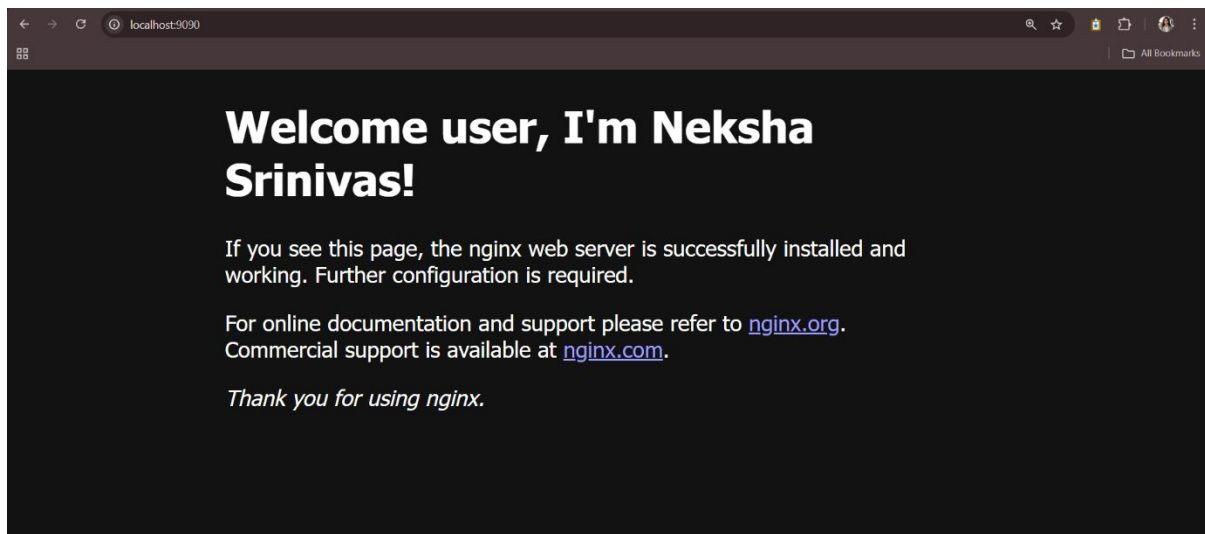


```
GNU nano 7.2 index.html
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome user, I'm Neksha Srinivas!</h1>
<p>If you see this page, the nginx web server is successfully installed and
working. Further configuration is required.</p>

<p>For online documentation and support please refer to
<a href="http://nginx.org/">nginx.org</a>.<br/>
Commercial support is available at
<a href="http://nginx.com/">nginx.com</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
```

Step 5: Viewing the page from local host



6. Docker

DOCKER IMAGE CREATION:

Image can be created in two ways:

1. Using Docker commit
2. Using docker file

Step 1: Created a new container of ubuntu so image can be created on that container

```
PS C:\Users\NekshaSrinivas\SE-1> docker run -it -p 9090:80 --name ubuntu-cont-1 ubuntu:latest
root@74098c332e58:/# apt update
Get:1 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble InRelease [256 kB]
Get:3 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [1137 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:5 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [2066 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:7 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1363 kB]
Get:8 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Packages [23.0 kB]
Get:9 http://archive.ubuntu.com/ubuntu noble/restricted amd64 Packages [117 kB]
Get:10 http://archive.ubuntu.com/ubuntu noble/universe amd64 Packages [19.3 MB]
Get:11 http://archive.ubuntu.com/ubuntu noble/main amd64 Packages [1808 kB]
Get:12 http://archive.ubuntu.com/ubuntu noble/multiverse amd64 Packages [331 kB]
Get:13 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Packa
```

Step 2: Using commit the image is being created:

```
PS C:\Users\NekshaSrinivas\SE-1> docker commit ubuntu-cont-1 img-commit-1
sha256:153126502820131f25f36cc59f7c4557275621bc7a54b48c8ffd4409685efd
PS C:\Users\NekshaSrinivas\SE-1> docker images
REPOSITORY          TAG                 IMAGE ID            CREATED             SIZE
img-commit-1        latest             153126502820       7 seconds ago      326MB
mynginx             latest             de77ca8d52cb       30 hours ago       279MB
mypythonapp         latest             8a39b6d82115       30 hours ago       1.63GB
nginx               latest             33e0bbc7ca9e       12 days ago        279MB
ubuntu              latest             7c06e91f61fa       3 weeks ago        117MB
PS C:\Users\NekshaSrinivas\SE-1> docker run -it img-commit-1
root@909ab066a51f:/# git --version
git version 2.43.0
root@909ab066a51f:/# docker tag img-commit-1 nekshasrinivas/img-commit-1
bash: docker: command not found
root@909ab066a51f:/# exit
exit
PS C:\Users\NekshaSrinivas\SE-1> docker tag img-commit-1 nekshasrinivas/img-commit-1
PS C:\Users\NekshaSrinivas\SE-1> docker push nekshasrinivas/img-commit-1
Using default tag: latest
The push refers to repository [docker.io/nekshasrinivas/img-commit-1]
4024494ad21b: Pushed
b71466b94f26: Mounted from library/ubuntu
latest: digest: sha256:153126502820131f25f36cc59f7c4557275621bc7a54b48c8ffd4409685efd size: 751
```

Step 3: Image creation using docker file

```
PS C:\Users\NekshaSrinivas\SE-1> mkdir image-creation

Directory: C:\Users\NekshaSrinivas\SE-1

Mode                LastWriteTime         Length Name
----                -
d-----            26-08-2025     18:26             image-creation

PS C:\Users\NekshaSrinivas\SE-1> ls

Directory: C:\Users\NekshaSrinivas\SE-1

Mode                LastWriteTime         Length Name
----                -
d-----            26-08-2025     18:26             image-creation
d-----            25-08-2025     12:25             static_site
-a-----            25-08-2025     12:15              36 app.py
-a-----            25-08-2025     12:15             100 Dockerfile
-a-----            23-08-2025     16:28        29739008 myapi.tar
-a-----            05-08-2025     16:33              38 README.md

PS C:\Users\NekshaSrinivas\SE-1> cd image-creation
PS C:\Users\NekshaSrinivas\SE-1\image-creation> notepad Dockerfile
PS C:\Users\NekshaSrinivas\SE-1\image-creation> ls

Directory: C:\Users\NekshaSrinivas\SE-1\image-creation

Mode                LastWriteTime         Length Name
----                -
-a-----            26-08-2025     18:27              59 Dockerfile.txt

PS C:\Users\NekshaSrinivas\SE-1\image-creation> ren Dockerfile.txt Dockerfile
```

Step 4: after writing the content in docker file use the command docker build

```
PS C:\Users\NekshaSrinivas\SE-1\image-creation> docker build -t img-dockerfile-1 .
[+] Building 44.0s (7/7) FINISHED
=> [internal] load build definition from Dockerfile 0.1s
=> => transferring dockerfile: 96B 0.0s
=> [internal] load metadata for docker.io/library/ubuntu:latest 0.1s
=> [internal] load .dockerignore 0.1s
=> => transferring context: 2B 0.0s
=> [1/3] FROM docker.io/library/ubuntu:latest@sha256:7c06e91f61fa88c 0.1s
=> => resolve docker.io/library/ubuntu:latest@sha256:7c06e91f61fa88c 0.0s
=> [2/3] RUN apt-get update 12.5s
=> [3/3] RUN apt-get install git -y 24.4s
=> exporting to image 6.5s
=> => exporting layers 4.5s
=> => exporting manifest sha256:99d816a6b717e709d838937a995f24d0121e 0.0s
=> => exporting config sha256:f021a40f65d4b684b65cd403292af90ec68210 0.0s
=> => exporting attestation manifest sha256:04dc38eee96b84e155b083e5 0.1s
=> => exporting manifest list sha256:9868ecb2df510b52e539c55076bf63c 0.0s
=> => naming to docker.io/library/img-dockerfile-1:latest 0.0s
=> => unpacking to docker.io/library/img-dockerfile-1:latest 1.8s
PS C:\Users\NekshaSrinivas\SE-1\image-creation> docker run -it img-dockerfile-1
root@adfe97a50685:/# docker --version
bash: docker: command not found
root@adfe97a50685:/# git --version
git version 2.43.0
root@adfe97a50685:/# exit
exit
PS C:\Users\NekshaSrinivas\SE-1\image-creation> docker tag img-dockerfile-1 nekshasrinivas/img-dockerfile-1
PS C:\Users\NekshaSrinivas\SE-1\image-creation> docker push nekshasrinivas/img-dockerfile-1
Using default tag: latest
The push refers to repository [docker.io/nekshasrinivas/img-dockerfile-1]
6a5ccfd4b031: Pushed
eddb67216c21: Pushed
b71466b94f26: Mounted from nekshasrinivas/img-commit-1
004a734bd8b1: Pushed
latest: digest: sha256:9868ecb2df510b52e539c55076bf63ccae47b54ab67e29de352ddbc3cb33b109 size: 855
```

Step 5: checking the images

```
PS C:\Users\NekshaSrinivas\SE-1> docker images
REPOSITORY              TAG         IMAGE ID      CREATED        SIZE
img-dockerfile-1        latest      9868ecb2df51  5 minutes ago  326MB
nekshasrinivas/img-dockerfile-1  latest      9868ecb2df51  5 minutes ago  326MB
img-commit-1            latest      153126502820 12 minutes ago 326MB
nekshasrinivas/img-commit-1      latest      153126502820 12 minutes ago 326MB
mynginx                 latest      de77ca8d52cb  30 hours ago   279MB
mypythonapp             latest      8a39b6d82115  30 hours ago   1.63GB
nginx                   latest      33e0bbc7ca9e  12 days ago    279MB
ubuntu                  latest      7c06e91f61fa  3 weeks ago    117MB
PS C:\Users\NekshaSrinivas\SE-1> |
```

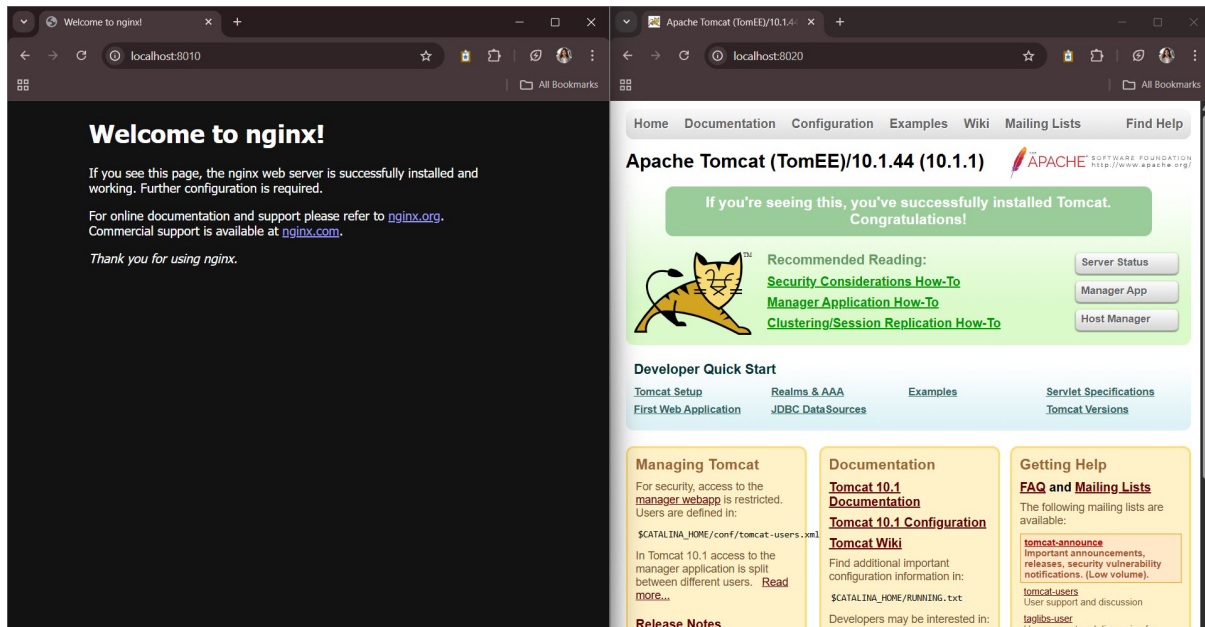
DOCKER COMPOSE FILE:

Docker Compose is a tool used to define and run multi-container Docker applications. It allows you to define services, networks, and volumes that your application needs, all in a single file. This makes it easier to manage complex applications that require multiple containers (e.g., a web server and a database).

Step 1: Running two servers at the same time on different ports

```
PS C:\Users\NekshaSrinivas\SE-1> docker run -d -p 8010:80 nginx
2ea4a201f197b93276310a7d23f2a46060ba9c7387f869e8a2a804931b66b2d9
PS C:\Users\NekshaSrinivas\SE-1> docker run -d -p 8020:8080 tomee
3a524036f6b212843be468585f80fb029aed07715a8e33a38e4eb306044765a2
PS C:\Users\NekshaSrinivas\SE-1> |
```

Step 2: Open the local host to view the pages



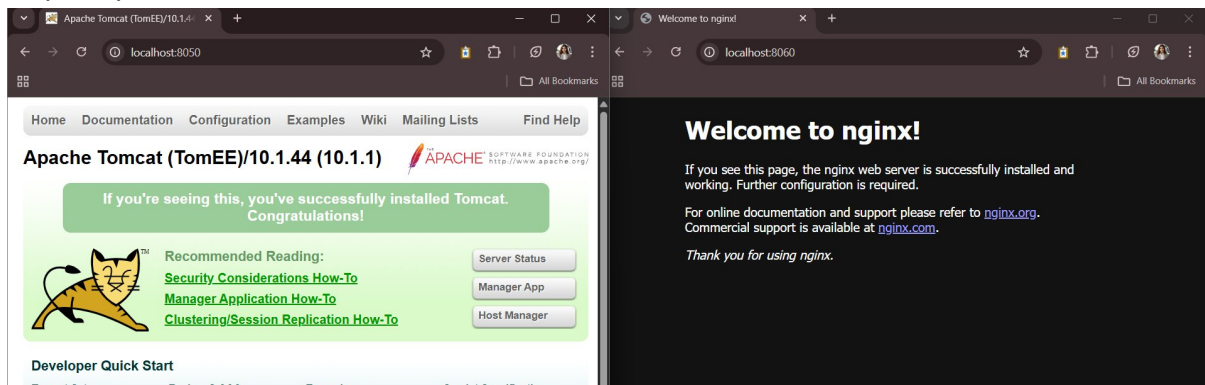
Step 3: Using docker file to run two servers parallely

```
docker-compose.yml X
C: > Users > NekshaSrinivas > SE-1 > comp-1-server > docker-compose.yml
1 services:
2   web:
3     image: nginx
4     ports:
5       - "8060:80"
6   db:
7     image: tomeet
8     ports:
9       - "8050:8080"
10
```

Step 4: Use the docker-compose up -d command to execute the docker file

```
no configuration file provided: not found
PS C:\Users\NekshaSrinivas\SE-1\comp-1-server> ren Dockerfile docker-compose
.yml
PS C:\Users\NekshaSrinivas\SE-1\comp-1-server> docker-compose up -d
[+] Running 3/3
 ✓ Network comp-1-server_default C... 0.1s
 ✓ Container comp-1-server-db-1 St... 0.6s
 ✓ Container comp-1-server-web-1 S... 0.7s
PS C:\Users\NekshaSrinivas\SE-1\comp-1-server> |
```

Step 5: Open the localhost to view the servers



WORD-PRESS:

Step 1: Create a docker-compose file and write the content for wordpress and mysql

```
PS C:\Users\NekshaSrinivas\SE-1> cd mysql
PS C:\Users\NekshaSrinivas\SE-1\mysql> notepad docker-compose
PS C:\Users\NekshaSrinivas\SE-1\mysql> ls

Directory: C:\Users\NekshaSrinivas\SE-1\mysql

Mode                LastWriteTime         Length Name
----                -
-a----             26-08-2025         18:48           672 docker-compose.txt

PS C:\Users\NekshaSrinivas\SE-1\mysql>
PS C:\Users\NekshaSrinivas\SE-1\mysql> ren docker-compose.txt docker-compose
.yml
PS C:\Users\NekshaSrinivas\SE-1\mysql> ls

Directory: C:\Users\NekshaSrinivas\SE-1\mysql

Mode                LastWriteTime         Length Name
----                -
-a----             26-08-2025         18:48           672 docker-compose.yml
```

Step 2: docker-compose.yml file:

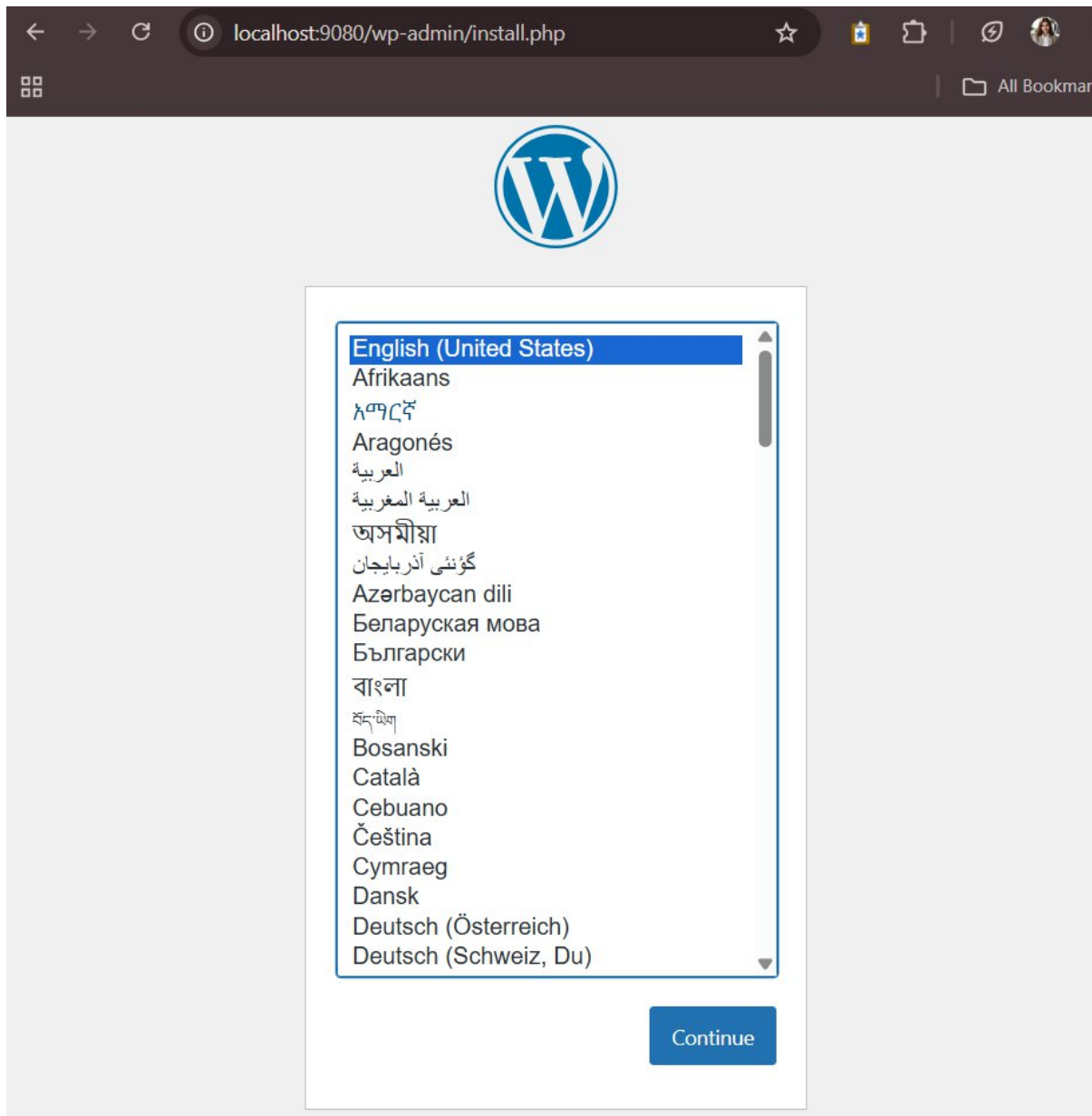
```
docker-compose.yml X
C: > Users > NekshaSrinivas > SE-1 > mysql > docker-compose.yml

1 services:
2   wordpress: # WordPress service
3     image: wordpress:latest
4     ports:
5       - "9080:80" # Map port 80 of the container to port 8080 of the host
6     environment:
7       WORDPRESS_DB_HOST: db:3306 # Database host
8       WORDPRESS_DB_USER: wordpress
9       WORDPRESS_DB_PASSWORD: wordpress
10      WORDPRESS_DB_NAME: wordpress
11     depends_on:
12       - db # Ensures the db service starts first
13
14   db: # MySQL service
15     image: mysql:5.7
16     environment:
17       MYSQL_ROOT_PASSWORD: rootpassword
18       MYSQL_DATABASE: wordpress
19       MYSQL_USER: wordpress
20       MYSQL_PASSWORD: wordpress
21
22
```

Step 3: Use the docker-compose up -d command to start the compose

```
PS C:\Users\NekshaSrinivas\SE-1\mysql> docker-compose up -d
[+] Running 3/3
✓ Network mysql_default          Created           0.1s
✓ Container mysql-db-1           Started          0.8s
✓ Container mysql-wordpress-1    Start...        1.0s
PS C:\Users\NekshaSrinivas\SE-1\mysql>
```

Step 4: Open in the local host and select the language



Step 5: Fill the details in the welcome page

Welcome

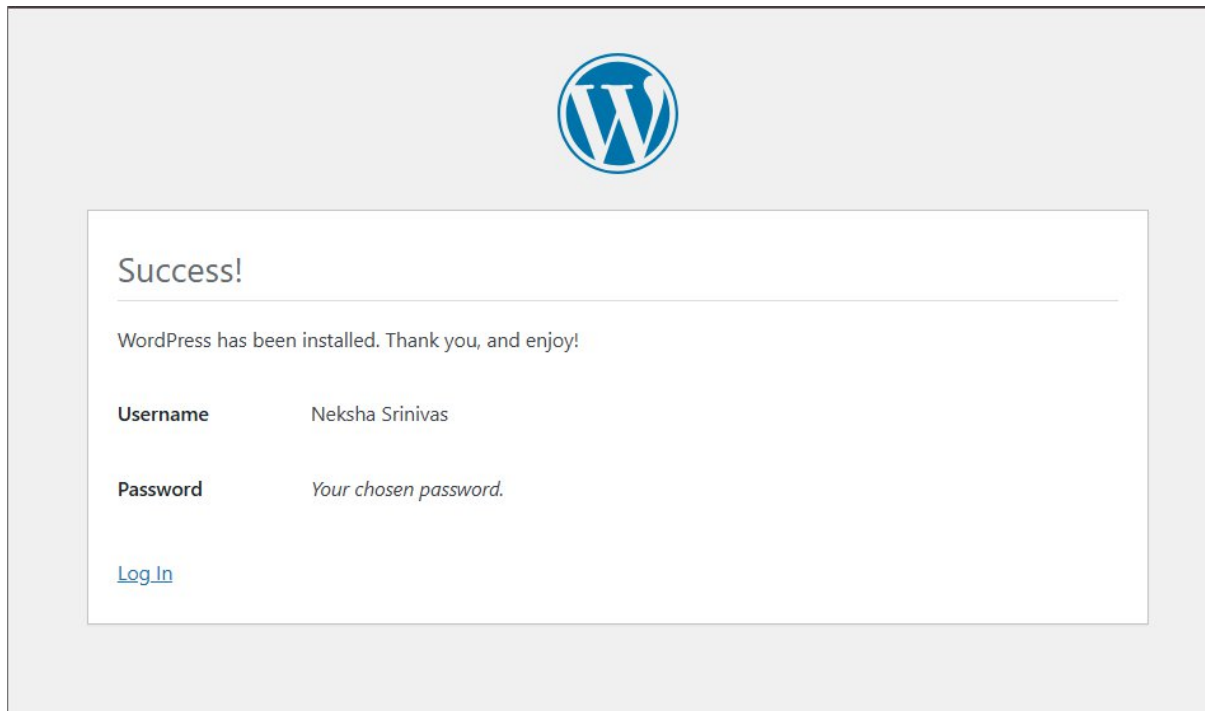
Welcome to the famous five-minute WordPress installation process! Just fill in the information below and you'll be on your way to using the most extendable and powerful personal publishing platform in the world.

Information needed

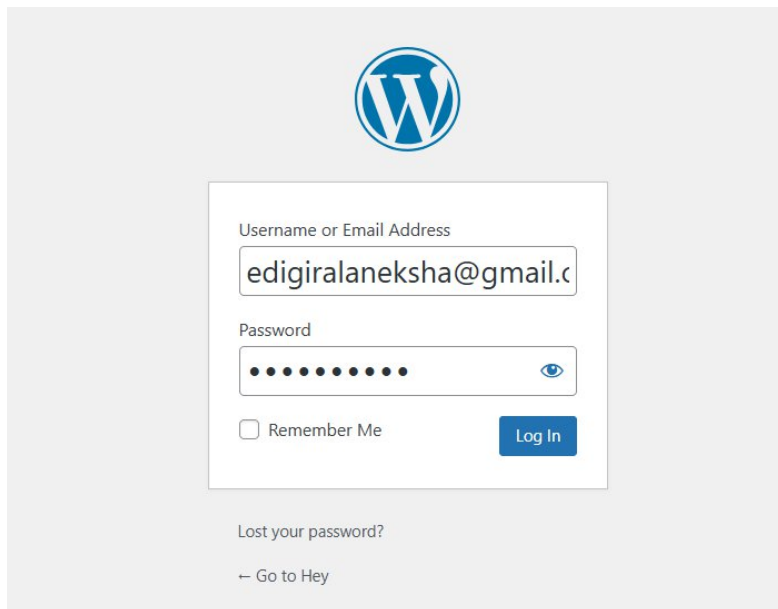
Please provide the following information. Do not worry, you can always change these settings later.

Site Title	Hey
Username	Neksha Srinivas Usernames can have only alphanumeric characters, spaces, underscores, hyphens, periods, and the @ symbol.
Password	Sri@121318  Hide Medium Important: You will need this password to log in. Please store it in a secure location.
Your Email	edigiralaneksha@gmail.com Double-check your email address before continuing.
Search engine visibility	☒ Discourage search engines from indexing this site It is up to search engines to honor this request.
Install WordPress	

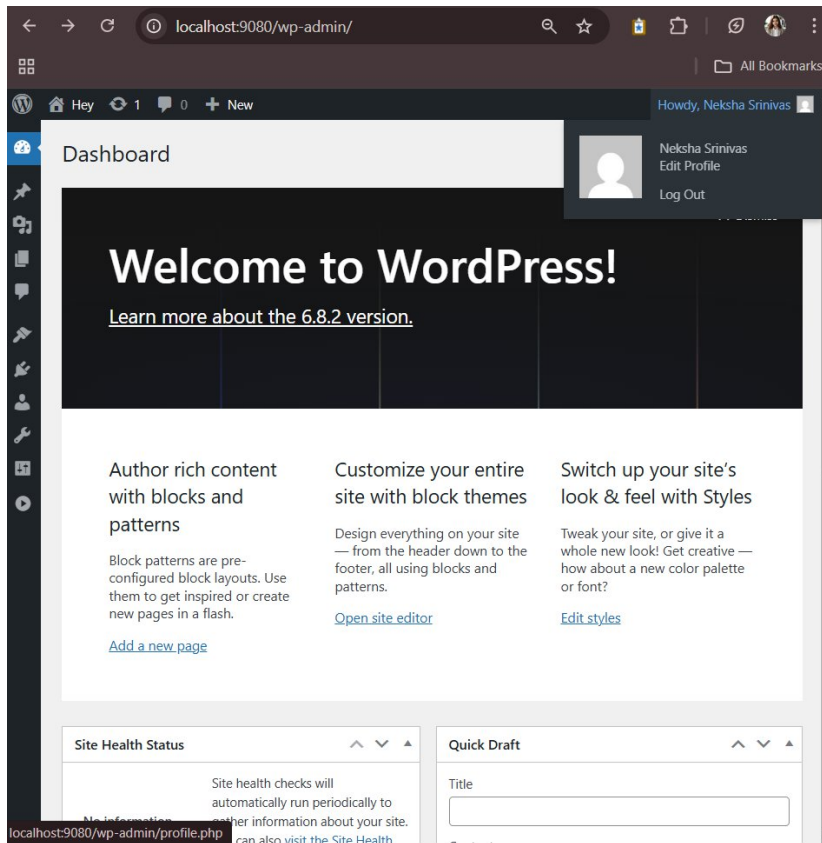
Step 6: Success message will be shown



Step 7: Use your credentials to log in



Step 7: The following page will be shown after login



Task:

Create a simple Flask app in app.py:

Step 1: create a separate folder

```
PS C:\Users\NekshaSrinivas\SE-1> mkdir custom_flask

Directory: C:\Users\NekshaSrinivas\SE-1

Mode                LastWriteTime         Length Name
----                -
d-----          28-08-2025   10:01             custom_flask

PS C:\Users\NekshaSrinivas\SE-1> cd custom_flask
PS C:\Users\NekshaSrinivas\SE-1\custom_flask> notepad app.py
PS C:\Users\NekshaSrinivas\SE-1\custom_flask> notepad Dockerfile
PS C:\Users\NekshaSrinivas\SE-1\custom_flask> ren Dockerfile.txt Dockerfile
PS C:\Users\NekshaSrinivas\SE-1\custom_flask> ls

Directory: C:\Users\NekshaSrinivas\SE-1\custom_flask

Mode                LastWriteTime         Length Name
----                -
-a-----          28-08-2025   10:02           187 app.py
-a-----          28-08-2025   10:02           105 Dockerfile

PS C:\Users\NekshaSrinivas\SE-1\custom_flask> notepad docker-compose.yml
```

Step 2: write the content of app.py , docker-compose.yml & Dockerfile

app.py:

```
app.py 1 X
C: > Users > NekshaSrinivas > SE-1 > custom_flask > app.py > home
1  from flask import Flask
2  app = Flask(__name__)
3  @app.route("/")
4  def home():
5      return "Hello from 24BD5A0503- NEKSHASRINIVAS"
6  if __name__ == "__main__":
7      app.run(host="0.0.0.0", port=5000)
8
```

docker-compose.yml:

```
docker-compose.yml X
C: > Users > NekshaSrinivas > SE-1 > custom_flask > docker-compose.yml
1  version: "3.9"
2
3  services:
4      web:
5          build: .
6          ports:
7              - "5000:5000"
8          depends_on:
9              - db
10
11     db:
12         image: mysql:8.0
13         restart: always
14         environment:
15             MYSQL_ROOT_PASSWORD: root
16             MYSQL_DATABASE: mydb
17         ports:
18             - "3306:3306"
19
```

Dockerfile:

```
FROM python:3.10-slim
WORKDIR /app
COPY app.py /app/
RUN pip install flask
CMD ["python", "app.py"]
```

Step 3: run the compose using the command docker compose up -build:

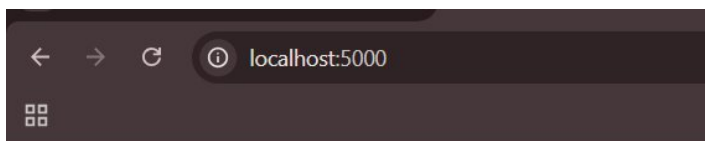
```
Mode                LastWriteTime         Length Name
----                -
-a-----          28-08-2025    10:02           187 app.py
-a-----          28-08-2025    10:03           82 docker-compose.yml
-a-----          28-08-2025    10:02          105 Dockerfile

PS C:\Users\NekshaSrinivas\SE-1\custom_flask> docker compose up --build
validating C:\Users\NekshaSrinivas\SE-1\custom_flask\docker-compose.yml: additional properties 'web' not allowed
PS C:\Users\NekshaSrinivas\SE-1\custom_flask> docker compose up --build
time="2025-08-28T10:24:45+05:30" level=warning msg="C:\\Users\\NekshaSrinivas\\SE-1\\custom_flask\\docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential confusion"
[+] Running 12/12
  ✓ db Pulled                                75.7s
  ✓ 04fa42a56901 Pull complete                1.8s
  ✓ 500d7b2546c4 Pull complete                38.2s
  ✓ ecc6cc933381 Pull complete                38.5s
  ✓ 5cd63fb67c17 Pull complete                1.7s
  ✓ 4d3eacc36b14 Pull complete                1.7s
  ✓ 9476b8faedba Pull complete                3.5s
  ✓ 789fa151603e Pull complete                1.8s
  ✓ 1756a372d796 Pull complete                1.9s
  ✓ bc0f5543b464 Pull complete               67.6s
  ✓ 131412d69359 Pull complete               42.5s
  ✓ 03ca01bc78d4 Pull complete                1.8s
#1 [internal] load local bake definitions
#1 reading from stdin 542B done
#1 DONE 0.0s

#2 [internal] load build definition from Dockerfile
#2 transferring dockerfile: 142B 0.0s done
#2 DONE 0.1s

#3 [internal] load metadata for docker.io/library/python:3.10-slim
```

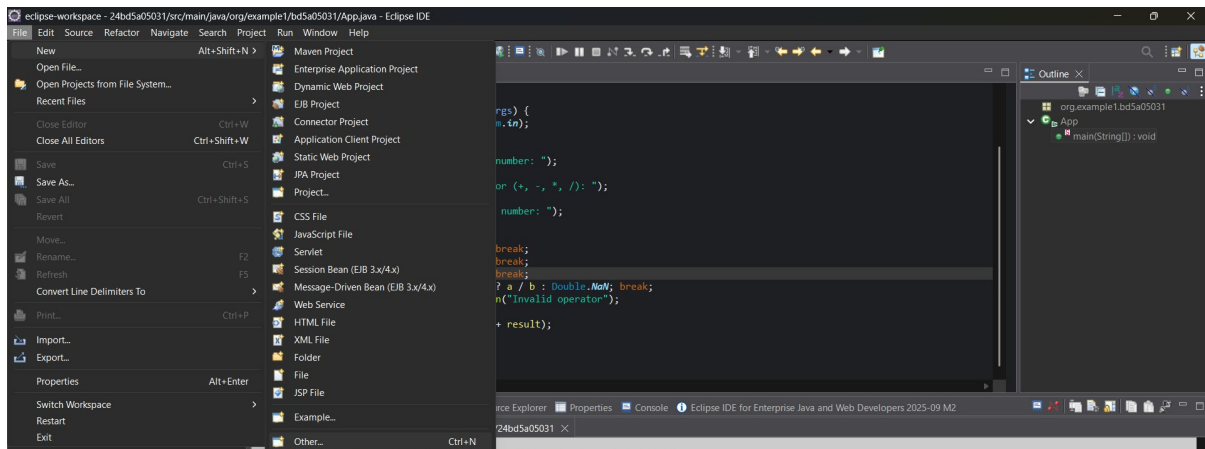
Step 4: Open the local host to view the custom page:



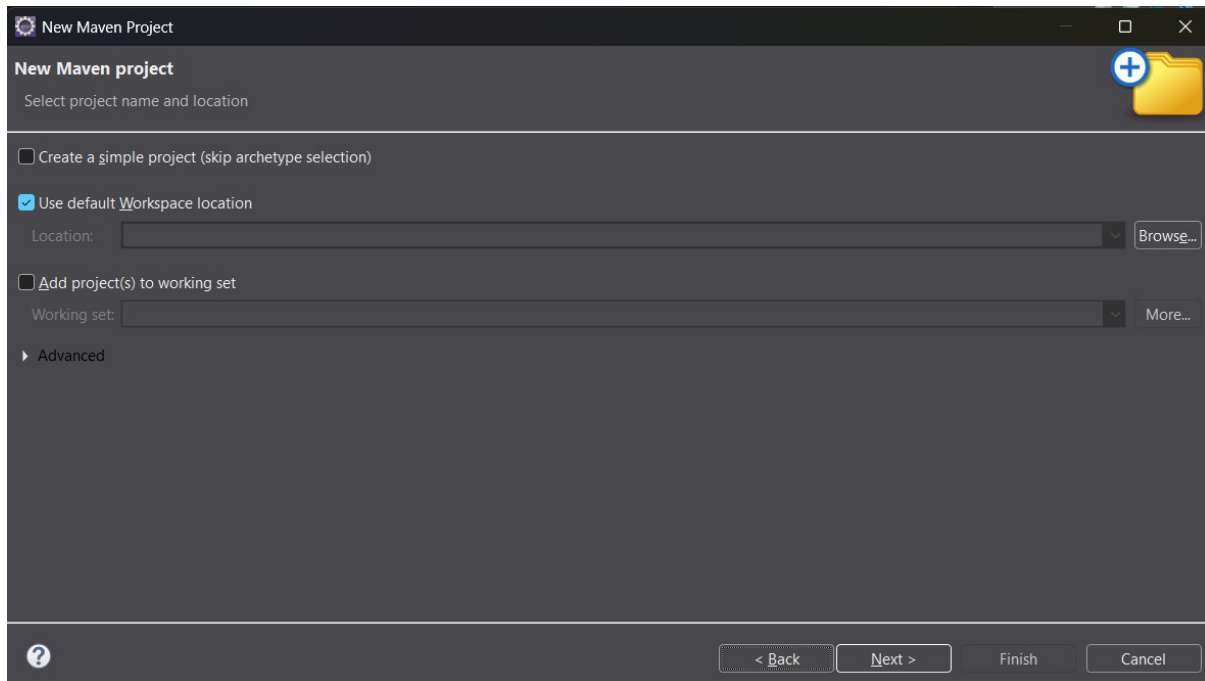
Hello from 24BD5A0503- NEKSHASRINIVAS

7. Creating a Multi-Module Maven Project

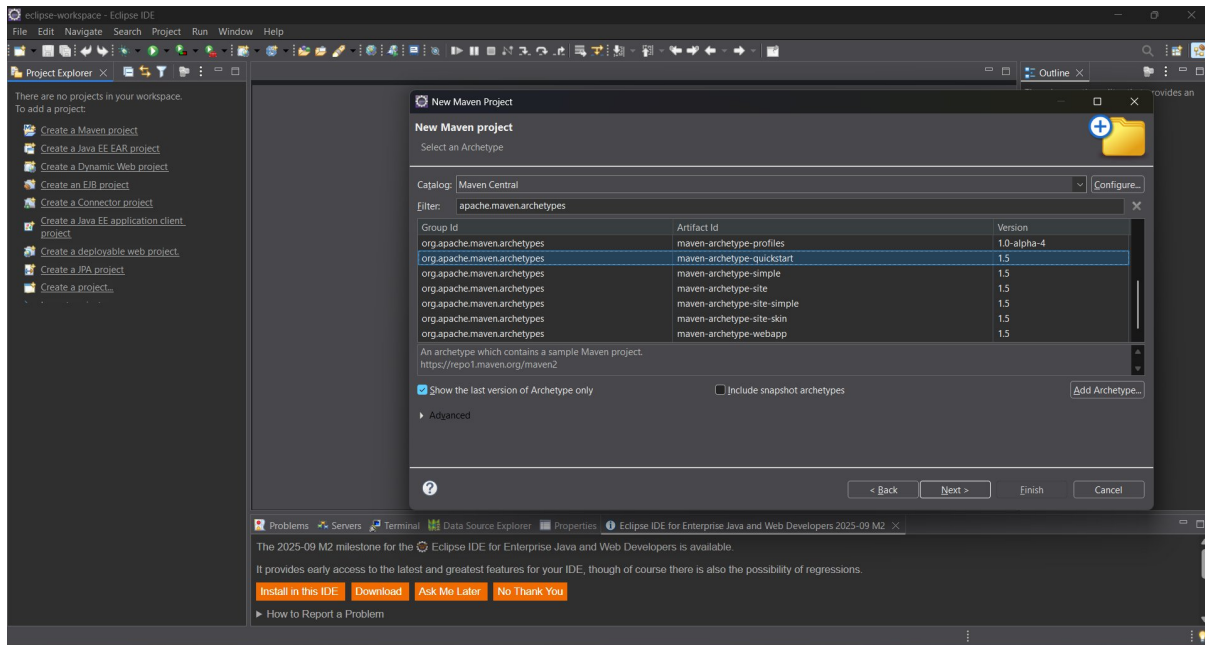
Step-1: Open the eclipse and click on file>new>Maven project



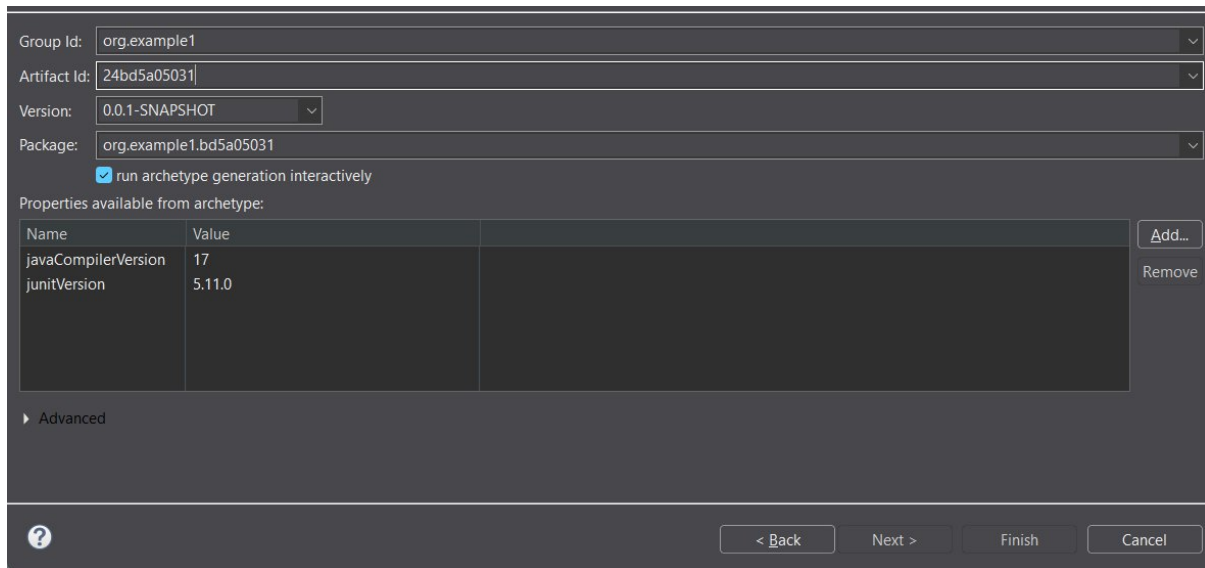
Step-2: select the default workspace and click on next



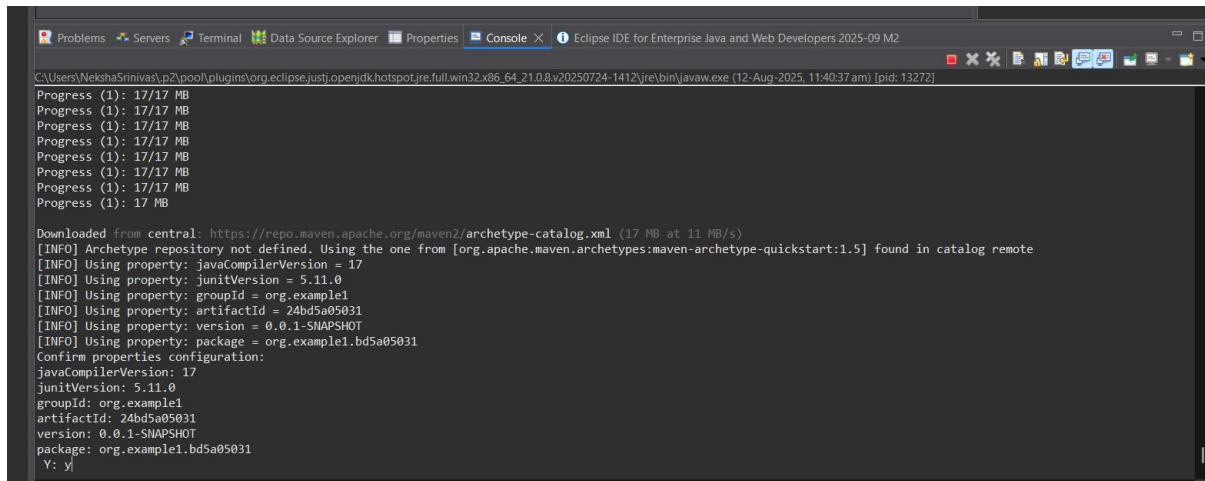
Step-3: in the filter option select the one maven-archetype-quickstart



Step-4: give the Group Id and Artifact Id and click on next



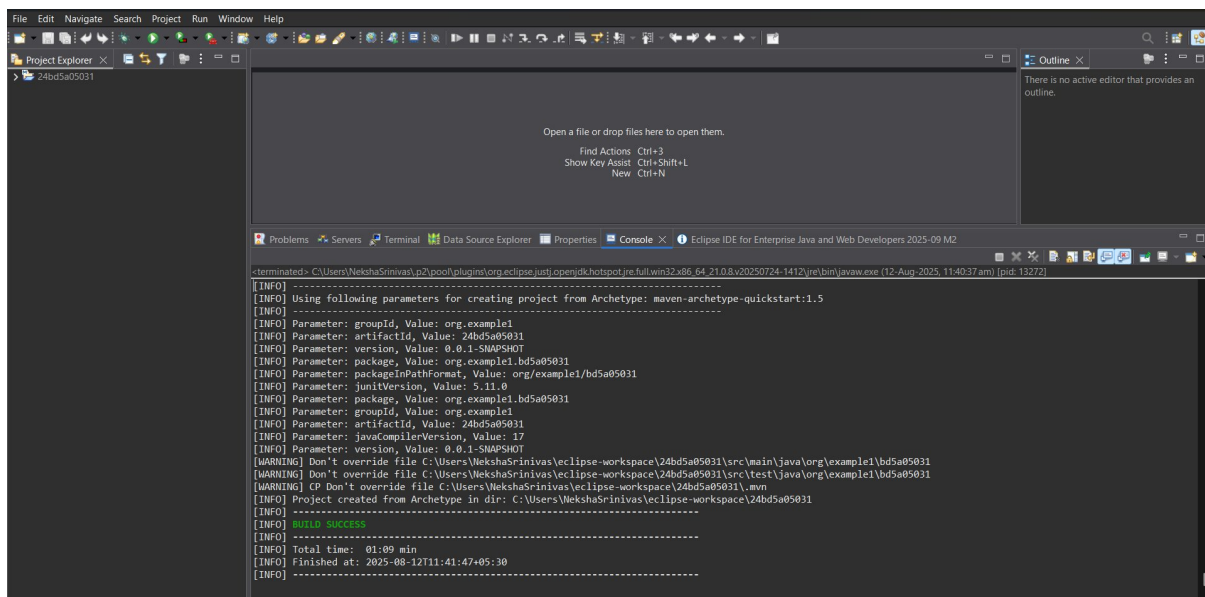
Step-5: In the console the progress will be showed type y (refers to yes) and press enter



```
C:\Users\NekshaSrinivas\p2\pool\plugins\org.eclipse.justi.openjdk.hotspot.jre.full.win32.x86_64.21.0.8.v20250724-1412\jre\bin\javaw.exe (12-Aug-2025, 11:40:37 am) [pid: 13272]
Progress (1): 17/17 MB
Progress (1): 17/17 MB
Progress (1): 17/17 MB
Progress (1): 17/17 MB
Progress (1): 17/17 MB
Progress (1): 17/17 MB
Progress (1): 17 MB

Downloaded from central: https://repo.maven.apache.org/maven2/archetype-catalog.xml (17 MB at 11 MB/s)
[INFO] Archetype repository not defined, Using the one from [org.apache.maven.archetypes:maven-archetype-quickstart:1.5] found in catalog remote
[INFO] Using property: javaCompilerVersion = 17
[INFO] Using property: junitVersion = 5.11.0
[INFO] Using property: groupId = org.example1
[INFO] Using property: artifactId = 24bd5a05031
[INFO] Using property: version = 0.0.1-SNAPSHOT
[INFO] Using property: package = org.example1.bd5a05031
Confirm properties configuration:
javaCompilerVersion: 17
junitVersion: 5.11.0
groupId: org.example1
artifactId: 24bd5a05031
version: 0.0.1-SNAPSHOT
package: org.example1.bd5a05031
Y: y
```

Step-6: BUILD SUCCESS will be shown

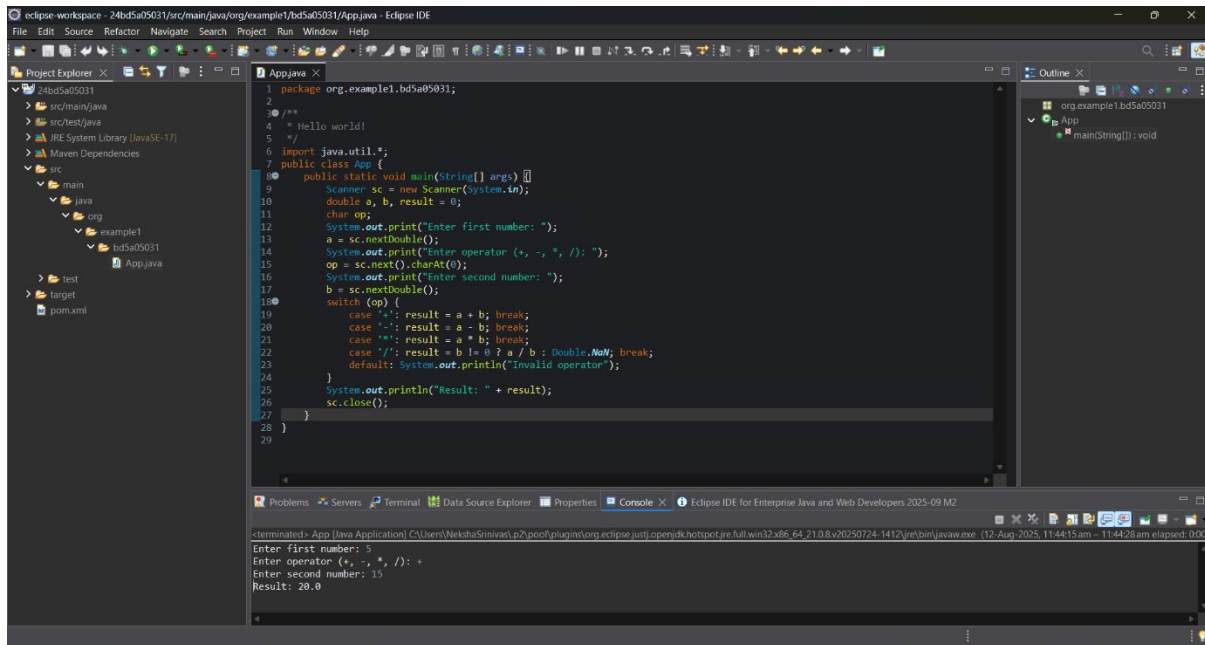


```
File Edit Navigate Search Project Run Window Help
Project Explorer x
> 24bd5a05031

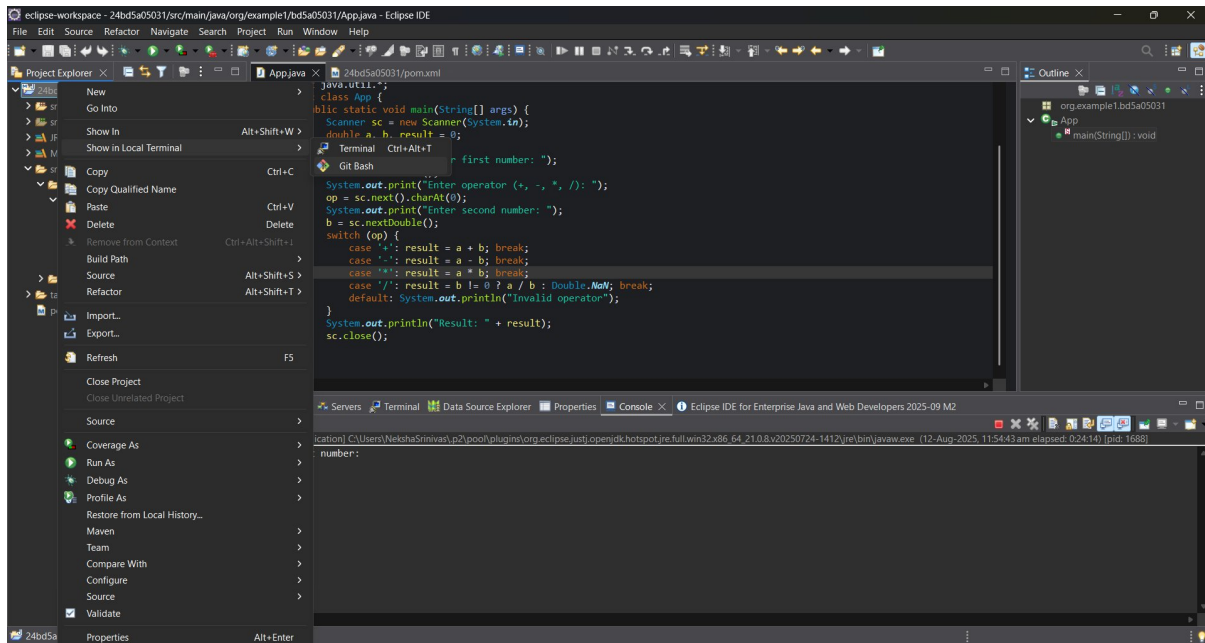
Open a file or drop files here to open them.
Find Actions Ctrl+3
Show Key Assist Ctrl+Shift+L
New Ctrl+N

c-terminated C:\Users\NekshaSrinivas\p2\pool\plugins\org.eclipse.justi.openjdk.hotspot.jre.full.win32.x86_64.21.0.8.v20250724-1412\jre\bin\javaw.exe (12-Aug-2025, 11:40:37 am) [pid: 13272]
[INFO] -----
[INFO] Using following parameters for creating project from Archetype: maven-archetype-quickstart:1.5
[INFO] -----
[INFO] Parameter: groupId, Value: org.example1
[INFO] Parameter: artifactId, Value: 24bd5a05031
[INFO] Parameter: version, Value: 0.0.1-SNAPSHOT
[INFO] Parameter: package, Value: org.example1.bd5a05031
[INFO] Parameter: packageInPathFormat, Value: org/example1/bd5a05031
[INFO] Parameter: junitVersion, Value: 5.11.0
[INFO] Parameter: package, Value: org.example1.bd5a05031
[INFO] Parameter: groupId, Value: org.example1
[INFO] Parameter: artifactId, Value: 24bd5a05031
[INFO] Parameter: javaCompilerVersion, Value: 17
[INFO] Parameter: version, Value: 0.0.1-SNAPSHOT
[WARNING] Don't override file C:\Users\NekshaSrinivas\workspace\24bd5a05031\src\main\java\org\example1\bd5a05031
[WARNING] CP Don't override file C:\Users\NekshaSrinivas\workspace\24bd5a05031\src\test\java\org\example1\bd5a05031
[INFO] Project created from Archetype in dir: C:\Users\NekshaSrinivas\workspace\24bd5a05031
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 01:09 min
[INFO] Finished at: 2025-08-12T11:41:47+05:30
[INFO] -----
```

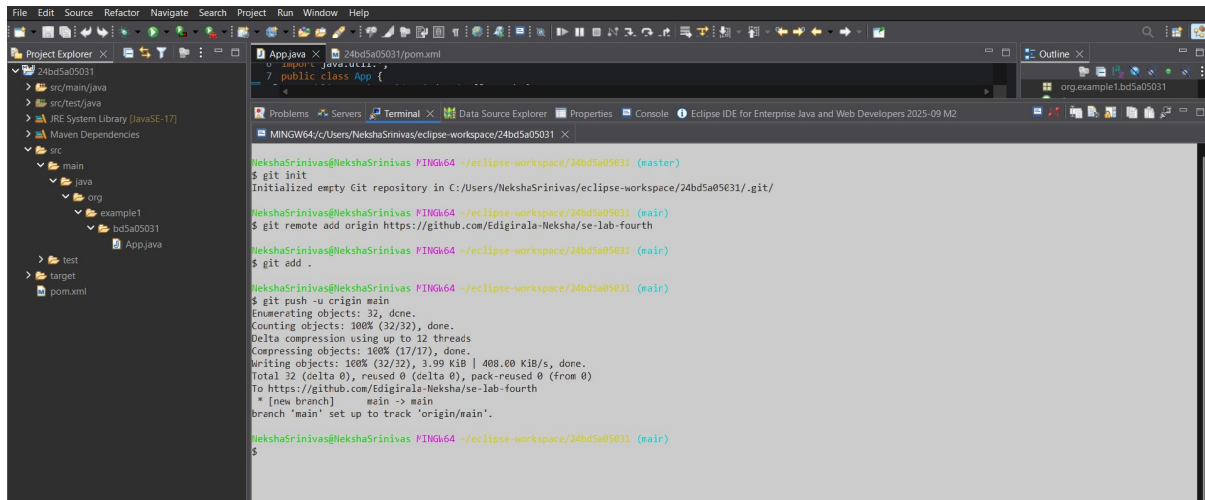
Step-6: write the code in the App.java file



Step-7: right click on the root folder and select show in git bash



Step-8: push to the git repo



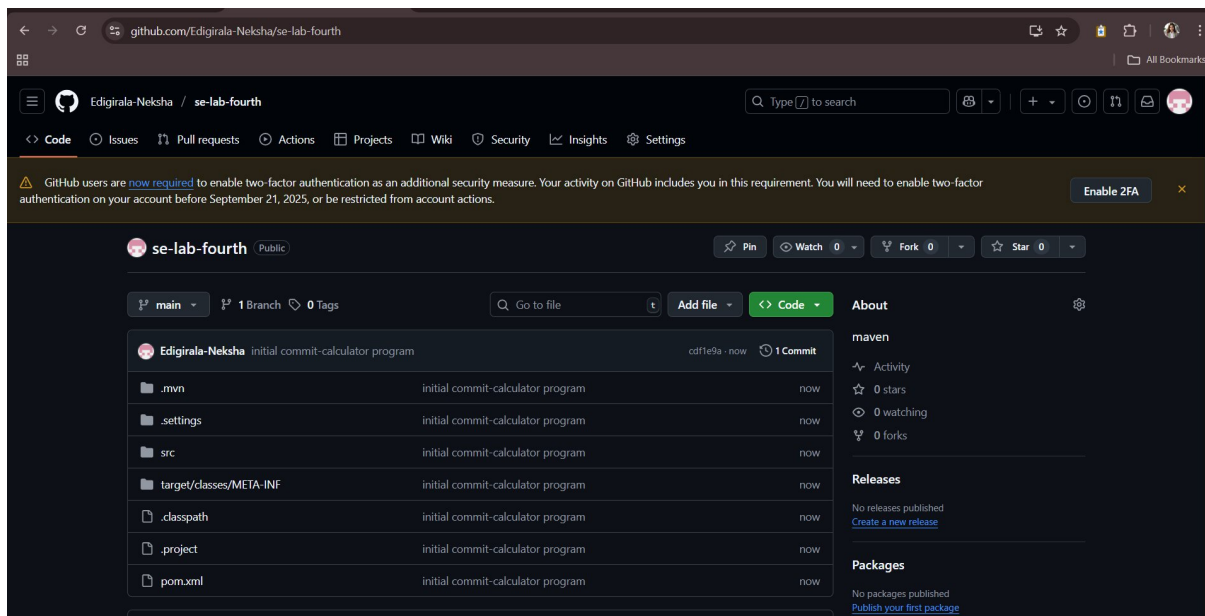
```
File Edit Source Refactor Navigate Search Project Run Window Help
Project Explorer
24bd5a05031
src/main/java
src/test/java
JRE System Library [JavaSE-17]
Maven Dependencies
src
main
java
org
example1
bd5a05031
App.java
test
target
pom.xml

App.java
24bd5a05031/pom.xml
import java.util.*;
public class App {
    ...
}

Terminal
MINGW64/C:/Users/NekshaSrinivas/eclipse-workspace/24bd5a05031
NekshaSrinivas@NekshaSrinivas: /eclipse-workspace/24bd5a05031 (master)
$ git init
Initialized empty Git repository in C:/Users/NekshaSrinivas/eclipse-workspace/24bd5a05031/.git/
NekshaSrinivas@NekshaSrinivas: /eclipse-workspace/24bd5a05031 (main)
$ git push -u origin main
$ git remote add origin https://github.com/Edigirala-Neksha/se-lab-fourth
NekshaSrinivas@NekshaSrinivas: /eclipse-workspace/24bd5a05031 (main)
$ git add .
NekshaSrinivas@NekshaSrinivas: /eclipse-workspace/24bd5a05031 (main)
$ git push -u origin main
Enumerating objects: 32, done.
Counting objects: 100% (32/32), done.
Delta compression using up to 12 threads
Compressing objects: 100% (17/17), done.
Writing objects: 100% (32/32), 3.99 KiB | 408.00 KiB/s, done.
Total 32 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/Edigirala-Neksha/se-lab-fourth
 * [new branch] main -> main
branch 'main' set up to track 'origin/main'.
NekshaSrinivas@NekshaSrinivas: /eclipse-workspace/24bd5a05031 (main)
$
```

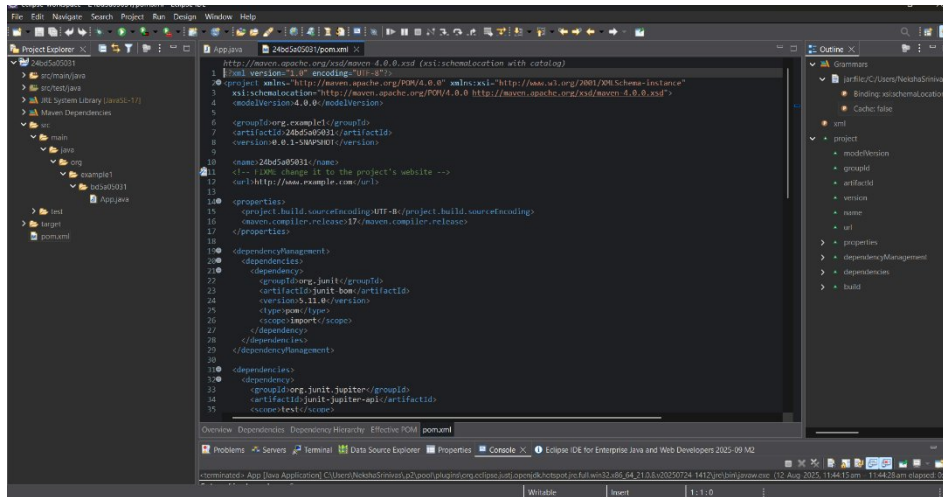
Git repo:

Git repo link: <https://github.com/Edigirala-Neksha/se-lab-fourth>



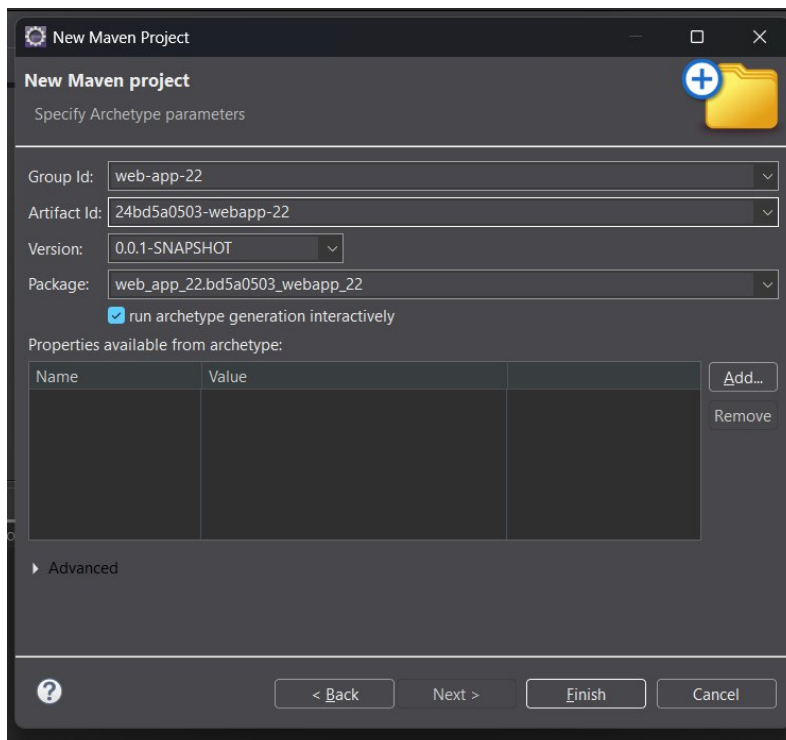
pom.xml file:

Shows the structure-



Creating maven-web project:

Step 1: Create a new maven project and give the details



Step 2: Click y to continue the creation of project

```
C:\Users\NekshaSrinivas\p2\pool\plugins\org.eclipse.justi.openjdk.hotspot.jre.full.win32.x86_64_21.0.8.v20250724-1412\jre\bin\javaw.exe (02-Sept-2025, 7:19:56 pm) [pid: 13772]
Progress (1): 17/17 MB
Progress (1): 17/17 MB
Progress (1): 17/17 MB
Progress (1): 17/17 MB
Progress (1): 17/17 MB
Progress (1): 17 MB

Downloaded from central: https://repo.maven.apache.org/maven2/archetype-catalog.xml (17 MB at 9.1 MB/s)
[INFO] Archetype repository not defined. Using the one from [org.apache.maven.archetypes:maven-archetype-webapp:1.5] found in catalog remote
[INFO] Using property: groupId = web-app-22
[INFO] Using property: artifactId = 24bd5a0503-webapp-22
[INFO] Using property: version = 0.0.1-SNAPSHOT
[INFO] Using property: package = web_app_22.bd5a0503_webapp_22
Confirm properties configuration:
groupId: web-app-22
artifactId: 24bd5a0503-webapp-22
version: 0.0.1-SNAPSHOT
package: web_app_22.bd5a0503_webapp_22
Y: y
```

Step 3: If the build is success it will show the message

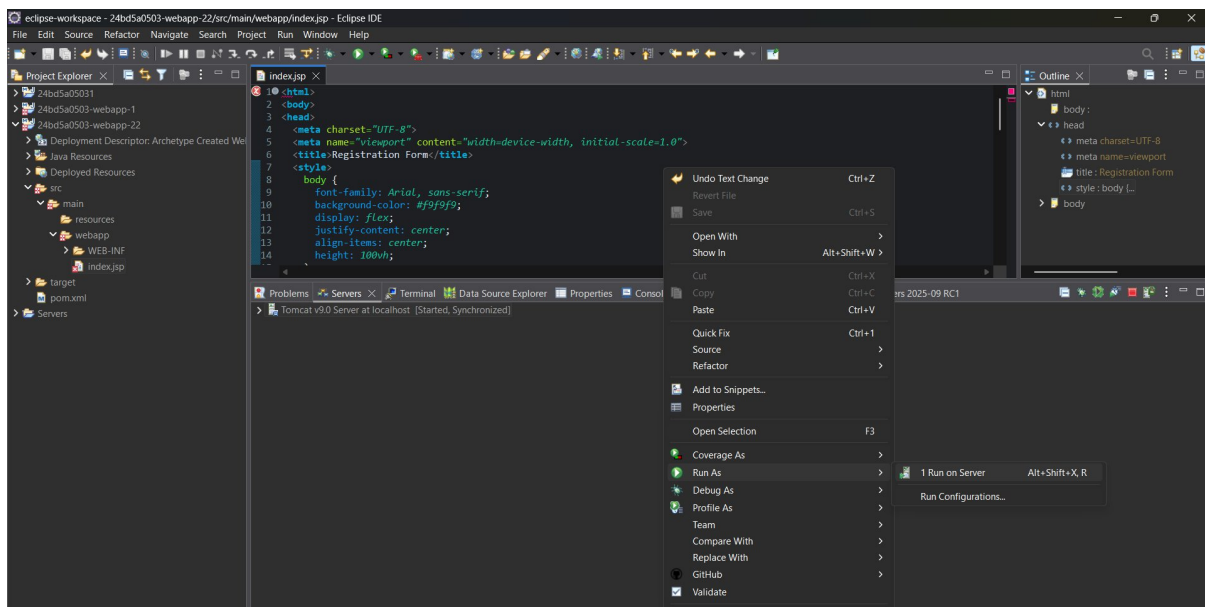
```
package: web_app_22.bd5a0503_webapp_22
Y: y
[INFO] -----
[INFO] Using following parameters for creating project from Old (1.x) Archetype: maven-archetype-webapp:1.0
[INFO] -----
[INFO] Parameter: basedir, Value: C:\Users\NekshaSrinivas\eclipse-workspace
[INFO] Parameter: package, Value: web_app_22.bd5a0503_webapp_22
[INFO] Parameter: groupId, Value: web-app-22
[INFO] Parameter: artifactId, Value: 24bd5a0503-webapp-22
[INFO] Parameter: packageName, Value: web_app_22.bd5a0503_webapp_22
[INFO] Parameter: version, Value: 0.0.1-SNAPSHOT
[INFO] project created from Old (1.x) Archetype in dir: C:\Users\NekshaSrinivas\eclipse-workspace\24bd5a0503-webapp-22
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 43.500 s
[INFO] Finished at: 2025-09-02T19:20:41+05:30
[INFO] -----
```

Step 4: write the html code for the web page:

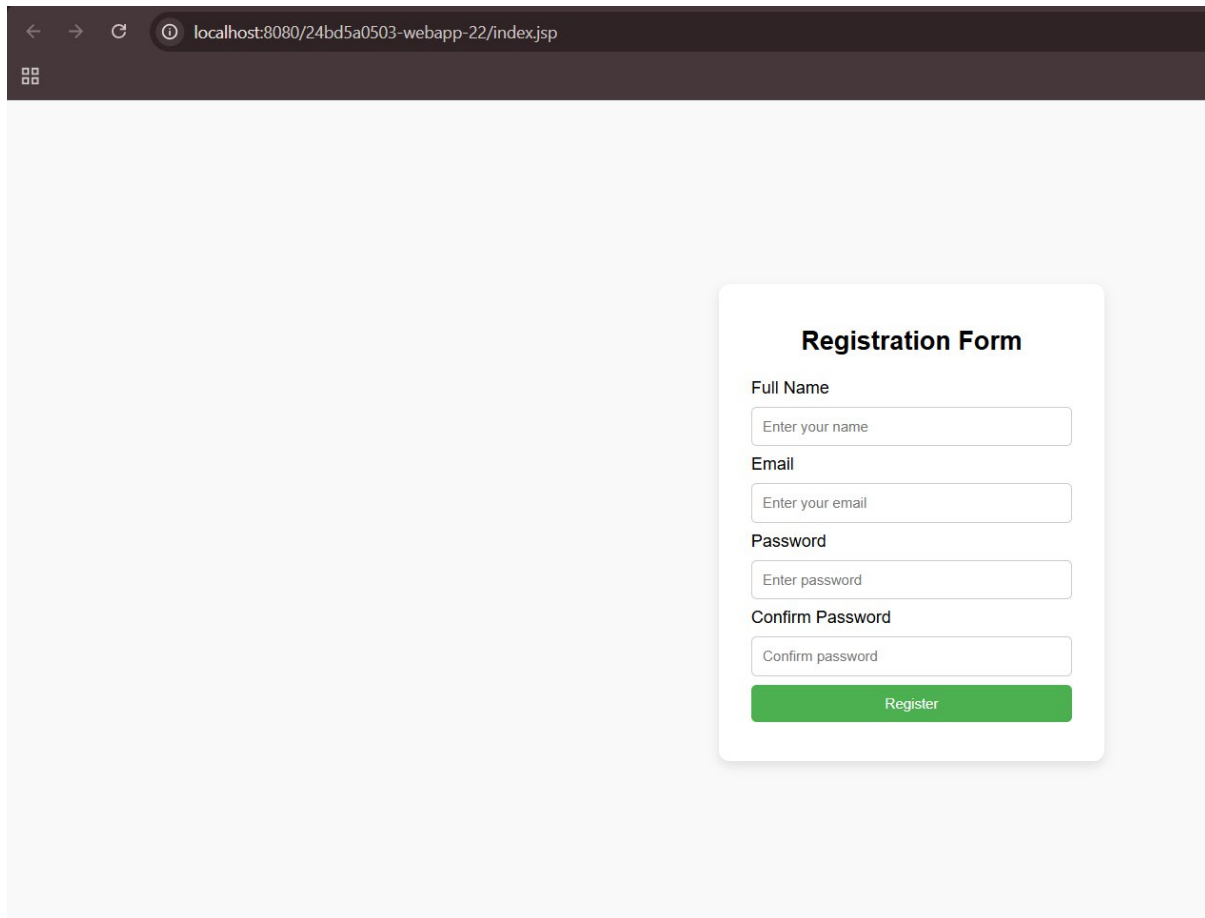
```
index.jsp
1 <html>
2 <body>
3 <head>
4 <meta charset="UTF-8">
5 <meta name="viewport" content="width=device-width, initial-scale=1.0">
6 <title>Registration Form</title>
7 <style>
8   body {
9     font-family: Arial, sans-serif;
10    background-color: #f9f9f9;
11    display: flex;
12    justify-content: center;
13    align-items: center;
14    height: 100vh;
15  }
16  .form-container {
17    background: #fff;
18    padding: 20px 30px;
19    border-radius: 10px;
20    box-shadow: 0 4px 10px rgba(0,0,0,0.1);
21    width: 300px;
22  }
23  .form-container h2 {
24    text-align: center;
25    margin-bottom: 20px;
26  }
27  .form-container input {
28    width: 100%;
29    padding: 10px;
30    margin: 8px 0;
31    border: 1px solid #ccc;
32    border-radius: 5px;
33  }
34  .form-container button {
35    width: 100%;
36    padding: 10px;
37    background: #4CAF50;
38  }
```

```
index.jsp X
32 border-radius: 5px;
33 }
34 .form-container button {
35 width: 100%;
36 padding: 10px;
37 background: #4CAF50;
38 color: white;
39 border: none;
40 border-radius: 5px;
41 cursor: pointer;
42 }
43 .form-container button:hover {
44 background: #45a049;
45 }
46 </style>
47 </head>
48 <body>
49 <div class="form-container">
50 <h2>Registration Form</h2>
51 <form action="#" method="post">
52 <label for="fullName">Full Name</label>
53 <input type="text" id="fullName" name="fullName" placeholder="Enter your name" required>
54
55 <label for="email">Email</label>
56 <input type="email" id="email" name="email" placeholder="Enter your email" required>
57
58 <label for="password">Password</label>
59 <input type="password" id="password" name="password" placeholder="Enter password" required>
60
61 <label for="confirm">Confirm Password</label>
62 <input type="password" id="confirm" name="confirm" placeholder="Confirm password" required>
63
64 <button type="submit">Register</button>
65 </form>
66 </div>
67 </body>
68 </html>
69
```

Step 5: Select run on server



Step 6: It will show the following output:



The screenshot shows a web browser window with the address bar displaying "localhost:8080/24bd5a0503-webapp-22/index.jsp". The browser's address bar has back, forward, and refresh buttons. Below the address bar, there is a grid icon. The main content area of the browser displays a registration form titled "Registration Form". The form contains four input fields: "Full Name" with placeholder text "Enter your name", "Email" with placeholder text "Enter your email", "Password" with placeholder text "Enter password", and "Confirm Password" with placeholder text "Confirm password". At the bottom of the form is a green button labeled "Register".

localhost:8080/24bd5a0503-webapp-22/index.jsp

Registration Form

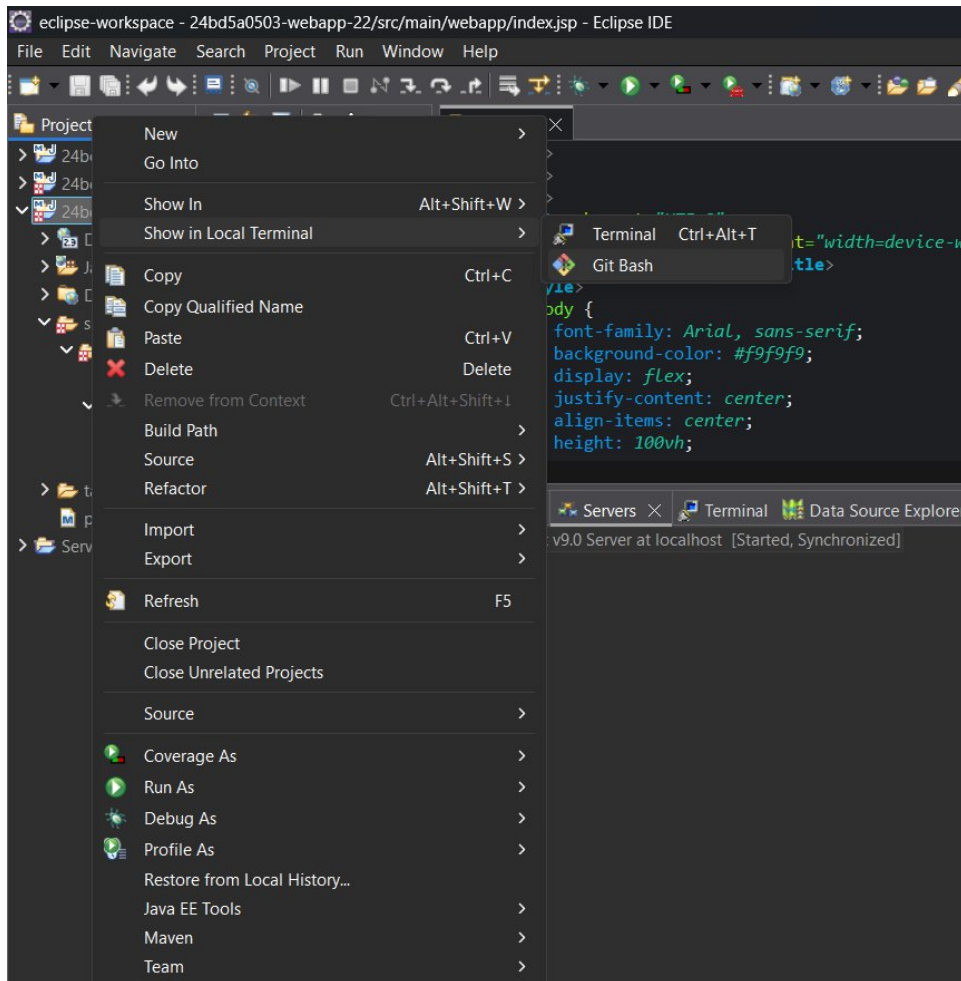
Full Name

Email

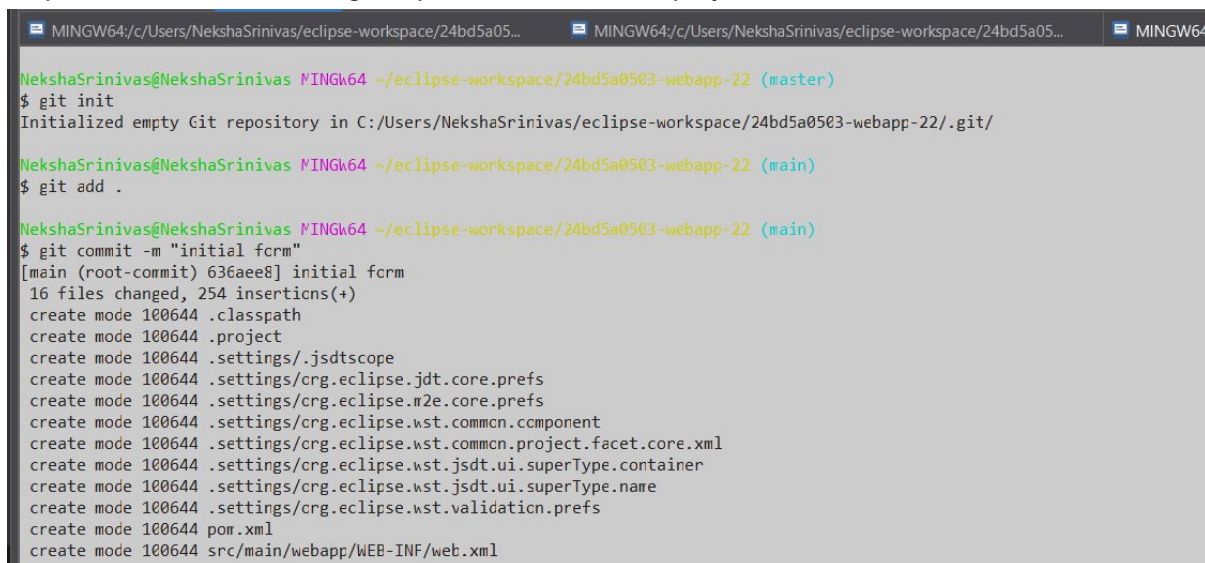
Password

Confirm Password

Step 7: To push it into git, select git bash from show in local terminal



Step 8: use the command of git to push the maven web project



```

NekshaSrinivas@NekshaSrinivas MINGW64 ~/eclipse-workspace/24bd5a0503-webapp-22 (main)
$ git branch
* main

NekshaSrinivas@NekshaSrinivas MINGW64 ~/eclipse-workspace/24bd5a0503-webapp-22 (main)
$ git push origin main
Enumerating objects: 29, done.
Counting objects: 100% (29/29), done.
Delta compression using up to 12 threads
Compressing objects: 100% (18/18), done.
Writing objects: 100% (29/29), 4.43 KiB | 283.00 KiB/s, done.
Total 29 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
rerote: Resolving deltas: 100% (1/1), done.
To https://github.com/Edigirala-Neksha/se-webapp-22.git
 * [new branch]      main -> main

NekshaSrinivas@NekshaSrinivas MINGW64 ~/eclipse-workspace/24bd5a0503-webapp-22 (main)
$

```

Step 9: verify the repo in git hub

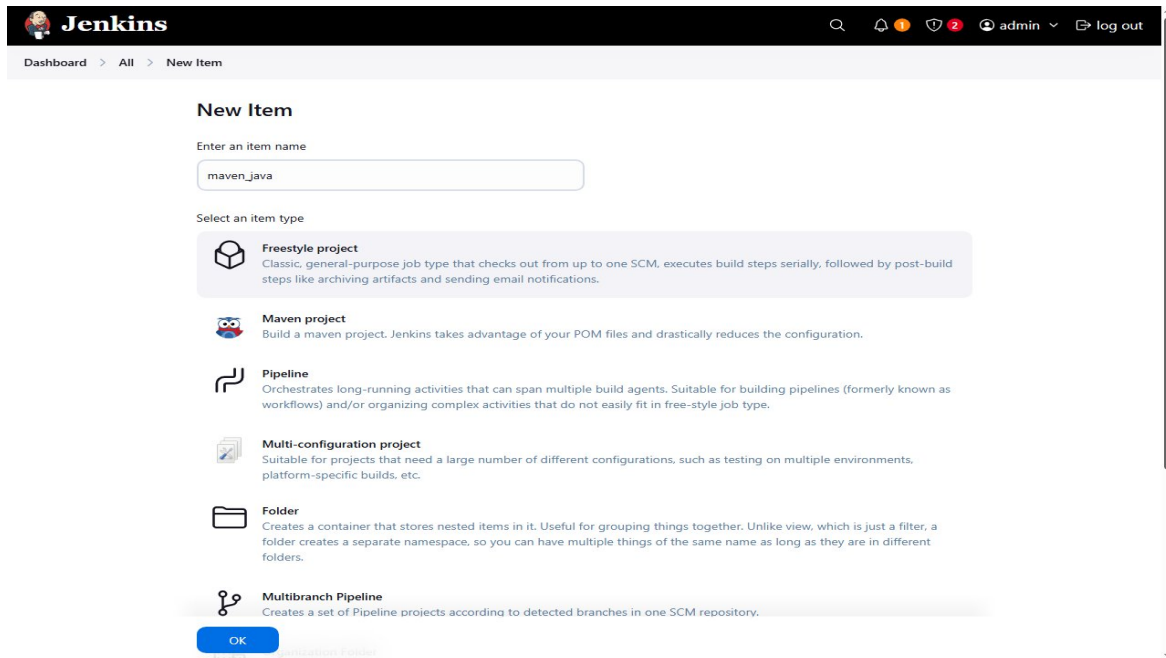
The screenshot shows the GitHub interface for a repository named 'se-webapp-22' owned by 'Edigirala-Neksha'. The repository is public and has 1 commit. The main branch is selected. The file list shows several files including .settings, src/main/webapp, target/m2e-wtp/web-resources/META-INF, .classpath, .project, and pom.xml, all with an 'initial form' status. The right sidebar contains sections for 'About' (se-lab-week7), 'Releases' (no releases published), 'Packages' (no packages published), and 'Languages'.

8. Jenkins Automation

Steps for MavenJava Automation

Step 1: Open Jenkins (localhost:8888)

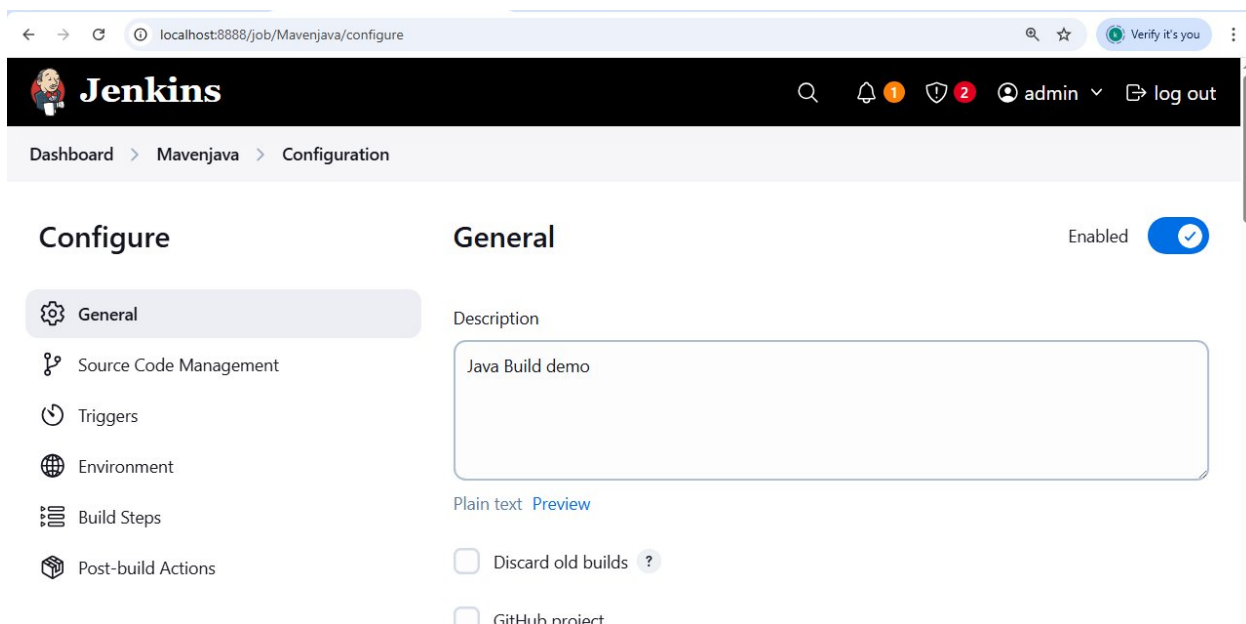
Click on "New Item" (left side menu) and name it as maven_java > select freestyle project > click on "OK"



The screenshot shows the Jenkins 'New Item' configuration page. At the top, the Jenkins logo and navigation links are visible. The breadcrumb trail is 'Dashboard > All > New Item'. The main heading is 'New Item'. Below it, there is a text input field for 'Enter an item name' with the value 'maven_java'. Underneath, a section titled 'Select an item type' lists several options: 'Freestyle project' (Classic, general-purpose job type), 'Maven project' (Build a maven project), 'Pipeline' (Orchestrates long-running activities), 'Multi-configuration project' (Suitable for projects that need a large number of different configurations), 'Folder' (Creates a container that stores nested items), and 'Multibranch Pipeline' (Creates a set of Pipeline projects). The 'Freestyle project' option is highlighted. At the bottom left, there is a blue 'OK' button.

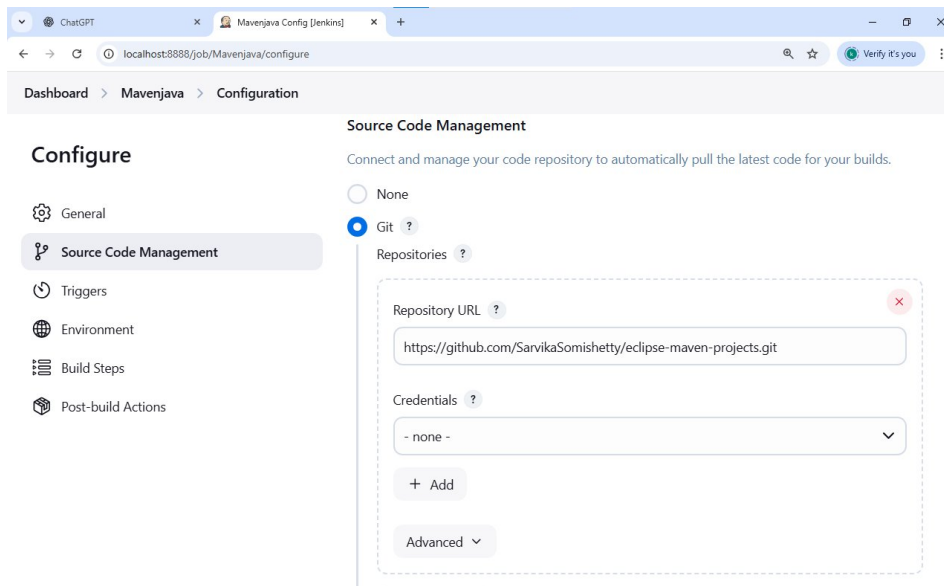
Step 2: Configuration of maven_java project

Give the description

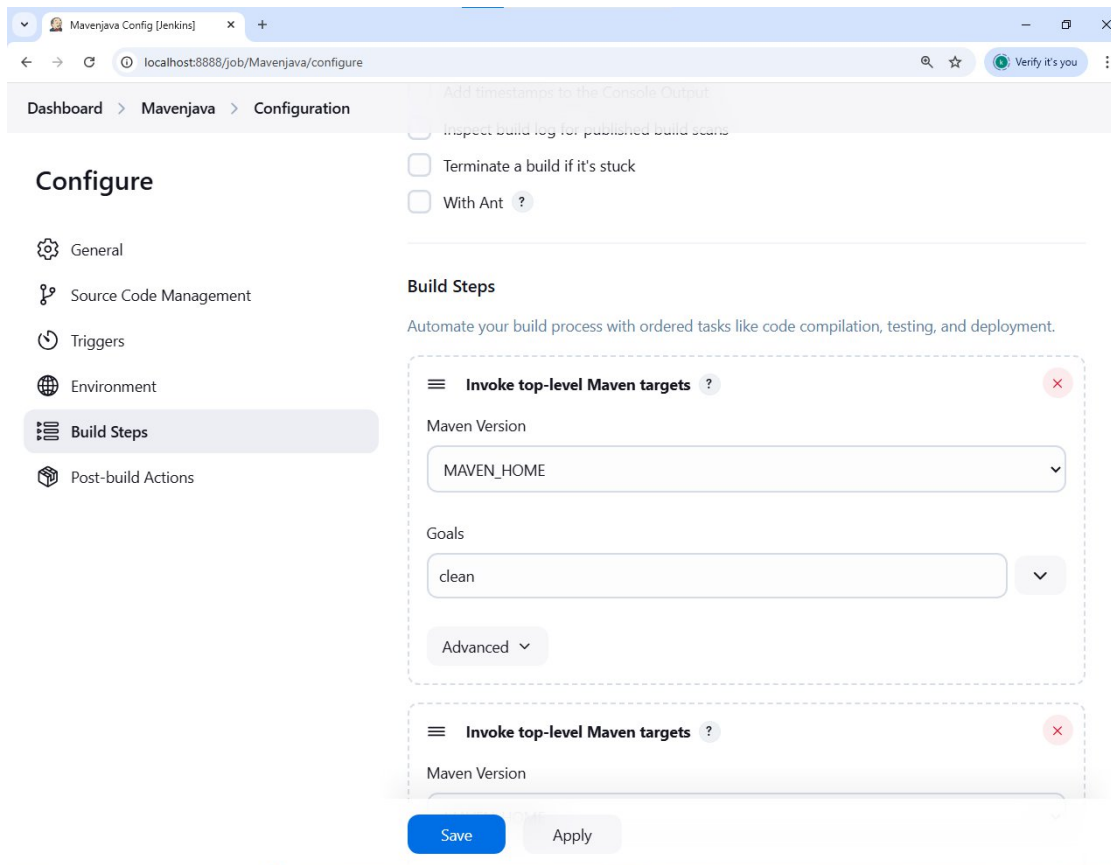


The screenshot shows the Jenkins 'Configuration' page for the 'Mavenjava' job. The breadcrumb trail is 'Dashboard > Mavenjava > Configuration'. The page is titled 'Configure' and has a 'General' tab selected. The 'General' tab is marked as 'Enabled' with a blue toggle switch. The 'Description' field contains the text 'Java Build demo'. Below the description, there are two checkboxes: 'Discard old builds' and 'GitHub project', both of which are currently unchecked. The left sidebar shows a list of configuration options: 'General', 'Source Code Management', 'Triggers', 'Environment', 'Build Steps', and 'Post-build Actions'.

In the source code management select git and give the git repo link



In the build steps click on add build step > give maven version as MAVEN_HOME > select invoke top-level maven targets > goals as clean



In the build steps click on add build step > give maven version as MAVEN_HOME > select invoke top-level maven targets > goals as install

The screenshot shows the Jenkins 'Configure' page for a job named 'Mavenjava Config'. The browser address bar indicates the URL is 'localhost:8888/job/Mavenjava/configure'. The left sidebar contains a 'Configure' section with a list of tabs: General, Source Code Management, Triggers, Environment, Build Steps (selected), and Post-build Actions. The main content area shows the 'Build Steps' configuration. A step named 'Invoke top-level Maven targets' is selected, with a red 'X' icon in the top right corner. The 'Maven Version' is set to 'MAVEN_HOME' and the 'Goals' are set to 'install'. Below the 'Goals' field is an 'Advanced' dropdown menu. At the bottom of the 'Build Steps' section is an 'Add build step' button. The 'Post-build Actions' section is visible below, with a description: 'Define what happens after a build completes, like sending notifications, archiving artifacts, or triggering other jobs.' At the bottom of the page are 'Save' and 'Apply' buttons.

Dashboard > Mavenjava > Configuration

Configure

- General
- Source Code Management
- Triggers
- Environment
- Build Steps**
- Post-build Actions

Invoke top-level Maven targets

Maven Version: MAVEN_HOME

Goals: install

Advanced

Add build step

Post-build Actions

Define what happens after a build completes, like sending notifications, archiving artifacts, or triggering other jobs.

Save Apply

In the post build actions > click on add post build action > select the archive the artifacts > in the file to archive give “**/*”

For the second post build action,

In the post build actions > click on add post build action > select build other projects > give projects to build as MavenJava_Test

Click on apply and save

The screenshot shows the Jenkins 'Configure' page for a job named 'Mavenjava Config [Jenkins]'. The breadcrumb navigation is 'Dashboard > Mavenjava > Configuration'. The left sidebar contains a 'Configure' section with a list of tabs: General, Source Code Management, Triggers, Environment, Build Steps, and Post-build Actions (which is currently selected). The main content area is titled 'Post-build Actions' and includes a description: 'Define what happens after a build completes, like sending notifications, archiving artifacts, or triggering other jobs.' There are two configured actions, each in a dashed box with a red 'X' icon in the top right corner. The first action is 'Archive the artifacts', with a 'Files to archive' field containing '**/*' and an 'Advanced' dropdown menu. The second action is 'Build other projects', with a 'Projects to build' field containing 'MavenJava_Test' and three radio button options: 'Trigger only if build is stable' (selected), 'Trigger even if the build is unstable', and 'Trigger even if the build fails'. At the bottom of the configuration area, there is an 'Add post-build action' dropdown, a blue 'Save' button, and a grey 'Apply' button.

If the build is success:

The screenshot displays the Jenkins web interface in a browser window. The address bar shows the URL `localhost:8888/job/maven_web_build/`. The Jenkins logo and navigation links are at the top. The main content area shows the build status for `maven_web_build` as successful (green checkmark). Below this, there are links for 'Changes', 'Workspace', 'Build Now', 'Configure', 'Delete Project', and 'Rename'. A 'Builds' list on the left shows two successful builds from today. The 'Downstream Projects' section lists `maven_web_test`. The 'Permalinks' section provides links to the last build, stable build, successful build, and completed build, all from 27 minutes ago. The bottom of the page shows the REST API endpoint and the Jenkins version 2.489. The Windows taskbar at the bottom includes the search bar and various application icons.

maven_web_build [Jenkins] x Edigirala-Neksha/se-lab-intern... x SSL connection issue fix x +

localhost:8888/job/maven_web_build/

Verify it's you Error

Jenkins

Dashboard > maven_web_build >

Status

Changes

Workspace

Build Now

Configure

Delete Project

Rename

Builds

Filter

Today

- #2 11:43 AM
- #1 11:42 AM

maven_web_build

web build demo

Last Successful Artifacts

Downstream Projects

- maven_web_test

Permalinks

- Last build (#2), 27 min ago
- Last stable build (#2), 27 min ago
- Last successful build (#2), 27 min ago
- Last completed build (#2), 27 min ago

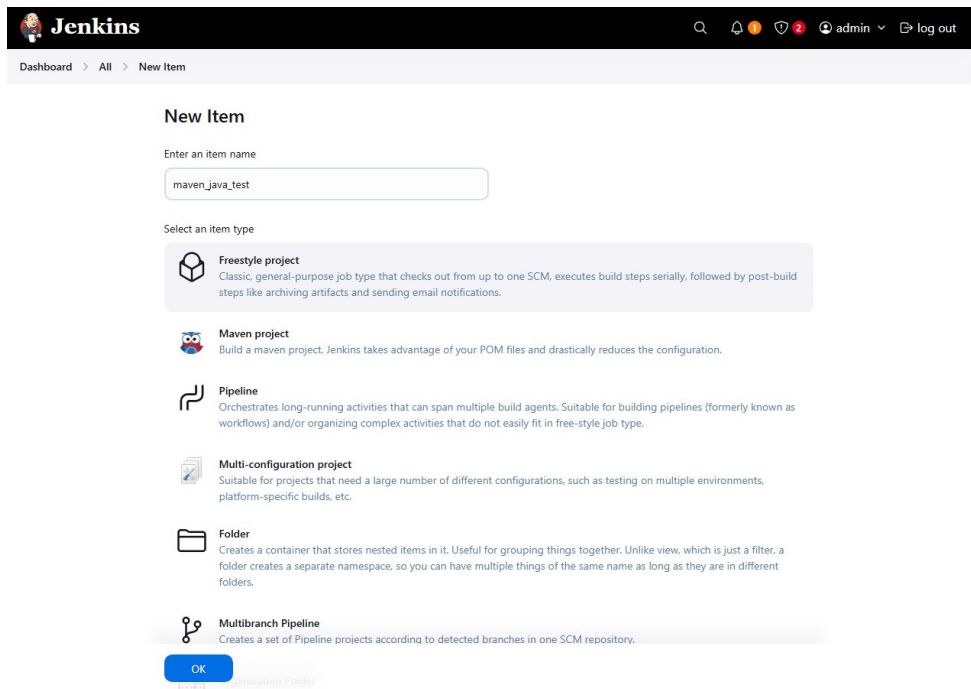
REST API Jenkins 2.489

Type here to search

12:11 07-10-2025

Step 3: Create Freestyle Project (e.g., MavenJava_Test)

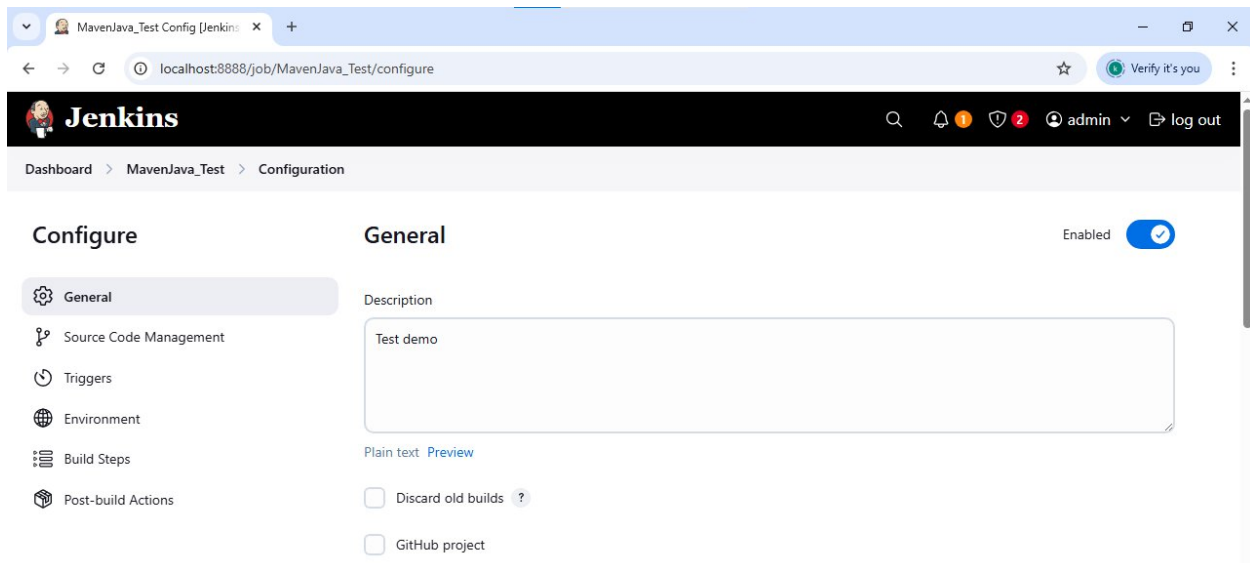
Click on new item> give item name as maven_java_test or MavenJava_Test and select free style project and click ok



The image shows the Jenkins 'New Item' form. At the top, there's a search bar and a user menu for 'admin'. Below the header, the breadcrumb is 'Dashboard > All > New Item'. The main section is titled 'New Item'. It has a text input field for 'Enter an item name' with the value 'maven_java_test'. Below that, there's a section 'Select an item type' with several options: 'Freestyle project' (selected), 'Maven project', 'Pipeline', 'Multi-configuration project', 'Folder', and 'Multibranch Pipeline'. Each option has a brief description. At the bottom, there's an 'OK' button.

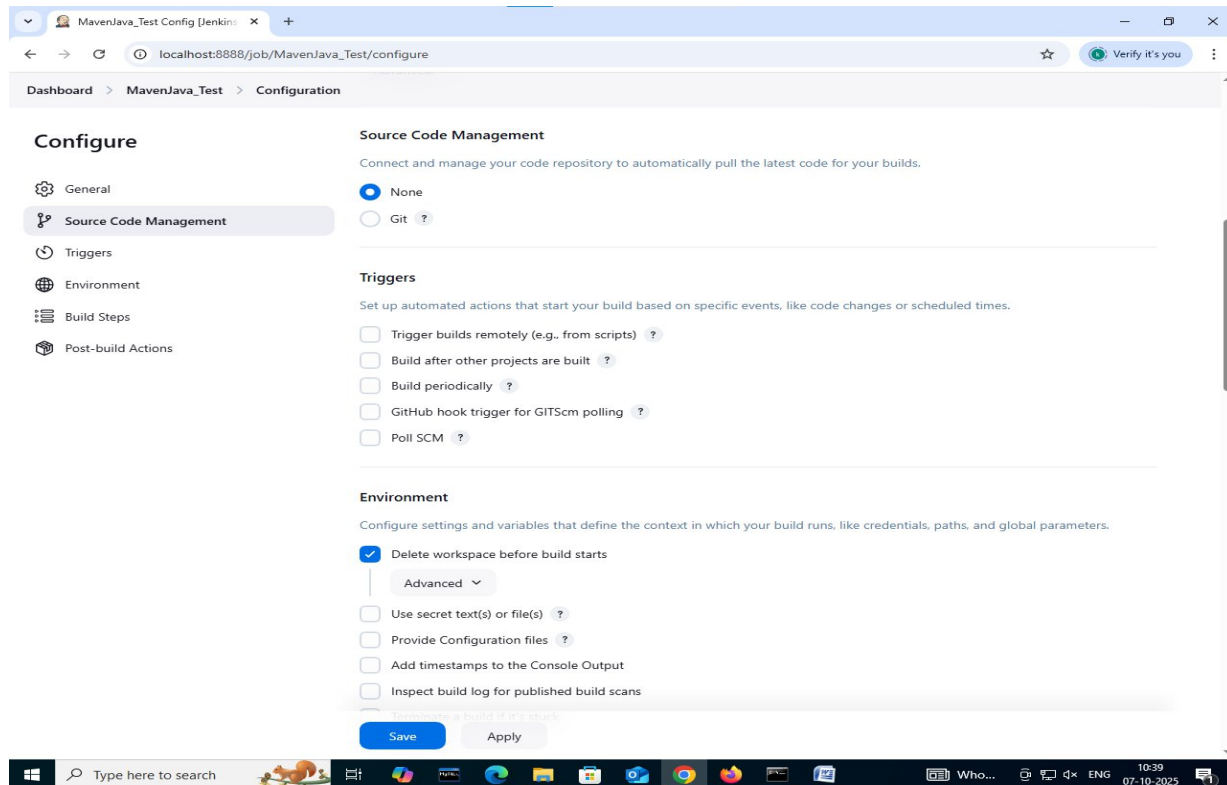
Step 4: Configuration of maven_java project

Give the description

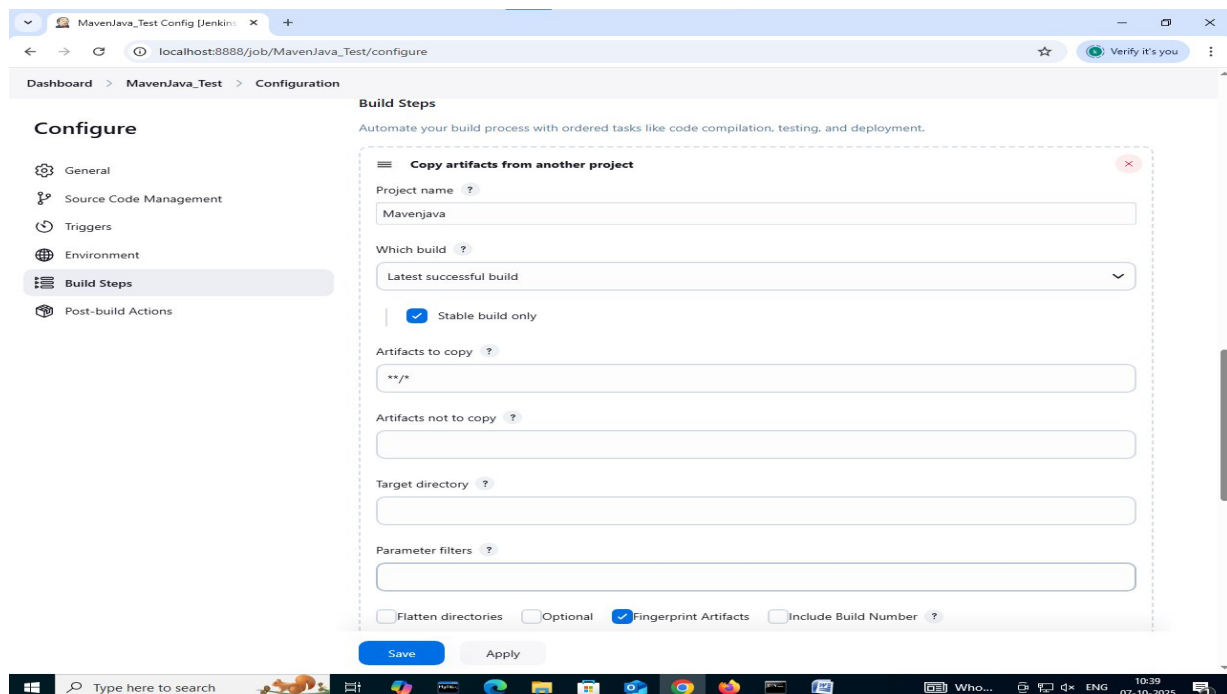


The image shows the Jenkins 'Configuration' page for the 'MavenJava_Test' job. The browser address bar shows 'localhost:8888/job/MavenJava_Test/configure'. The page has a sidebar on the left with a 'Configure' section containing links for 'General', 'Source Code Management', 'Triggers', 'Environment', 'Build Steps', and 'Post-build Actions'. The main area is titled 'General' and has an 'Enabled' toggle switch. It contains a 'Description' text area with the value 'Test demo'. Below the description, there are two checkboxes: 'Discard old builds' and 'GitHub project', both of which are currently unchecked.

In the source code management select none and in environment select “delete workspace before build starts”

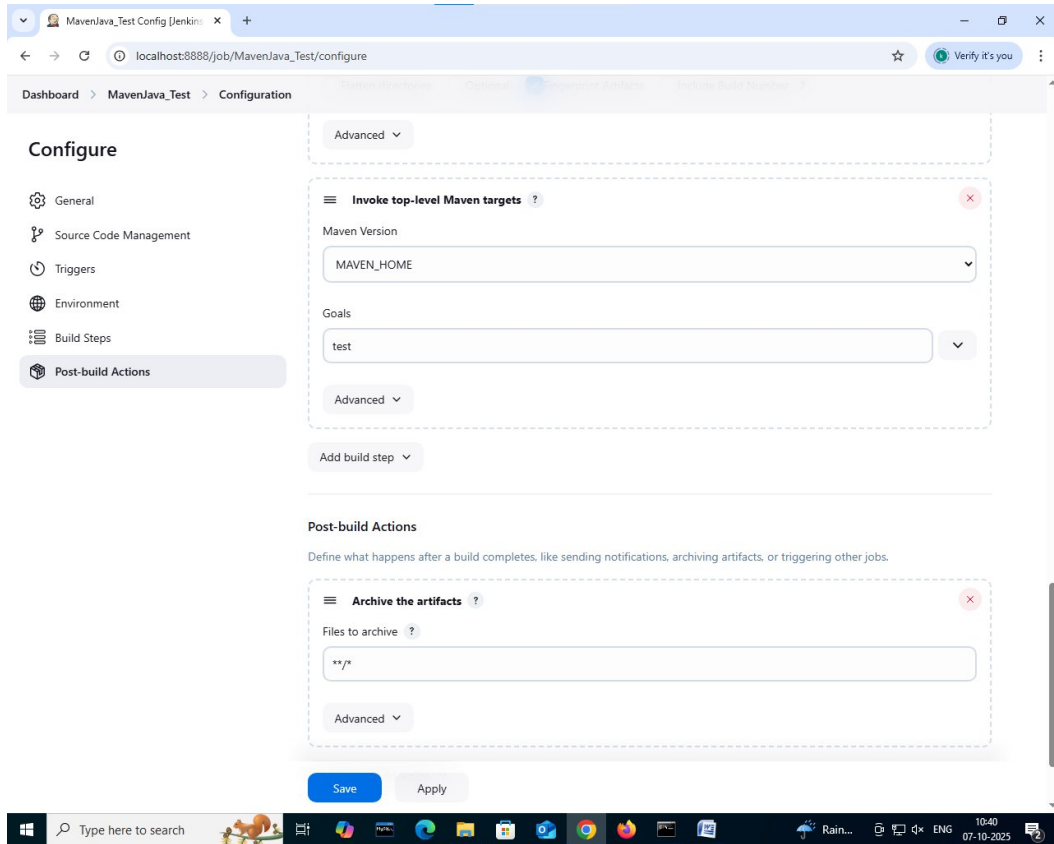


In the build steps> select add a build step> select “copy artifacts from another project” > give project name as Maven java and artifacts to copy as **/*

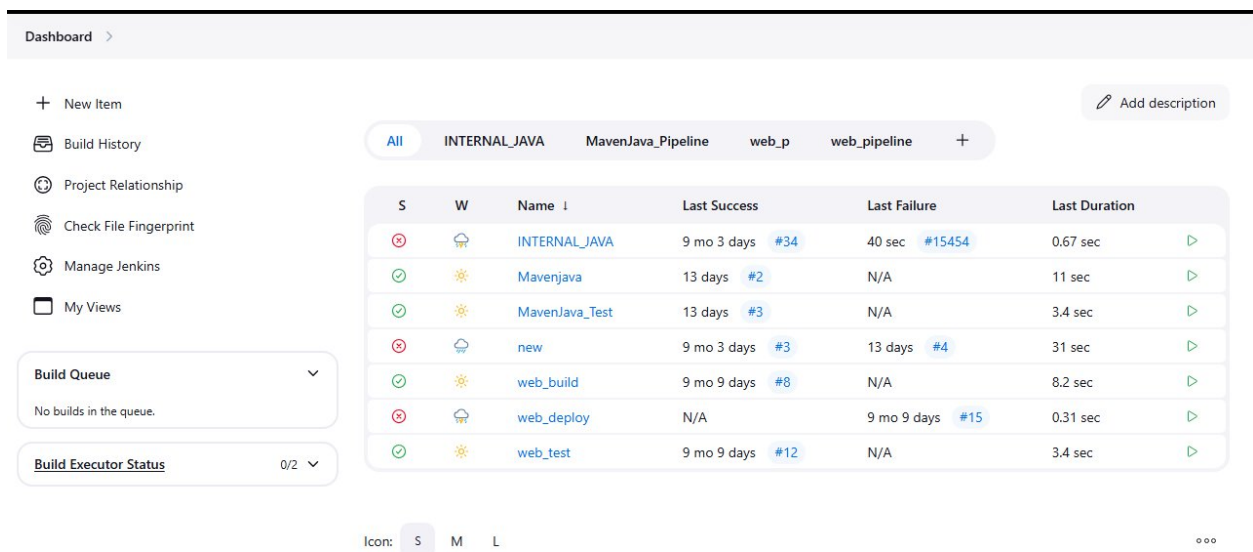


In the post build actions> select archive the artifacts and enter files as **/*

Click on apply and save



In the dashboard you will find MavenJava and MavenJava_Test



If you open the MavenJava file the following will be shown in case on no errors

The screenshot shows the Jenkins web interface for the 'MavenJava' job. The job status is 'Success' (green checkmark). The left sidebar contains navigation links: Status, Changes, Workspace, Build Now, Configure, Delete Project, and Rename. The main content area displays 'Last Successful Artifacts' as a table with columns for file name, size, and a 'view' link. Below this, it shows 'Downstream Projects' (MavenJava_Test) and 'Permalinks' (Last build #21, 13 days ago).

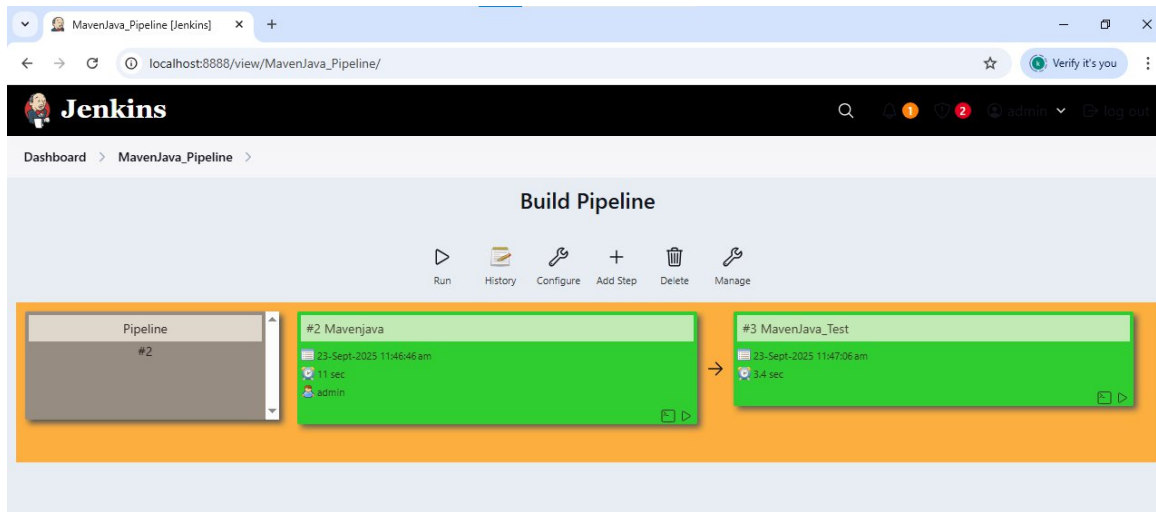
File Name	Size	View
.classpath	1.65 KiB	view
.project	1.06 KiB	view
.jsdtscope	639 B	view
org.eclipse.jdt.core.prefs	616 B	view
org.eclipse.m2e.core.prefs	90 B	view
org.eclipse.wst.common.component	665 B	view
org.eclipse.wst.common.project.facet.core.xml	252 B	view
org.eclipse.wst.jsdt.ui.superType.container	49 B	view
org.eclipse.wst.jsdt.ui.superType.name	6 B	view
org.eclipse.wst.validation.prefs	50 B	view
Dockerfile	131 B	view
pom.xml	738 B	view
webapp/index.jsp	57 B	view
webapp/WEB-INF/web.xml	222 B	view
pom.properties	71 B	view
org.demo/index.jsp	57 B	view
org.demo/WEB-INF/web.xml	222 B	view
org.demo.war	1.64 KiB	view

If you open the MavenJava_Test file the following will be shown in case on no errors

The screenshot shows the Jenkins web interface for the 'MavenJava_Test' job. The job status is 'Success' (green checkmark). The left sidebar contains navigation links: Status, Changes, Workspace, Build Now, Configure, Delete Project, and Rename. The main content area displays 'Last Successful Artifacts' as a table with columns for file name, size, and a 'view' link. Below this, it shows 'Upstream Projects' (MavenJava) and 'Permalinks' (Last build #3, 13 days ago; Last stable build #3, 13 days ago).

File Name	Size	View
.classpath	1.65 KiB	view
.project	1.06 KiB	view
.jsdtscope	639 B	view
org.eclipse.jdt.core.prefs	616 B	view
org.eclipse.m2e.core.prefs	90 B	view
org.eclipse.wst.common.component	665 B	view
org.eclipse.wst.common.project.facet.core.xml	252 B	view
org.eclipse.wst.jsdt.ui.superType.container	49 B	view
org.eclipse.wst.jsdt.ui.superType.name	6 B	view
org.eclipse.wst.validation.prefs	50 B	view
Dockerfile	131 B	view
pom.xml	738 B	view
webapp/index.jsp	57 B	view
webapp/WEB-INF/web.xml	222 B	view
pom.properties	71 B	view
org.demo/index.jsp	57 B	view
org.demo/WEB-INF/web.xml	222 B	view
org.demo.war	1.64 KiB	view

MavenJava_pipeline

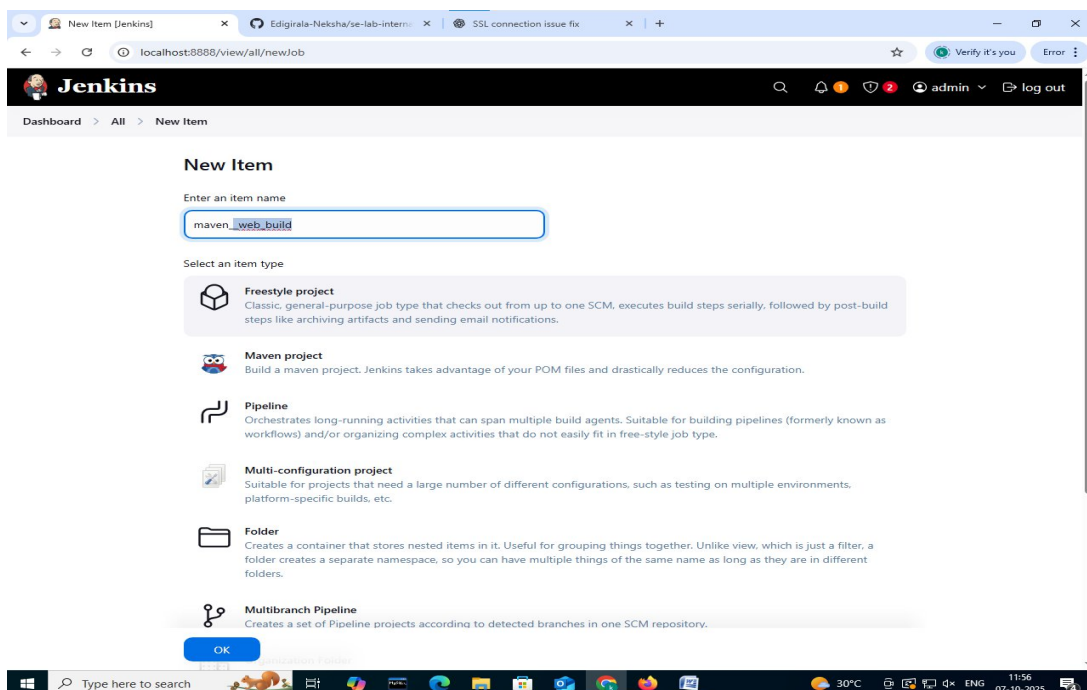


II. Maven Web Automation Steps:

Create Freestyle Project (e.g., MavenWeb_Build)

Step 1: Open Jenkins (localhost:8888)

Click on "New Item" (left side menu) and name it as maven_web_build> select freestyle project > click on "OK"



Step 2: Configuration of maven_web_build project

Give the description

The screenshot shows the Jenkins web interface in a browser. The address bar indicates the URL is `localhost:8888/job/maven_web_build/configure`. The Jenkins logo and navigation links are at the top. The breadcrumb trail shows `Dashboard > maven_web_build > Configuration`. On the left, the 'Configure' sidebar lists various settings: General (selected), Source Code Management, Triggers, Environment, Build Steps, and Post-build Actions. The main area is titled 'General' and shows a toggle switch for 'Enabled' which is turned on. Below this, there is a 'Description' text area containing the text 'web build demo'. A 'Plain text' link and a 'Preview' link are visible. A list of checkboxes follows: 'Discard old builds', 'GitHub project', 'Permission to Copy Artifact', 'This project is parameterized', 'Throttle builds', and 'Execute concurrent builds if necessary'. An 'Advanced' dropdown menu is also present. At the bottom, the 'Source Code Management' section is partially visible, with a description: 'Connect and manage your code repository to automatically pull the latest code for your builds.' Two buttons, 'Save' and 'Apply', are at the bottom of the configuration area.

Configure

- General
- Source Code Management
- Triggers
- Environment
- Build Steps
- Post-build Actions

General

Enabled

Description

web build demo

Plain text Preview

- ☐ Discard old builds
- ☐ GitHub project
- ☐ Permission to Copy Artifact
- ☐ This project is parameterized
- ☐ Throttle builds
- ☐ Execute concurrent builds if necessary

Advanced

Source Code Management

Connect and manage your code repository to automatically pull the latest code for your builds.

Save Apply

In the source code management select git and give the git repo link

The screenshot shows the Jenkins configuration interface for a job named 'maven_web_build'. The 'Source Code Management' section is active, and 'Git' is selected as the provider. The 'Repository URL' is configured with the GitHub link 'https://github.com/Edigirala-Neksha/se-lab-internal-1.git'. Under 'Branches to build', the 'Branch Specifier' is set to '*/main'. The 'Save' button is highlighted in blue at the bottom of the configuration panel.

Dashboard > maven_web_build > Configuration

Configure

- General
- Source Code Management**
- Triggers
- Environment
- Build Steps
- Post-build Actions

Source Code Management

Connect and manage your code repository to automatically pull the latest code for your builds.

☐ None ☒ **Git** ?

Repositories ?

Repository URL ?

https://github.com/Edigirala-Neksha/se-lab-internal-1.git

Credentials ?

- none -

+ Add

Advanced ▾

Add Repository

Branches to build ?

Branch Specifier (blank for 'any') ?

*/main

Add Branch

Save Apply

Windows taskbar: Type here to search, 30°C, 11:57, 07-10-2025

In the build steps click on add build step > give maven version as MAVEN_HOME > select invoke top-level maven targets > goals as clean

For the second build step,

In the build steps click on add build step > give maven version as MAVEN_HOME > select invoke top-level maven targets > goals as install

The screenshot shows the Jenkins 'Configure' page for a job named 'maven_web_build'. The 'Build Steps' section is active, showing two build steps. The first step is 'Invoke top-level Maven targets' with 'Maven Version' set to 'MAVEN_HOME' and 'Goals' set to 'clean'. The second step is also 'Invoke top-level Maven targets' with 'Maven Version' set to 'MAVEN_HOME' and 'Goals' set to 'install'. The 'Add build step' button is visible at the bottom of the list. The 'Save' and 'Apply' buttons are at the bottom of the page. The left sidebar shows the 'Configure' menu with options: General, Source Code Management, Triggers, Environment, Build Steps (selected), and Post-build Actions. The top of the page shows the browser address bar with 'localhost:8888/job/maven_web_build/configure' and the Jenkins logo.

In the post build actions > click on add post build action > select the archive the artifacts > in the file to archive give “**/*”

For the second post build action,

In the post build actions > click on add post build action > select build other projects > give projects to build as maven_web_test

Click on apply and save

The screenshot shows the Jenkins 'Configure' page for the 'maven_web_build' job. The 'Post-build Actions' section is active, showing two configured actions:

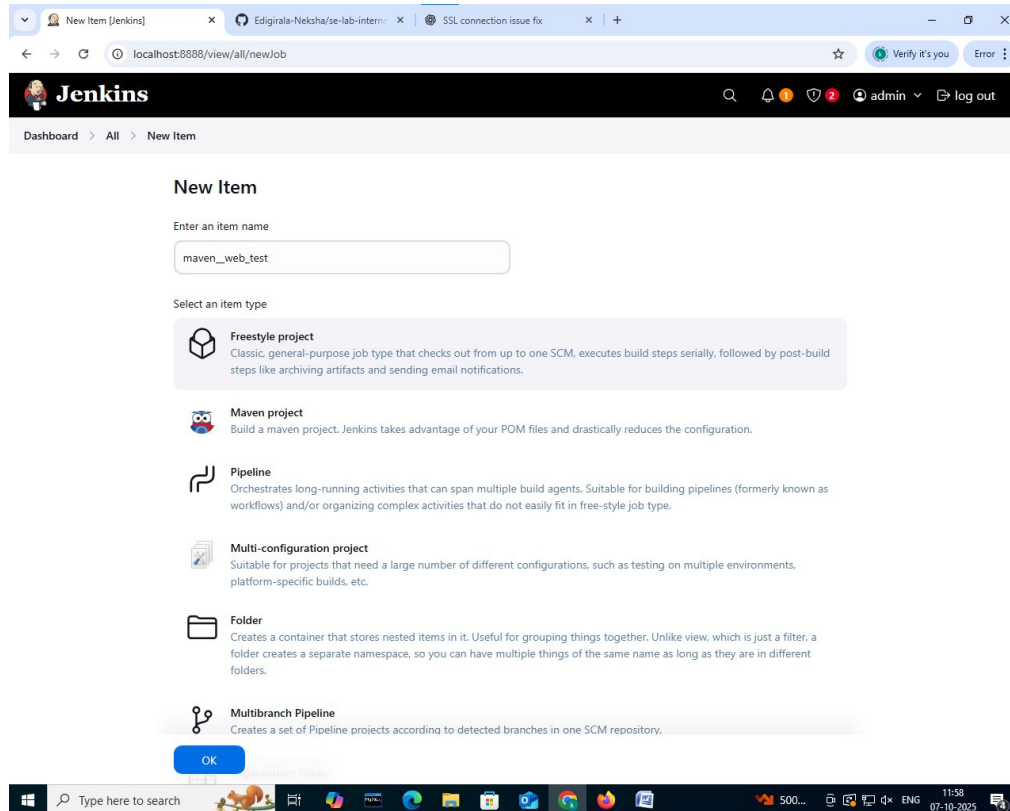
- Archive the artifacts**: Files to archive are set to `**/*`.
- Build other projects**: Projects to build are set to `maven_web_test`. The trigger condition is set to 'Trigger only if build is stable'.

The 'Save' button is highlighted in blue, indicating it has been clicked. The bottom of the screen shows the Windows taskbar with the search bar and various application icons.

Create Freestyle Project (e.g., MavenWeb_Test):

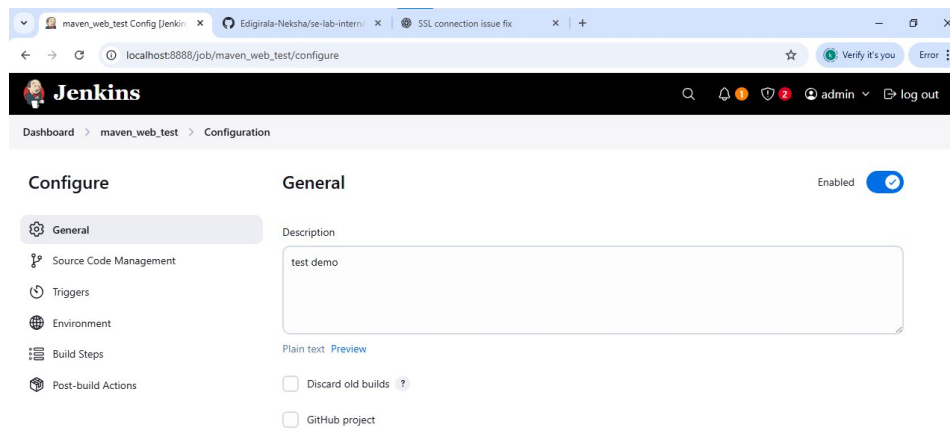
Step 1: Open Jenkins (localhost:8888)

Click on "New Item" (left side menu) and name it as maven_web_test> select freestyle project > click on "OK"



Step 2: Configuration of maven_web_test project

Give the description



In the source code management select none and in environment select “delete workspace before build starts”

The screenshot shows the Jenkins web interface in a browser. The address bar indicates the URL is `localhost:8888/job/maven_web_test/configure`. The breadcrumb navigation shows `Dashboard > maven_web_test > Configuration`. On the left, the 'Configure' sidebar has tabs for General, Source Code Management (selected), Triggers, Environment, Build Steps, and Post-build Actions. The main content area is divided into three sections: Source Code Management, Triggers, and Environment. In the Source Code Management section, 'None' is selected under 'Connect and manage your code repository'. In the Triggers section, all checkboxes are unchecked. In the Environment section, 'Delete workspace before build starts' is checked, and the 'Advanced' dropdown is visible. At the bottom of the configuration area are 'Save' and 'Apply' buttons. The Windows taskbar at the bottom shows the search bar, task view button, and several application icons, with the system clock displaying 11:59 on 07-10-2025.

Dashboard > maven_web_test > Configuration

Configure

- General
- Source Code Management**
- Triggers
- Environment
- Build Steps
- Post-build Actions

Source Code Management

Connect and manage your code repository to automatically pull the latest code for your builds.

☒ None

☐ Git ?

Triggers

Set up automated actions that start your build based on specific events, like code changes or scheduled times.

☐ Trigger builds remotely (e.g., from scripts) ?

☐ Build after other projects are built ?

☐ Build periodically ?

☐ GitHub hook trigger for GITScm polling ?

☐ Poll SCM ?

Environment

Configure settings and variables that define the context in which your build runs, like credentials, paths, and global parameters.

☒ Delete workspace before build starts

Advanced ▾

☐ Use secret text(s) or file(s) ?

☐ Provide Configuration files ?

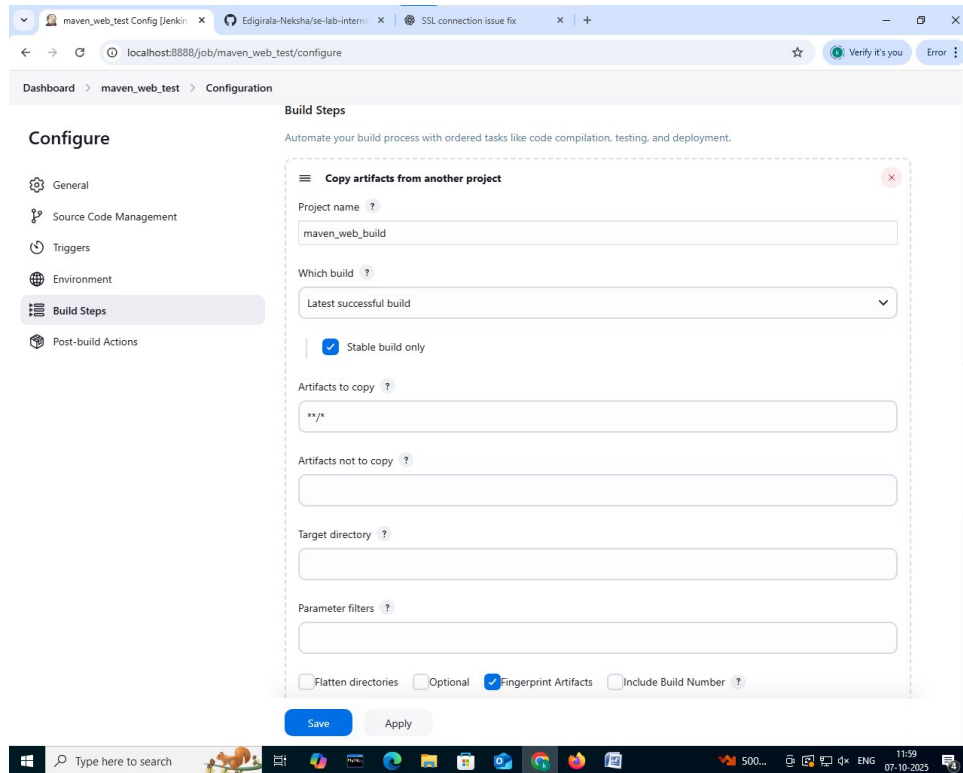
☐ Add timestamps to the Console Output

☐ Inspect build log for published build scans

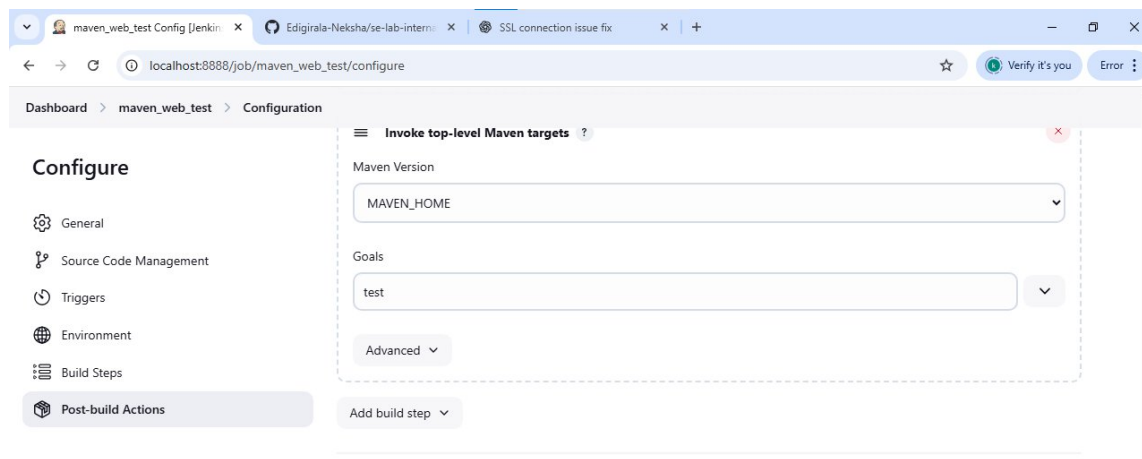
☐ Terminate a build if it's stuck

Save Apply

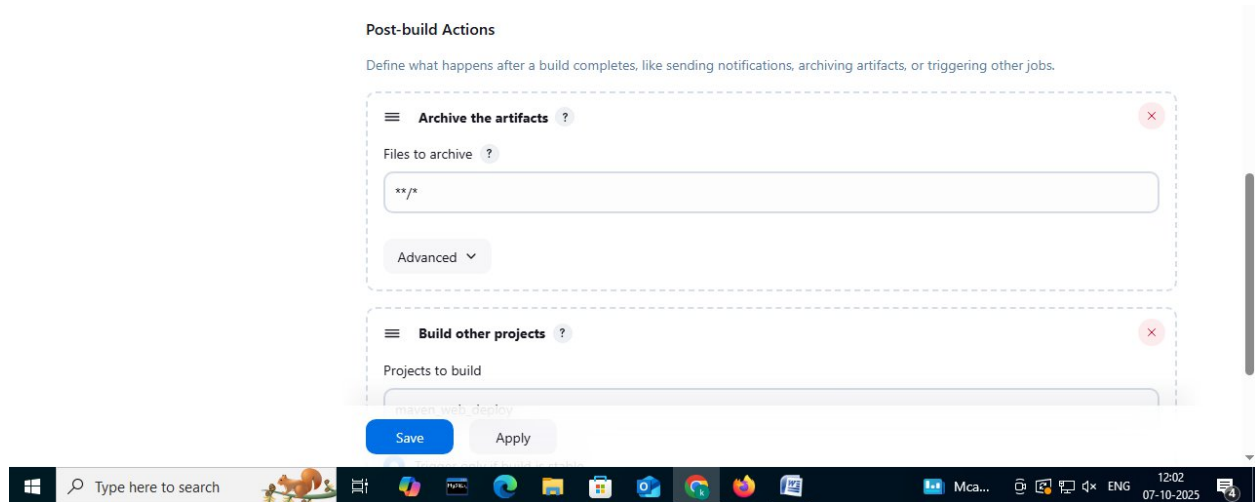
In the build steps click on add build step > select copy artifacts from another project > give project name as maven_web_build > give artifacts to copy as **/*



In the build steps click on add build step > give maven version as MAVEN_HOME > select invoke top-level maven targets > goals as test



In the post build actions > click on add post build action > select the archive the artifacts > in the file to archive give **/*



In the post build actions > click on add post build action > select build other projects > give name as maven_web_deploy> select “trigger only if build is stable”



If the build is success:

The screenshot displays the Jenkins web interface in a browser. The address bar shows the URL `localhost:8888/view/webpipeline/job/maven_web_test/`. The Jenkins logo and navigation links are at the top. The main content area shows the job `maven_web_test` with a green checkmark indicating a successful build. Below the job name, there's a section for 'Upstream Projects' with `maven_web_build` and 'Downstream Projects' with `maven_web_deploy`. A 'Builds' list on the left shows four builds, all successful, with the most recent at 12:36 PM. A 'Permalinks' section on the right provides links to the last build, stable build, successful build, and completed build, all from 1 minute and 30 seconds ago. The bottom of the screen shows a Windows taskbar with various application icons and a system tray with the date and time.

Dashboard > webpipeline > maven_web_test >

Status ✓ maven_web_test [Edit description](#)

</> Changes

Workspace

Build Now

Configure

Delete Project

Rename

Builds

Filter

Today

- ✓ #4 12:36 PM
- ✓ #3 12:36 PM
- ✓ #2 11:43 AM
- ✓ #1 11:43 AM

test demo

Upstream Projects

- ✓ maven_web_build

Downstream Projects

- ✓ maven_web_deploy

Permalinks

- Last build (#4), 1 min 30 sec ago
- Last stable build (#4), 1 min 30 sec ago
- Last successful build (#4), 1 min 30 sec ago
- Last completed build (#4), 1 min 30 sec ago

REST API Jenkins 2.489

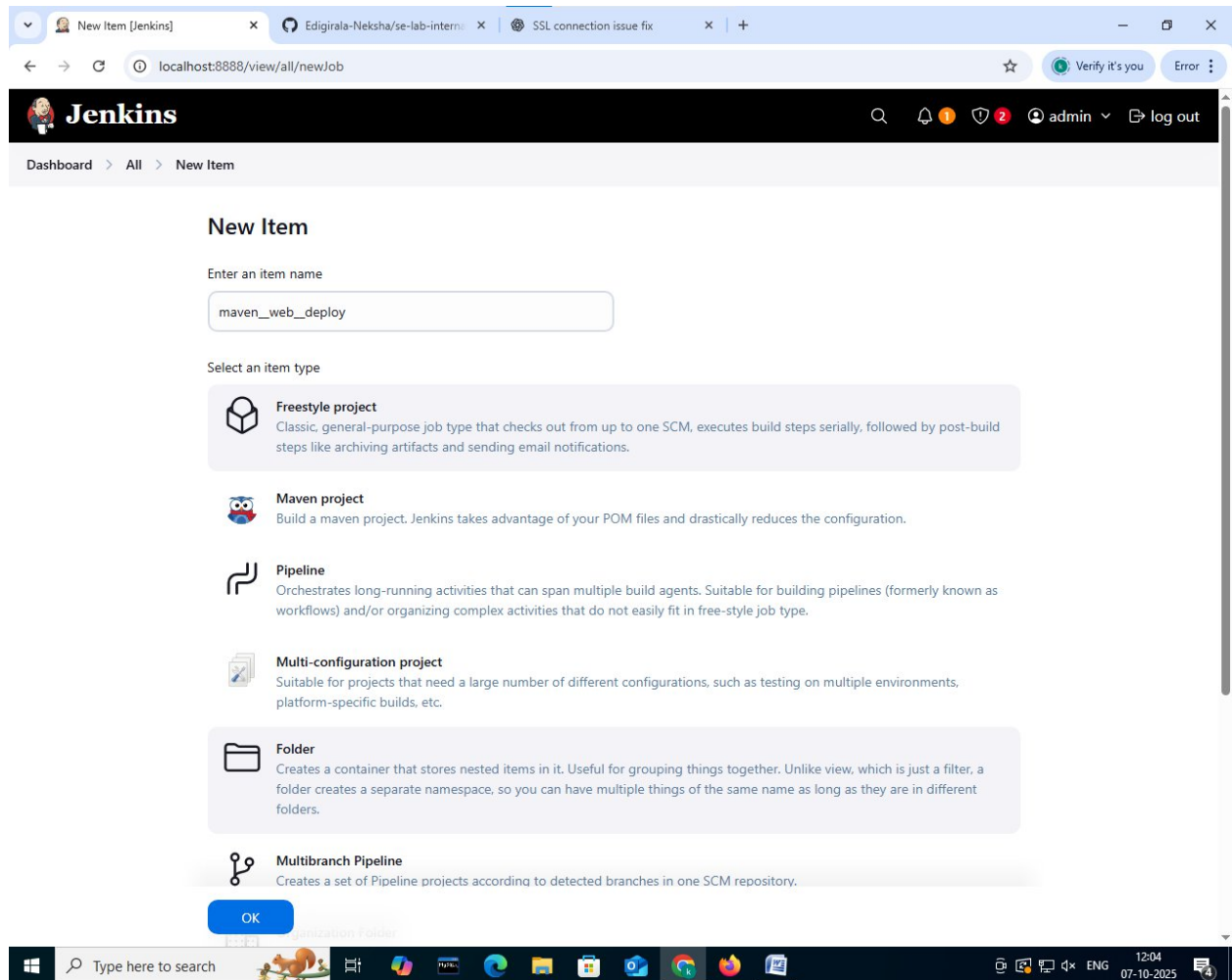
Type here to search

30°C 12:37 07-10-2025

Create Freestyle Project (e.g., MavenWeb_Deploy):

Step 1: Open Jenkins (localhost:8888)

Click on "New Item" (left side menu) and name it as maven_web_deploy > select freestyle project > click on "OK"



Step 2: Configuration of maven_web_deploy project

Give the description

The screenshot shows the Jenkins web interface in a browser. The address bar indicates the URL is `localhost:8888/job/maven_web_deploy/configure`. The Jenkins logo and navigation links are at the top. The breadcrumb trail shows `Dashboard > maven_web_deploy > Configuration`. On the left, the 'Configure' sidebar lists various settings: General (selected), Source Code Management, Triggers, Environment, Build Steps, and Post-build Actions. The main area is titled 'General' and shows a 'Description' text area containing the text 'deploy demo'. Below this are several unchecked checkboxes: 'Discard old builds', 'GitHub project', 'Permission to Copy Artifact', 'This project is parameterized', 'Throttle builds', and 'Execute concurrent builds if necessary'. An 'Advanced' dropdown menu is visible. At the bottom, the 'Source Code Management' section is partially visible. The 'Save' and 'Apply' buttons are at the bottom of the configuration area. The Windows taskbar at the bottom shows the search bar, task view button, and several application icons. The system clock indicates the time is 12:04 on 07-10-2025.

Dashboard > maven_web_deploy > Configuration

Configure

- General
- Source Code Management
- Triggers
- Environment
- Build Steps
- Post-build Actions

General

Enabled ☒

Description

deploy demo

Plain text [Preview](#)

- ☐ Discard old builds ?
- ☐ GitHub project
- ☐ Permission to Copy Artifact
- ☐ This project is parameterized ?
- ☐ Throttle builds ?
- ☐ Execute concurrent builds if necessary ?

Advanced ▾

Source Code Management

Connect and manage your code repository to automatically pull the latest code for your builds.

[Save](#) [Apply](#)

In the source code management select none and in environment select “delete workspace before build starts”

The screenshot shows the Jenkins Configuration page for a job named 'maven_web_deploy'. The browser address bar indicates the URL is 'localhost:8888/job/maven_web_deploy/configure'. The left sidebar contains a 'Configure' section with a list of tabs: General, Source Code Management (selected), Triggers, Environment, Build Steps, and Post-build Actions. The main content area is divided into three sections: 'Source Code Management', 'Triggers', and 'Environment'. In the 'Source Code Management' section, the 'None' radio button is selected. In the 'Triggers' section, all checkboxes are unchecked. In the 'Environment' section, the 'Delete workspace before build starts' checkbox is checked, and the 'Advanced' dropdown is visible. At the bottom of the configuration area are 'Save' and 'Apply' buttons. The Windows taskbar at the bottom shows the search bar, task view button, and several application icons, including NIFTY, with the system clock displaying 12:07 on 07-10-2025.

Dashboard > maven_web_deploy > Configuration

Configure

- General
- Source Code Management**
- Triggers
- Environment
- Build Steps
- Post-build Actions

Source Code Management

Connect and manage your code repository to automatically pull the latest code for your builds.

☒ None

☐ Git ?

Triggers

Set up automated actions that start your build based on specific events, like code changes or scheduled times.

☐ Trigger builds remotely (e.g., from scripts) ?

☐ Build after other projects are built ?

☐ Build periodically ?

☐ GitHub hook trigger for GITScm polling ?

☐ Poll SCM ?

Environment

Configure settings and variables that define the context in which your build runs, like credentials, paths, and global parameters.

☒ Delete workspace before build starts

Advanced ▾

☐ Use secret text(s) or file(s) ?

☐ Provide Configuration files ?

☐ Add timestamps to the Console Output

☐ Inspect build log for published build scans

☐ Terminate a build if it's stuck

Save **Apply**

In the build steps click on add build step > select copy artifacts from another project > give project name as maven_web_test > give artifacts to copy as **/*

The screenshot shows the Jenkins web interface. The browser address bar indicates the URL is `localhost:8888/job/maven_web_deploy/configure`. The page title is "Configure" and the breadcrumb navigation shows "Dashboard > maven_web_deploy > Configuration". On the left sidebar, the "Build Steps" option is selected. The main content area is titled "Build Steps" with the subtitle "Automate your build process with ordered tasks like code compilation, testing, and deployment." Below this, a dashed box contains the configuration for the "Copy artifacts from another project" build step. The configuration fields are: "Project name" (text input with value "maven_web_test"), "Which build" (dropdown menu with value "Latest successful build"), "Stable build only" (checked checkbox), "Artifacts to copy" (text input with value "**/*"), "Artifacts not to copy" (empty text input), "Target directory" (empty text input), "Parameter filters" (empty text input), and "Flatten directories" (unchecked checkbox). At the bottom of the configuration box, there are checkboxes for "Optional" (unchecked), "Fingerprint Artifacts" (checked), and "Include Build Number" (unchecked). Below the configuration box are "Save" and "Apply" buttons. The Windows taskbar at the bottom shows the search bar, task view button, and several application icons. The system tray on the right shows the date and time as "07-10-2025 12:07".

Dashboard > maven_web_deploy > Configuration

Configure

- General
- Source Code Management
- Triggers
- Environment
- Build Steps**
- Post-build Actions

Build Steps

Automate your build process with ordered tasks like code compilation, testing, and deployment.

Copy artifacts from another project

Project name ?
maven_web_test

Which build ?
Latest successful build

☒ Stable build only

Artifacts to copy ?
**/*

Artifacts not to copy ?

Target directory ?

Parameter filters ?

☐ Flatten directories ☐ Optional ☒ Fingerprint Artifacts ☐ Include Build Number ?

Save Apply

In the post build actions > click on add post build actions > select deploy war/ear to a container > enter war/ear files as `**/*.war`> context path as `webpath` > give the credentials and tomcat URL

The screenshot shows the Jenkins configuration interface for a job named 'maven_web_deploy'. The 'Post-build Actions' tab is selected in the left sidebar. Under 'Post-build Actions', the 'Deploy war/ear to a container' action is configured. The 'WAR/EAR files' field contains '**/*.war'. The 'Context path' field contains 'webpath'. Under the 'Containers' section, the 'Tomcat 9.x Remote' container is selected. The 'Credentials' field contains 'admin/*****'. The 'Tomcat URL' field contains 'https://localhost:8080/'. The 'Advanced' options are expanded, showing a 'Save' button and an 'Apply' button. The bottom of the screen shows the Windows taskbar with the search bar and various application icons.

If the build is success:

The screenshot shows the Jenkins web interface in a browser. The browser's address bar displays 'localhost:8888/job/maven_web_deploy/'. The Jenkins header includes the logo, a search icon, notification icons, a user profile for 'admin', and a 'log out' link. The breadcrumb trail shows 'Dashboard > maven_web_deploy >'. On the left sidebar, the 'Status' tab is selected, showing options like 'Changes', 'Workspace', 'Build Now', 'Configure', 'Delete Project', and 'Rename'. The main content area features a green checkmark icon and the job name 'maven_web_deploy', with an 'Edit description' button. Below this, the 'Upstream Projects' section lists 'maven_web_test'. The 'Permalinks' section provides links to various build types: Last build (#13), Last stable build (#13), Last successful build (#13), Last failed build (#11), Last unsuccessful build (#11), and Last completed build (#13). A 'Builds' panel on the left lists recent builds from #4 to #13, with the last three marked as successful (green checkmarks) and the previous ones as failed (red X's). The Windows taskbar at the bottom shows the system clock as 12:38 on 07-10-2025.

Dashboard > maven_web_deploy >

Status

maven_web_deploy

deploy demo

Upstream Projects

maven_web_test

Permalinks

- Last build (#13), 1 min 54 sec ago
- Last stable build (#13), 1 min 54 sec ago
- Last successful build (#13), 1 min 54 sec ago
- Last failed build (#11), 6 min 58 sec ago
- Last unsuccessful build (#11), 6 min 58 sec ago
- Last completed build (#13), 1 min 54 sec ago

Builds

Filter

Today

- #13 12:36 PM
- #12 12:35 PM
- #11 12:31 PM
- #10 12:28 PM
- #9 12:20 PM
- #8 12:19 PM
- #7 12:12 PM
- #6 11:50 AM
- #5 11:50 AM
- #4 11:48 AM

Type here to search

30°C

12:38

07-10-2025

Create Pipeline View for MavenWeb

Click "+" beside "All" on the dashboard and Enter name as maven_web_pipeline

Select type as build pipeline view

New view [Jenkins] x Edigirala-Neksha/se-lab-intern x SSL connection issue fix x +

localhost:8888/newView

Verify it's you Error

Jenkins 🔍 🔔 1 🛡️ 2 👤 admin 🚪 log out

Dashboard > New view

+ New Item

- 📁 Build History
- 👤 Project Relationship
- 🔍 Check File Fingerprint
- ⚙️ Manage Jenkins
- 📁 My Views

Build Queue ▼
No builds in the queue.

Build Executor Status 0/2 ▼

New view

Name

maven_web_pipeline

Type

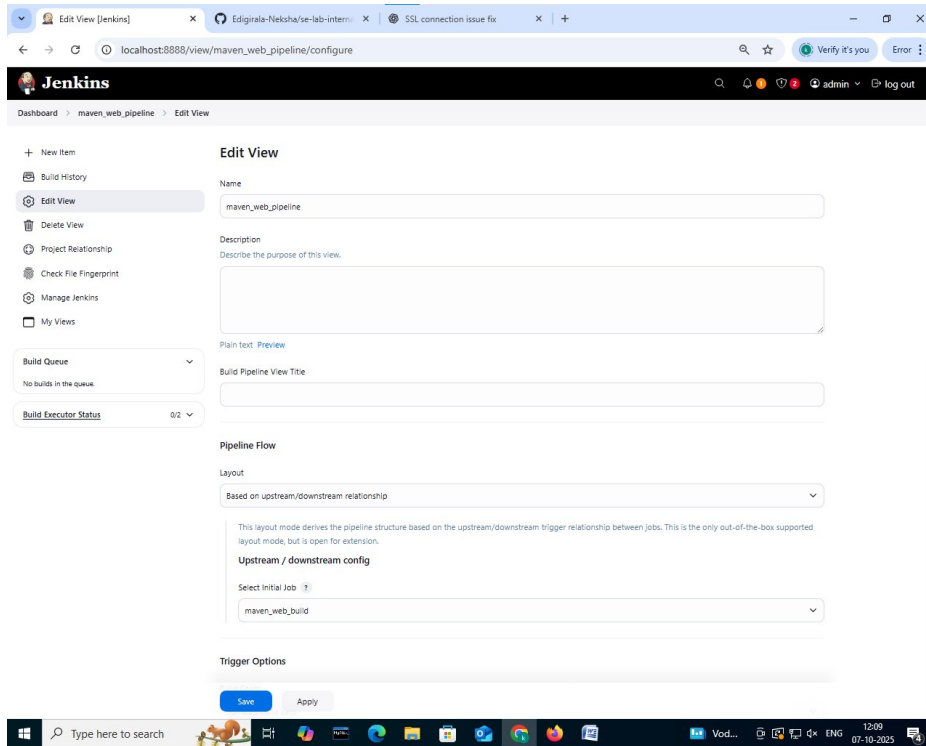
☒ **Build Pipeline View**
Shows the jobs in a build pipeline view. The complete pipeline of jobs that a version propagates through are shown as a row in the view.

☐ **List View**
Shows items in a simple list format. You can choose which jobs are to be displayed in which view.

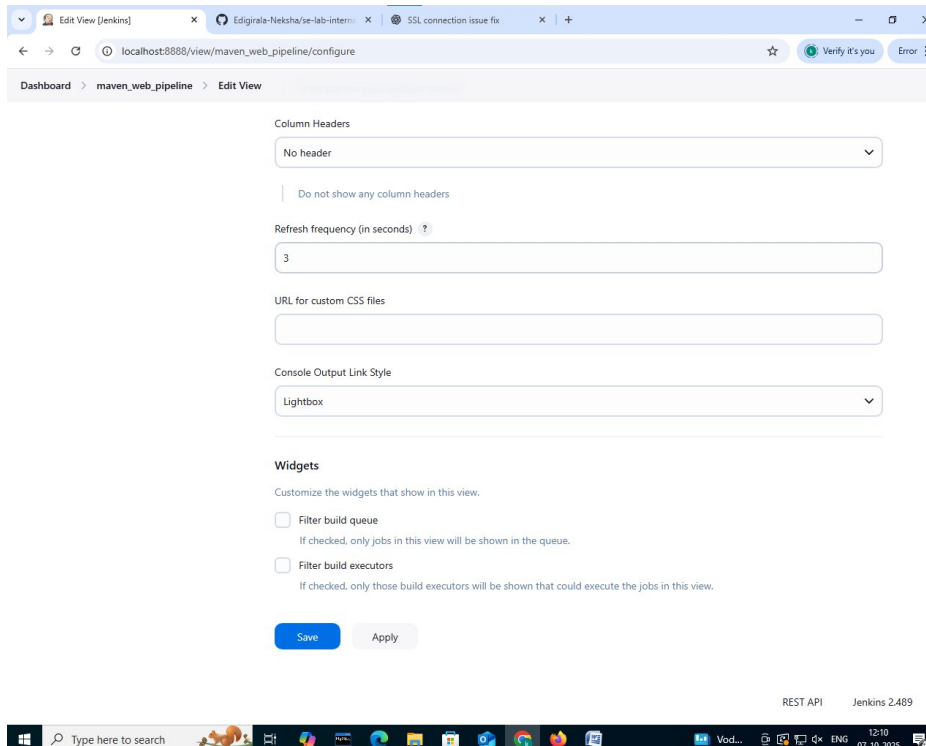
☐ **My View**
This view automatically displays all the jobs that the current user has an access to.

Create

Give the description and in the upstream directly the maven_web_build will be shown



Click on apply and save



In the stage view it we be shown as:

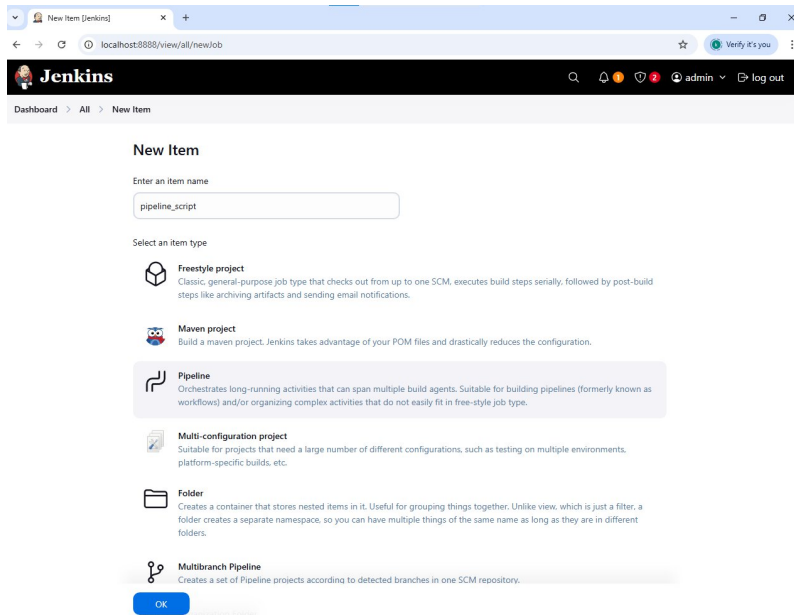
The screenshot displays the Jenkins web interface in a browser window. The address bar shows 'localhost:8888/view/webpipeline/'. The Jenkins logo and navigation links are at the top. The main section is titled 'Build Pipeline' and contains a sequence of three build steps:

- Pipeline #4**: A summary box on the left.
- #4 maven_web_build**: A green box indicating a successful build at 07-Oct-2025 12:36:06 pm, taking 11 seconds, by user 'admin'.
- #4 maven_web_test**: A green box indicating a successful test at 07-Oct-2025 12:36:23 pm, taking 4.1 seconds, by user 'admin'.
- #14 maven_web_deploy**: A green box indicating a successful deployment at 07-Oct-2025 12:42:24 pm, taking 0.9 seconds, by user 'admin'.

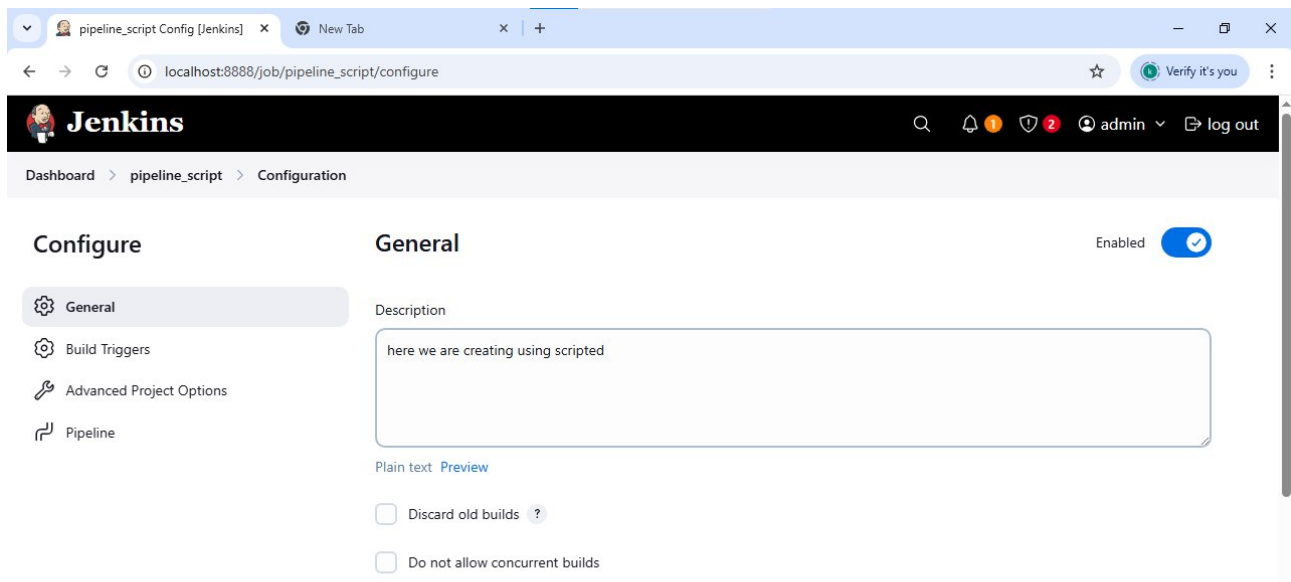
Arrows connect the build steps, showing the flow of the pipeline. At the bottom of the interface, the text 'REST API' and 'Jenkins 2.489' are visible. The Windows taskbar is at the very bottom, showing the search bar and various application icons.

9. Pipeline Creation using script

Step 1: In the Jenkins select the new item and give the name as pipeline_script and select pipeline and click ok



Step 2: In the configuration, give the description



Step 3: In the pipeline section give definition as pipeline script and enter the script with git reop link and project name

The screenshot shows the Jenkins web interface for configuring a pipeline named 'pipeline_script'. The browser address bar indicates the URL is 'localhost:8888/job/pipeline_script/configure'. The left sidebar contains a 'Configure' section with sub-links: 'General' (selected), 'Build Triggers', 'Advanced Project Options', and 'Pipeline'. The main content area is divided into several sections: 'Build Triggers' with checkboxes for 'This project is parameterized', 'Throttle builds', 'Build after other projects are built', 'Build periodically', 'Build whenever a SNAPSHOT dependency is built', 'GitHub hook trigger for GITScm polling', 'Poll SCM', 'Quiet period', and 'Trigger builds remotely (e.g., from scripts)'. Below this is the 'Advanced Project Options' section with a dropdown set to 'Advanced'. The 'Pipeline' section is at the bottom, featuring a 'Definition' dropdown menu set to 'Pipeline script' and a 'Script' text area containing a YAML pipeline definition. The script starts with '1 pipeline {' and '2 agent any'. At the bottom of the configuration page are 'Save' and 'Apply' buttons. The Windows taskbar at the very bottom shows the search bar, task view icon, and several application icons, with the system clock displaying '10:58 07-10-2025'.

pipeline_script Config [Jenkins] x New Tab x +

localhost:8888/job/pipeline_script/configure

Dashboard > pipeline_script > Configuration

Configure

- General
- Build Triggers
- Advanced Project Options
- Pipeline

☐ This project is parameterized ?

☐ Throttle builds ?

Build Triggers

- ☐ Build after other projects are built ?
- ☐ Build periodically ?
- ☐ Build whenever a SNAPSHOT dependency is built ?
- ☐ GitHub hook trigger for GITScm polling ?
- ☐ Poll SCM ?
- ☐ Quiet period ?
- ☐ Trigger builds remotely (e.g., from scripts) ?

Advanced Project Options

Advanced ▾

Pipeline

Definition

Pipeline script ▾

Script ?

```
1 pipeline {  
2   agent any  
3   tools {
```

Save Apply

Type here to search

How... ENG 10:58 07-10-2025

Step 4: click on apply and then save

The screenshot shows the Jenkins web interface for configuring a pipeline. The browser address bar indicates the URL is `localhost:8888/job/pipeline_script/configure`. The page title is "pipeline_script Config [Jenkins]". The breadcrumb navigation shows "Dashboard > pipeline_script > Configuration".

On the left, the "Configure" section has four tabs: "General", "Build Triggers", "Advanced Project Options" (which is selected), and "Pipeline".

The main content area is titled "Advanced Project Options" and has a sub-tab "Advanced". Under the "Pipeline" section, the "Definition" dropdown is set to "Pipeline script".

A "Script" section contains a text area with the following Groovy code:

```
1 pipeline {  
2   agent any  
3   tools{  
4     maven 'MAVEN-HOME'  
5   }  
6   stages {  
7     stage('git repo & clean') {  
8       steps {  
9         //bat "rmdir /s /q mavenjava"  
10        bat "git clone https://github.com/SarvikaSomishetty/eclipse-maven-projects.git"  
11        bat "mvn clean -f eclipse-maven-projects"  
12      }  
13    }  
14    stage('install') {  
15      steps {  
16        bat "mvn install -f eclipse-maven-projects"  
17      }  
18    }  
19  }  
20 }
```

Below the script area, the checkbox "Use Groovy Sandbox" is checked. A "Pipeline Syntax" link is also present.

At the bottom of the configuration area are two buttons: "Save" (in blue) and "Apply" (in grey).

The bottom of the page shows the Jenkins footer with "REST API" and "Jenkins 2.489". The Windows taskbar at the very bottom shows the search bar, task view button, and several application icons, including the Jenkins icon.

Step 8: Check the stage view. If is successful.

The screenshot shows the Jenkins web interface for the 'pipeline_script' job. The top navigation bar includes the Jenkins logo, search, notifications, and user 'admin'. The breadcrumb trail is 'Dashboard > pipeline_script >'. The left sidebar contains links for Status, Changes, Build Now, Configure, Delete Pipeline, Full Stage View, Stages, Rename, and Pipeline Syntax. The main content area shows the 'pipeline_script' job status as 'Success' with a green checkmark. Below the status, it says 'here we are creating using scripted'. The 'Stage View' section displays a table of stage durations for the current build (#2) and average stage times. The stages are: Declarative: Tool Install (296ms), git repo & clean (5s), install (9s), test (3s), and package (4s). The full run time is approximately 26 seconds. The bottom left shows the 'Builds' section with a filter and a list of builds, including the current successful build (#2) at 11:02 AM. The bottom right shows 'Permalinks' for the build.

pipeline_script [Jenkins]

localhost:8888/job/pipeline_script/

Jenkins

Dashboard > pipeline_script >

Status

Changes

Build Now

Configure

Delete Pipeline

Full Stage View

Stages

Rename

Pipeline Syntax

pipeline_script

here we are creating using scripted

Stage View

Average stage times:
(full run time: ~26s)

	Declarative: Tool Install	git repo & clean	install	test	package
#2 Oct 07 11:02 No Changes	296ms	5s	9s	3s	4s

Builds

Filter

Today

#2 11:02 AM

Permalinks

- Last build (#2), 4 min 29 sec ago
- Last stable build (#2), 4 min 29 sec ago
- Last successful build (#2), 4 min 29 sec ago
- Last completed build (#2), 4 min 29 sec ago

10. Kubernetes Using Minikube:

Step -1:

Start Minikube : Command- minikube start

- First, you need to start your Kubernetes cluster using Minikube.
- When you start it, Minikube sets up a lightweight virtual machine on your system and runs a local Kubernetes node inside it.

Step-2:Then check for the status Minikube status

Step-3:Create an image

```
PS C:\Users\User>
PS C:\Users\User> kubectl delete deployment mynginx
deployment.apps "mynginx" deleted
PS C:\Users\User> kubectl create deployment mynginx --image=nginx
deployment.apps/mynginx created
PS C:\Users\User> kubectl expose deployment mynginx --type=NodePort --port=80
service/mynginx exposed
PS C:\Users\User> kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
mynginx-79bb8756c7-wpslj            1/1     Running   0           34s
```

Step-4: Check the NGINX Service Details

- After creating the service, check its details to see which port Kubernetes assigned to it.

```
DownwardAPI: true
QoS Class:   BestEffort
Node-Selectors:
Tolerations: node.kubernetes.io/not-ready:NoExecute op=Exists for 300s
              node.kubernetes.io/unreachable:NoExecute op=Exists for 300s
Events:
  Type     Reason      Age   From          Message
  ----     -
  Normal   Scheduled   68s   default-scheduler   Successfully assigned default/mynginx-79bb8756c7-wpslj to minikube
  Normal   Pulling     67s   kubelet         Pulling image "nginx"
  Normal   Pulled      65s   kubelet         Successfully pulled image "nginx" in 2.416s (2.416s including waiting). Image size: 159974475 bytes.
  Normal   Created     65s   kubelet         Created container nginx
  Normal   Started     64s   kubelet         Started container nginx
PS C:\Users\User> kubectl scale deployment mynginx --replicas=4
deployment.apps/mynginx scaled
PS C:\Users\User> kubectl get service myngnix
Error from server (NotFound): services "myngnix" not found
PS C:\Users\User> kubectl port-forward svc/mynginx 8081:80
Forwarding from 127.0.0.1:8081 -> 80
Forwarding from [::1]:8081 -> 80
```

Step-5:check the detail of the kubectl .

```
PS C:\Users\User> kubectl describe pods
Name:          mynginx-79bb8756c7-wpslj
Namespace:     default
Priority:       0
Service Account: default
Node:          minikube/192.168.49.2
Start Time:    Tue, 14 Oct 2025 12:38:19 +0530
Labels:        app=mynginx
               pod-template-hash=79bb8756c7
Annotations:   <none>
Status:        Running
IP:            10.244.0.16
IPs:
  IP:          10.244.0.16
Controlled By: ReplicaSet/mynginx-79bb8756c7
Containers:
  nginx:
    Container ID:  docker://675066efbd98a54ba39177103943b196de2c61f01d820ede859b48578f3e245e
    Image:         nginx
    Image ID:      docker-pullable://nginx@sha256:3b7732505933ca591ce4a6d860cb713ad96a3176b82f7979a8dfa9973486a0d6
    Port:          <none>
    Host Port:     <none>
    State:         Running
      Started:     Tue, 14 Oct 2025 12:38:22 +0530
    Ready:         True
    Restart Count: 0
    Environment:   <none>
    Mounts:
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-nh2rw (ro)
Conditions:
  Type                               Status
  PodReadyToStartContainers          True
  Initialized                         True
  Ready                              True
  ContainersReady                    True
  PodScheduled                       True
Volumes:
  kube-api-access-nh2rw:
    Type:              Projected (a volume that contains injected data from multiple sources)
    TokenExpirationSeconds: 3607
    ConfigMapName:       kube-root-ca.crt
    ConfigMapOptional:   <nil>
    DownwardAPI:         true
QoS Class:               BestEffort
Node-Selectors:           <none>
Tolerations:              node.kubernetes.io/not-ready:NoExecute op=Exists for 300s
                           node.kubernetes.io/unreachable:NoExecute op=Exists for 300s
Events:
```

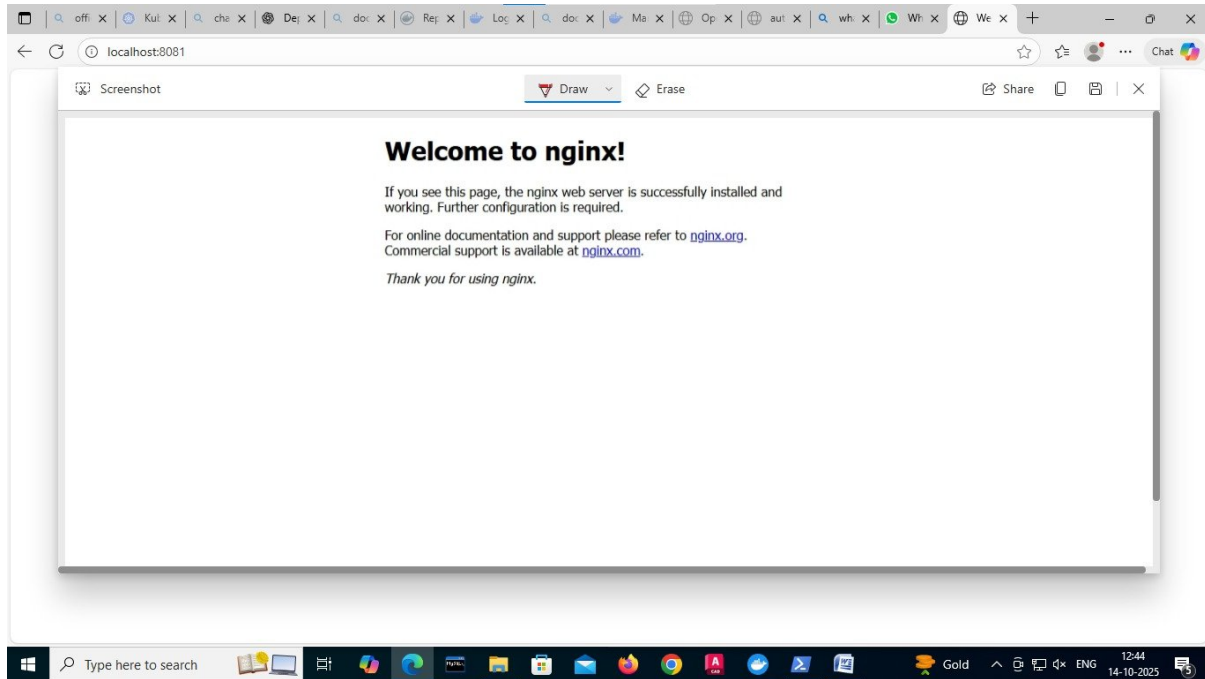
Step-6:Check the NGINX Service Details

- After creating the service, check its details to see which port Kubernetes assigned to it.

```
PS C:\Users\User> kubectl port-forward svc/mynginx 8081:80
forwarding from 127.0.0.1:8081 -> 80
forwarding from [::1]:8081 -> 80
handling connection for 8081
handling connection for 8081
handling connection for 8081
handling connection for 8081
```

Step-7: Open NGINX in the Browser

- Now that your service is exposed, you can open NGINX in your browser.



11. Jenkins-CI/CD

Setting Up Jenkins CI-----using GitHub Webhook with Jenkins

Step 1: Take the authentication key from the ngrok and setup in ngrok terminal

```
tcp          start a TCP tunnel
tls          start a TLS endpoint
update       update ngrok to the latest version
version      print the version string

EXAMPLES:
# forward http traffic from assigned public URL to local port 80
ngrok http 80
# port 8080 available at baz.ngrok.dev
ngrok http --url baz.ngrok.dev 8080
# tunnel arbitrary TCP traffic to port 22
ngrok tcp 22
# secure your app with oauth
ngrok http 80 --oauth=google --oauth-allow-email=foo@foo.com

Paid Features:
  ngrok http 80 --url mydomain.com           # run ngrok with your own custom domain
  ngrok http 80 --cidr-allow 2600:8c00::a03c:91ee:fe69:9695/32 # run ngrok with IP policy restrictions
  Upgrade your account at https://dashboard.ngrok.com/billing/subscription to access paid features

Upgrade your account at https://dashboard.ngrok.com/billing/subscription to access paid features

Flags:
  -h, --help      help for ngrok

Use "ngrok [command] --help" for more information about a command.

ngrok is a command line application, try typing 'ngrok.exe http 80'
at this terminal prompt to expose port 80.
C:\Windows\System32>ngrok config add-authtoken 34gKWhQDcoITj34K6eN73XoYG6J_58fBgmpjM5ikZVdKVDyCe|
```

Step-2: Execute the following command using the port number on which Jenkins is running

```
C:\Windows\System32>ngrok.exe http 8888|
```

- Following output will be given:

```
ngrok (Ctrl+C to quit)
♦ Block threats before they reach your services with new WAF actions → https://ngrok.com/r/waf

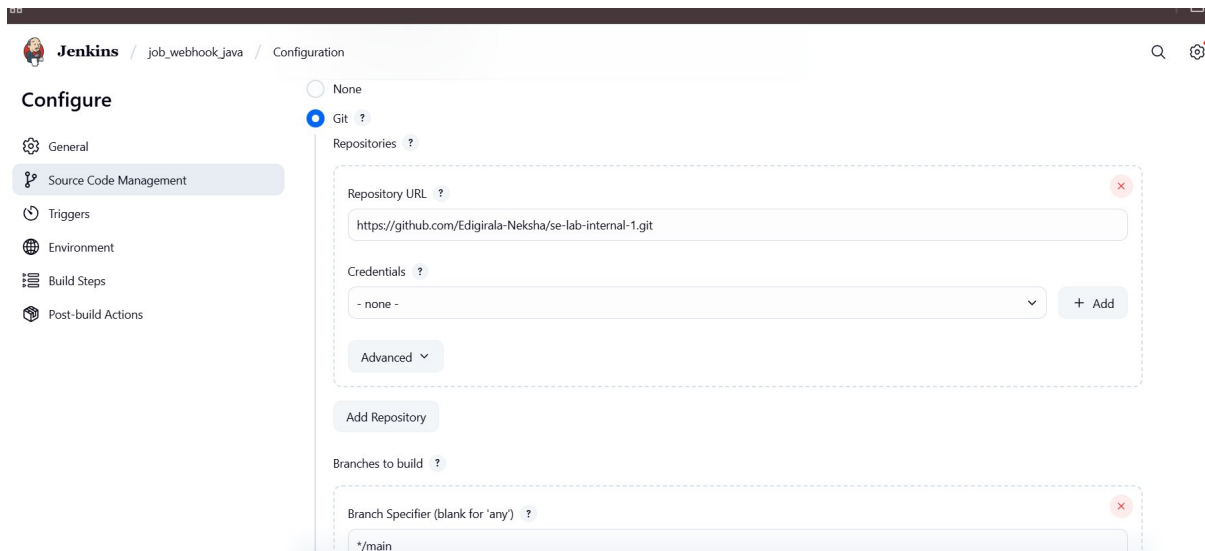
Session Status      online
Account             Neksha Edigirala (Plan: Free)
Update              update available (version 3.32.0, Ctrl-U to update)
Version             3.24.0-msix
Region              India (in)
Latency             147ms
Web Interface       http://127.0.0.1:4040
Forwarding           https://corkier-darla-handsome.ngrok-free.dev -> http://localhost:8888

Connections
  ttl   opn   rt1   rt5   p50   p90
    2     0    0.00  0.00  30.28  30.47

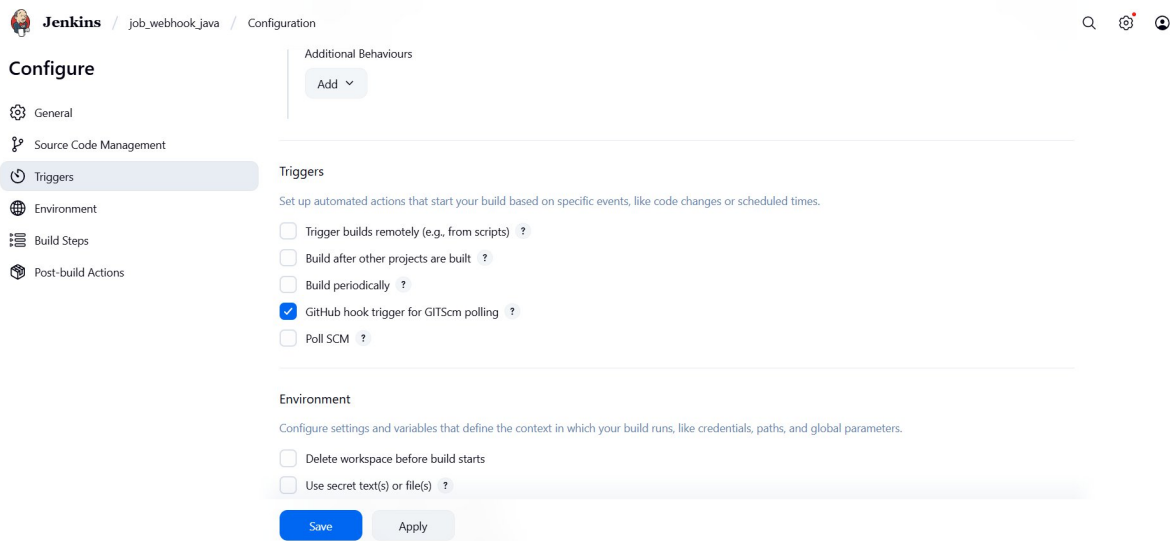
HTTP Requests
-----
11:35:59.377 IST POST /github-webhook/ 200 OK
11:34:29.479 IST POST /github-webhook/ 200 OK
```

Go to Jenkins:

Step-3: Create the Jenkins job in the source code management select the git and enter git repo url and make sure the branch is same (i.e., main)

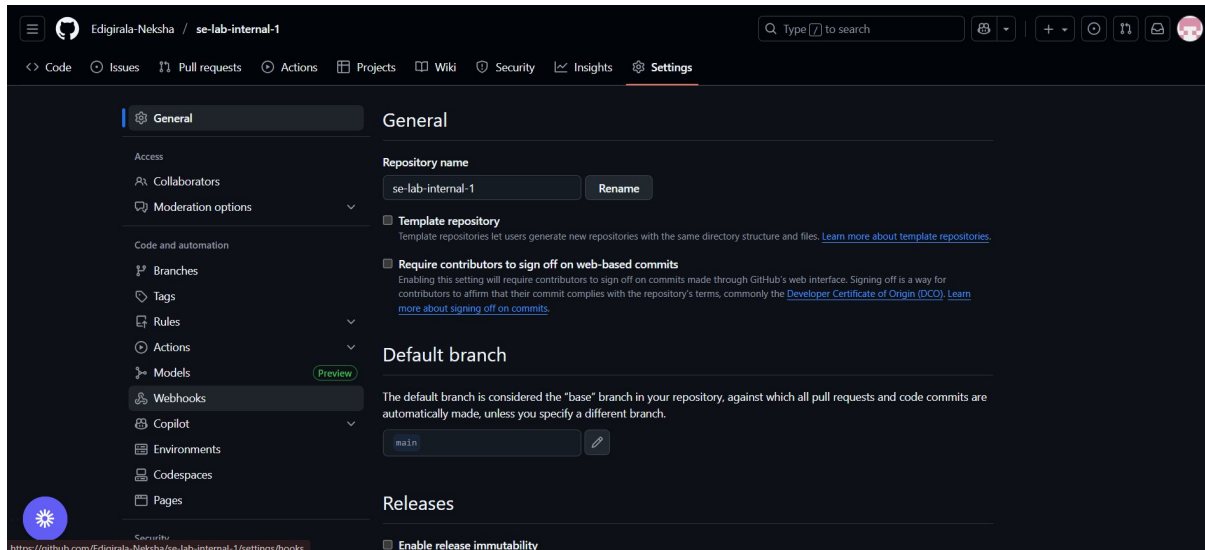


Step-4: In the triggers section select “Github hook trigger for GITScm polling”



Click on apply and save

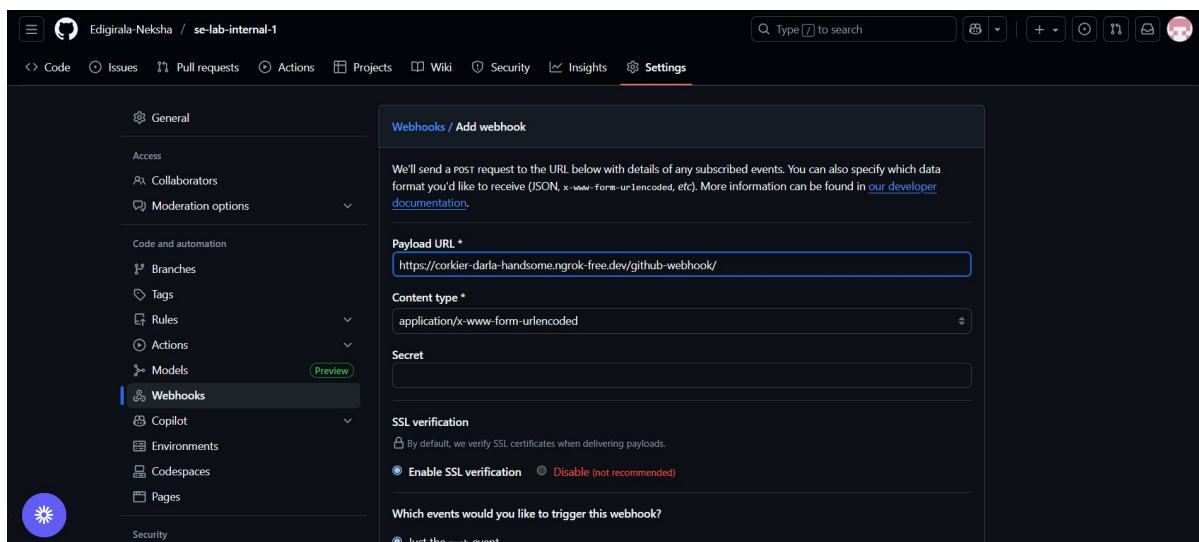
Step-6: open the git hub repo open setting of repo and then go to webhooks



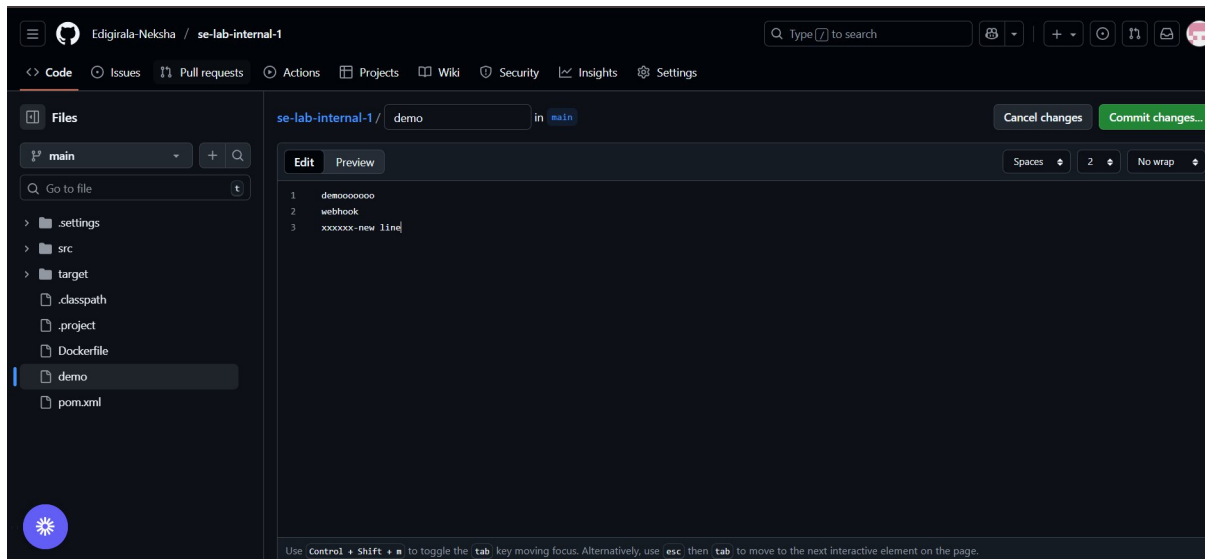
Step-7: Click on add a webhook and take the forwarding URL from ngrok and paste in payload URL and add /github-webhook/ along with the forwarding url

Forwarding URL: <https://corkier-darla-handsome.ngrok-free.dev>

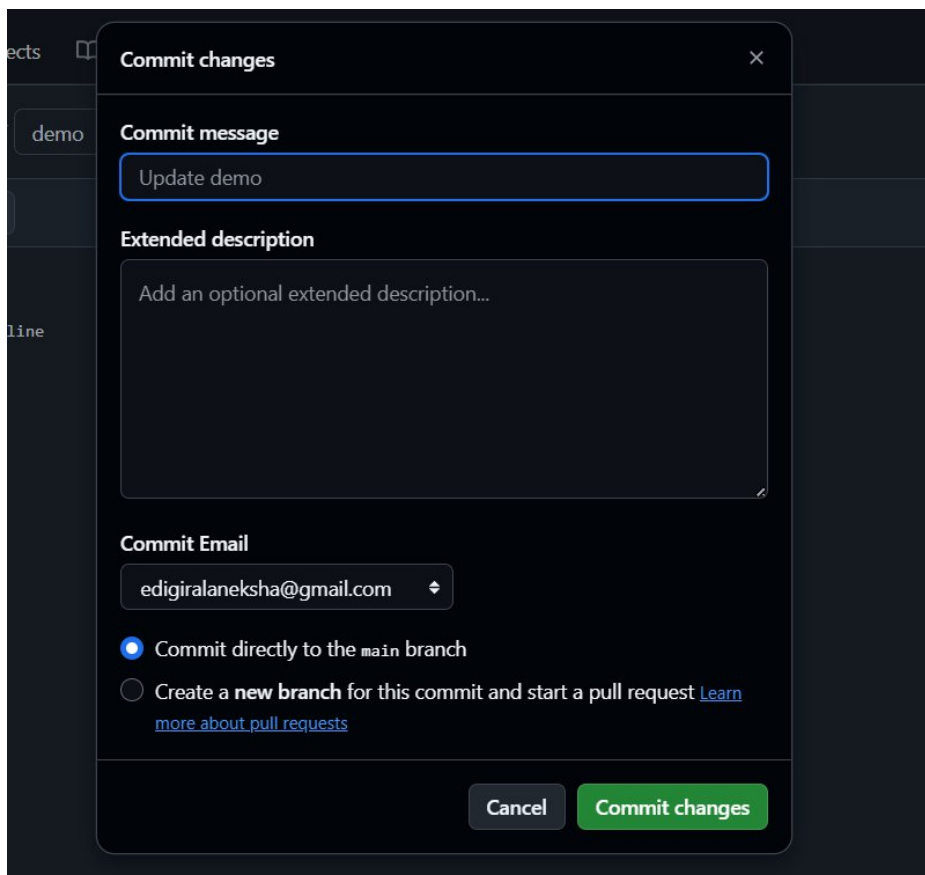
Payload url: <https://corkier-darla-handsome.ngrok-free.dev/github-webhook/>



Step 8: make changes in the files in github



Step 9: click on commit changes



Step 10: open Jenkins the build will start automatically

 **Jenkins** / job_webhook_java

Status

</> Changes

Workspace

▶ Build Now

⚙️ Configure

🗑️ Delete Project

📄 GitHub Hook Log

✎ Rename

👤 Credentials

Builds

Filter

Pending

⋯ #3

In the quiet period. Expires in 2.9 sec

Today

✔️ #2 11:36 am

✔️ job_webhook_java

Permalinks

- [Last build \(#2\), 3 min 46 sec ago](#)
- [Last stable build \(#2\), 3 min 46 sec ago](#)
- [Last successful build \(#2\), 3 min 46 sec ago](#)
- [Last completed build \(#2\), 3 min 46 sec ago](#)



Jenkins / job_webhook_java

Status

Changes

Workspace

Build Now

Configure

Delete Project

GitHub Hook Log

Rename

Credentials



job_webhook_java

Permalinks

- Last build (#2), 3 min 46 sec ago
- Last stable build (#2), 3 min 46 sec ago
- Last successful build (#2), 3 min 46 sec ago
- Last completed build (#2), 3 min 46 sec ago

Builds

Filter

Today

- ✓ #3 11:41 am
- ✓ #2 11:36 am
- ✓ #1 11:35 am

You can check status : started by git hub push



Jenkins / job_webhook_java / #3

Status

Changes

Console Output

Edit Build Information

Delete build '#3'

Polling Log

Timings

Git Build Data

Previous Build



#3 (28-Oct-2025, 11:41:02 am)

Add description

Keep this build forever



Started by GitHub push by Edigirala-Neksha



This run spent:

- 7.8 sec waiting;
- 2.1 sec build duration;
- 10 sec total from scheduled to completion.



Revision: bc52c46b2c311be243984889d49707f7839687de

Repository: <https://github.com/Edigirala-Neksha/se-lab-internal-1.git>

- refs/remotes/origin/main



Changes

1. Update demo (details / [githubweb](#))

Started 20 sec ago
Took 2.1 sec

Setting Up Jenkins Email Notification Setup (Using Gmail with AppPassword)

Step-1: Creation of app password

Gmail: Enable App Password (for 2-Step Verification)

ii. Enable 2-Step Verification

iii. Generate App Password for Jenkins

- Go to:
 - Security → App passwords
- Select:
 - **App:** Other (Custom name)
 - **Name:** Jenkins-Demo
- Click **Generate**
- Copy the **16-digit app password**
 - Save it in a secure location (e.g., Notepad)

2. Jenkins Plugin Installation

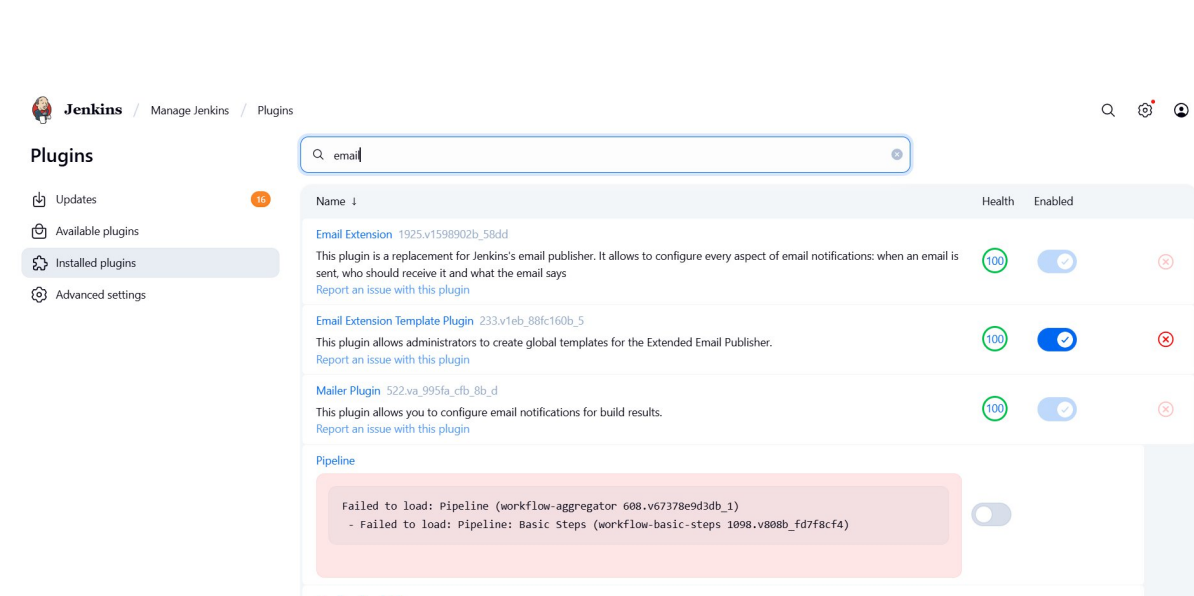
i. Open Jenkins Dashboard

ii. Navigate to:

- Manage Jenkins → Manage Plugins

iii. Install Plugin:

- Search for and install:
 - Email Extension Plugin



3. Configure Jenkins Global Email Settings

Go to:

- Manage Jenkins → Configure System

A. E-mail Notification Section

Field	Value
SMTP Server	smtp.gmail.com
Use SMTP Auth	☒ Enabled
User Name	Your Gmail ID (e.g., archanareddykmit@gmail.com)
Password	Paste the 16-digit App Password
Use SSL	☒ Enabled
SMTP Port	465

Reply-To Address Your Gmail ID (same as above)

► Test Configuration

- Click: Test configuration by sending test e-mail
- Provide a valid email address to receive a test mail
- ☒ Should receive email from Jenkins


Jenkins / Manage Jenkins / System





E-mail Notification

SMTP server

Default user e-mail suffix ?

Advanced ^

Edited

☒ Use SMTP Authentication ?

User Name

Password

 Concealed

Change Password

☒ Use SSL ?

☐ Use TLS

SMTP Port ?

Reply-To Address

Save

Apply

B. Extended E-mail Notification Section

Field	Value
SMTP Server	smtp.gmail.com
SMTP Port	465
Use SSL	✓ Enabled
Credentials	Add Gmail ID and App Password as Jenkins credentials
Default Content Type	text/html or leave default

Field	Value
Default Recipients	Leave empty or provide default emails
Triggers	Select as per needs (e.g., Failure)

Extended E-mail Notification

SMTP server

smtp.gmail.com

SMTP Port

465

Advanced ^

Edited

Credentials

edigiralaneksha@gmail.com/***** (fst)

+ Add

☒ Use SSL

☐ Use TLS

☐ Use OAuth 2.0

Advanced Email Properties

Save

Apply

Default Triggers ^

Default Triggers ?

☐ Aborted

☒ Always

☐ Before Build

☐ Failure - 1st

☐ Failure - 2nd

☒ Failure - Any

☐ Failure - Still

☒ Failure - X

☐ Failure -> Unstable (Test Failures)

☐ Fixed

☐ Not Built

☐ Script - After Build

☐ Script - Before Build

☒ Status Changed

☒ Success

☒ Test Improvement

☒ Test Regression

☐ Unstable (Test Failures)

☐ Unstable (Test Failures) - 1st

☐ Unstable (Test Failures) - Still

☒ Unstable (Test Failures)/Failure -> Success

Content Token Reference ?

4. Configure Email Notifications for a Jenkins Job

i. Go to:

- Jenkins → Select a Job → Configure

Jenkins / job_webhook_java / Configuration

Configure

- General
- Source Code Management
- Triggers
- Environment
- Build Steps
- Post-build Actions

General

Enabled

Description

java webhook

Plain text Preview

- ☐ Discard old builds ?
- ☐ GitHub project
- ☐ Notify when Job configuration changes
- ☐ This project is parameterized ?
- ☐ Throttle builds ?
- ☐ Execute concurrent builds if necessary ?

Advanced

Source Code Management

Connect and manage your code repository to automatically pull the latest code for your builds.

- ☐ None
- ☒ Git ?

Repositories ?

Save Apply

ii. In the Post-build Actions section:

- Click: Add post-build action → **Editable Email Notification**

A. Fill in the fields:

Field	Value
Project Recipient List	Add recipient email addresses (comma-separated)
Content Type	Default (text/plain) or text/html
Triggers	Select events (e.g., Failure, Success, etc.)
Attachments	(Optional) Add logs, reports, etc.

iii. Click Save

Post-build Actions

Define what happens after a build completes, like sending notifications, archiving artifacts, or triggering other jobs.

Editable Email Notification

?

×

Allows the user to disable the publisher, while maintaining the settings

☐

Disable Extended Email Publisher

?

Project From

Project Recipient List

?

Comma-separated list of email address that should receive notifications for this project.

edigiralaneksha@gmail.com,nekshasri99@gmail.com

Project Reply-To List

?

Comma-separated list of email address that should be in the Reply-To header for this project.

\$DEFAULT_REPLYTO

Save

Apply



job_webhook_log

Rename

Credentials

Builds

More Actions

Filter

Today

#4 11:45 am

28 October 2025

#3 11:41 am

#2 11:36 am

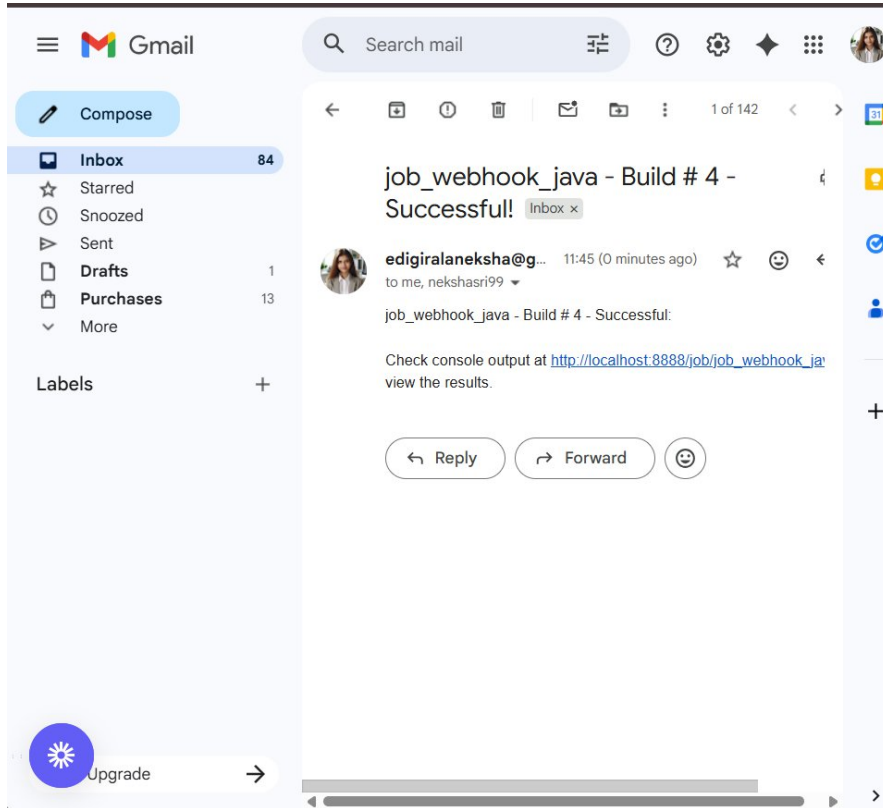
#1 11:35 am

job_webhook_java

Add description

Permalinks

- Last build (#3), 3 days 0 hr ago
- Last stable build (#3), 3 days 0 hr ago
- Last successful build (#3), 3 days 0 hr ago
- Last completed build (#3), 3 days 0 hr ago

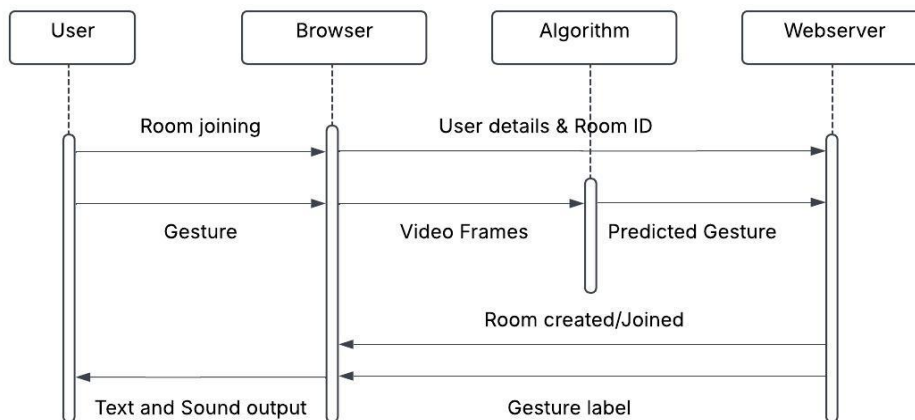


TUNEORA - A Music Web

App

1. Sequence Diagram:

A sequence diagram shows how objects interact in a particular scenario of a use case. It focuses on the time order of messages exchanged between different components in a system.



2. Class Diagram:

A class diagram represents the static structure of a system by showing classes, their attributes, methods, and relationships.

It is mainly used for object-oriented design and modeling data structures.

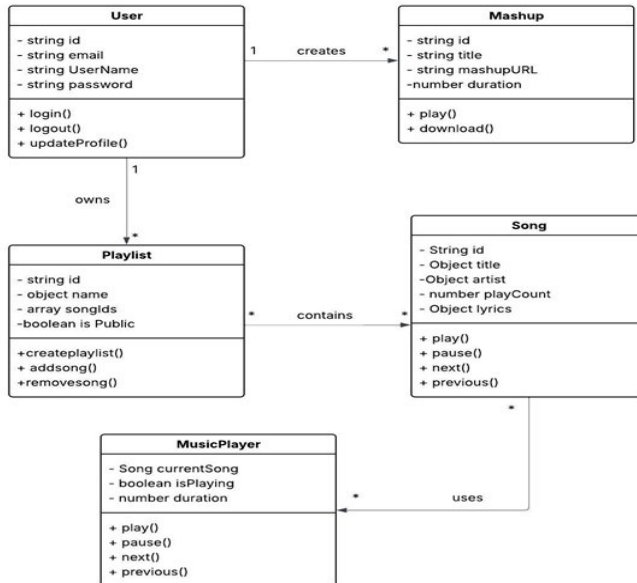
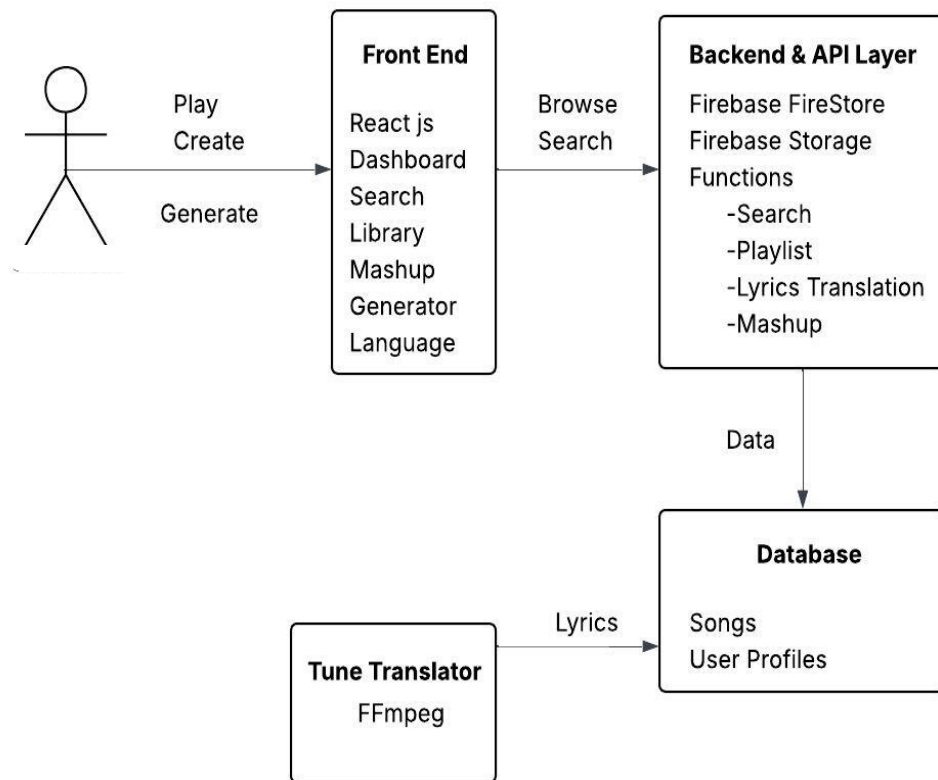


Fig 5: Class Diagram for TuneOra

3. Component Diagram:

A component diagram illustrates how different software components are connected and interact to form a complete system.

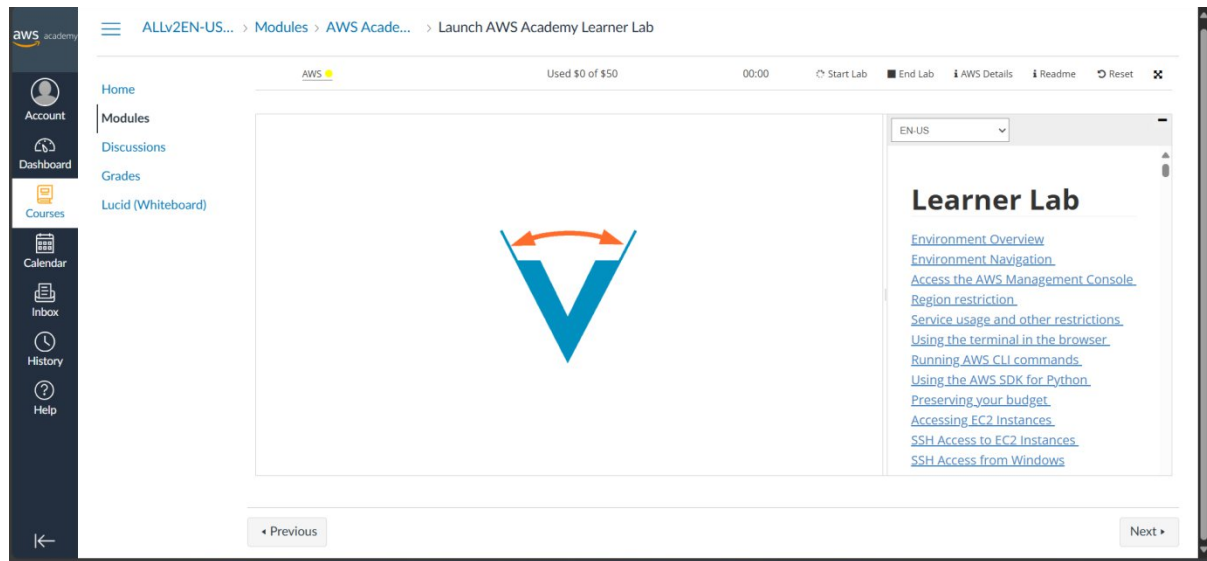
It helps visualize the organization and dependencies among modules or subsystems.



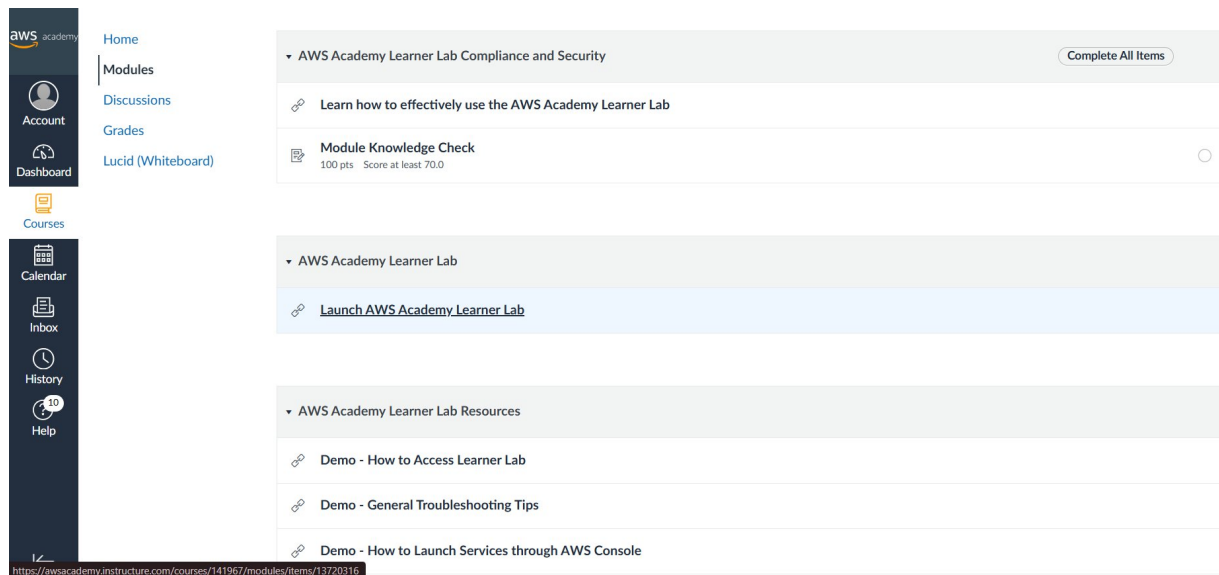
12. Creation of virtual machine for Ubuntu OS and Deploying the web application

DEPLOYMENT OF INDEX.HTML USING EC2 INSTANCE in AWS

Step 1: Click on Modules.



Step 2: Scroll down and select Lunch AWS Academy Lab



Step 3: click on start lab

aws academy ALLv2EN-US-LT113-141967 > Modules > AWS Academy Learner Lab > Launch AWS Academy Learner Lab

Account Dashboard Courses Calendar Inbox History Help

Home Modules Discussions Grades Lucid (Whiteboard)

AWS Used \$0 of \$50 00:00 Start Lab End Lab AWS Details Readme Reset

eee_W_5353255@runweb195092:~\$

EN-US

Learner Lab

- [Environment Overview](#)
- [Environment Navigation](#)
- [Access the AWS Management Console](#)
- [Region restriction](#)
- [Service usage and other restrictions](#)
- [Using the terminal in the browser](#)
- [Running AWS CLI commands](#)
- [Using the AWS SDK for Python](#)
- [Preserving your budget](#)
- [Accessing EC2 Instances](#)
- [SSH Access to EC2 Instances](#)
- [SSH Access from Windows](#)

Previous Next

Step 4: click on AWS and in the services select EC2

ALLv2EN-US... > Modules > AWS Acade... > Launch AWS Academy Learner Lab

Account Dashboard Courses Calendar Inbox History Help

Home Modules Discussions Grades Lucid (Whiteboard)

AWS Used \$0 of \$50 04:00 Start Lab End Lab AWS Details Readme Reset

eee_W_3940257@runweb155453:~\$

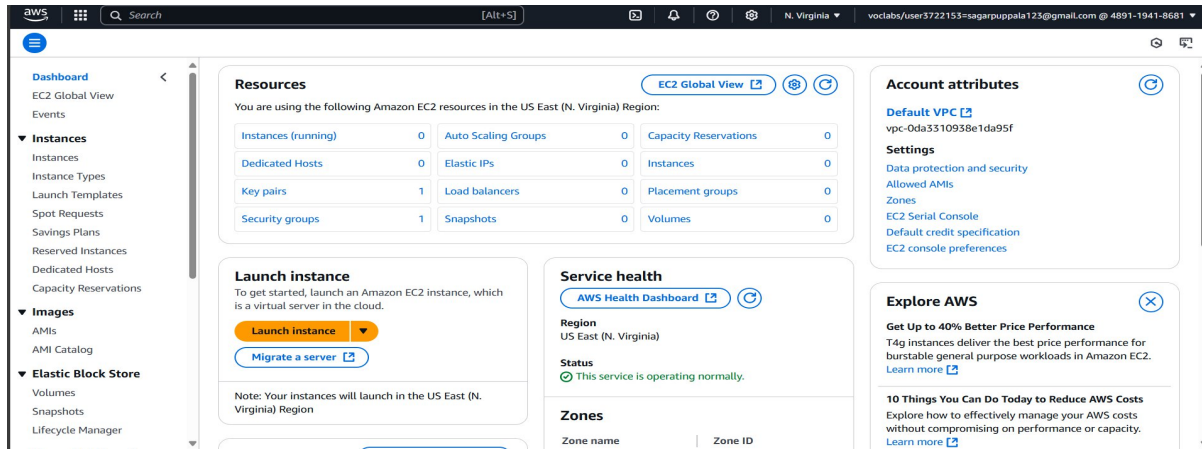
EN-US

Learner Lab

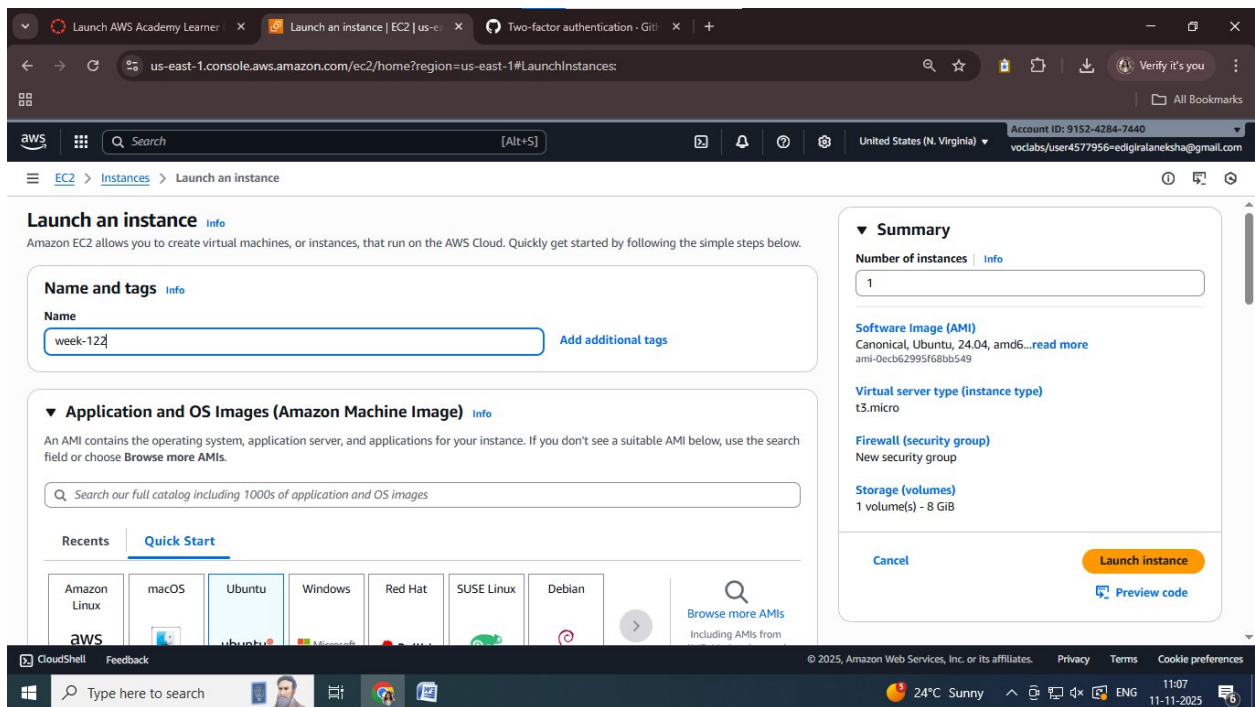
- [Environment Overview](#)
- [Environment Navigation](#)
- [Access the AWS Management Console](#)
- [Region restriction](#)
- [Service usage and other restrictions](#)
- [Using the terminal in the browser](#)
- [Running AWS CLI commands](#)
- [Using the AWS SDK for Python](#)
- [Preserving your budget](#)
- [Accessing EC2 Instances](#)
- [SSH Access to EC2 Instances](#)
- [SSH Access from Windows](#)

Previous Next

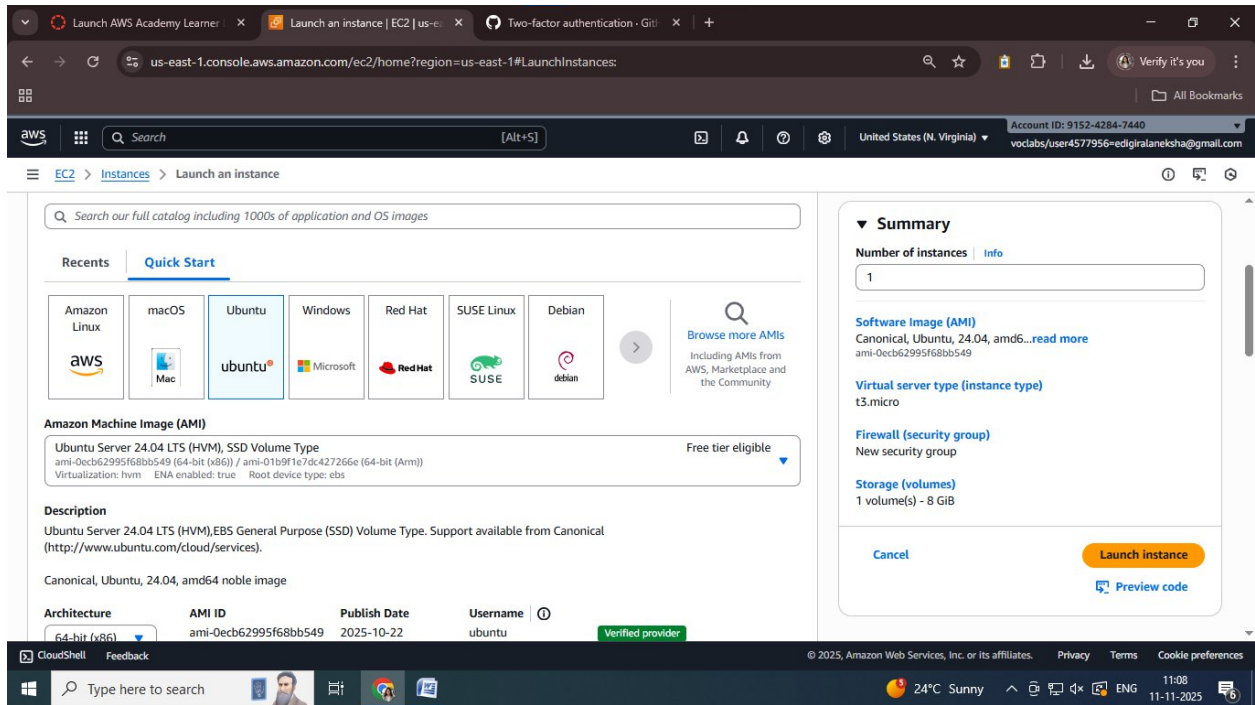
Step 5: select instances and select instance click on launch instance



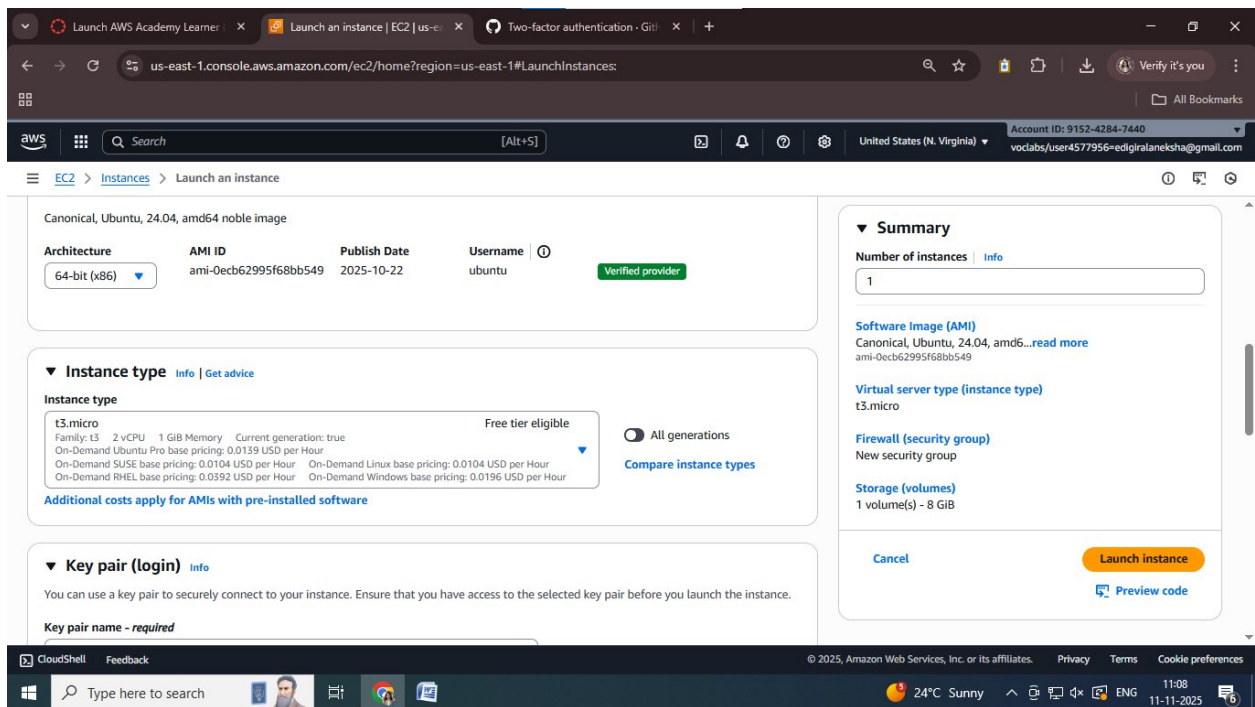
Step 6: Give the name of the machine “week-122”



Step 6: Select the ubuntu server



Step 7: select architecture as 64-bit and instance type as t3.micro(i.e., they are free)



Step 8: Create a new keypair and select type as RSA and .pem option and click on create key pair

and Windows base pricing: 0.0196 USD per Hour

ware

Sur

ember

ftwar

monica

-0ecb

tual s

micro

ewall

w sec

rage

olum

Canc

your in

© 2025, Amazon We

Create key pair

×

Key pair name

Key pairs allow you to connect to your instance securely.

week-122

The name can include up to 255 ASCII characters. It can't include leading or trailing spaces.

Key pair type

☒ **RSA**
RSA encrypted private and public key pair

☐ **ED25519**
ED25519 encrypted private and public key pair

Private key file format

☒ **.pem**
For use with OpenSSH

☐ **.ppk**
For use with PuTTY

⚠ When prompted, store the private key in a secure and accessible location on your computer. **You will need it later to connect to your instance.** [Learn more](#) ↗

Cancel

Create key pair

Step 9: In network settings select “create security group” and give the security group name

▼ Network settings [Info](#)

VPC - required [Info](#)

vpc-05a9ef3852073b114 (default) 

Subnet [Info](#)

No preference   [Create new subnet](#)

Availability Zone [Info](#)

No preference   [Enable additional zones](#)

Auto-assign public IP [Info](#)

Enable 

Firewall (security groups) [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Create security group ☐ Select existing security group

Security group name - required

week-122

This security group will be added to all network interfaces. The name can't be edited after the security group is created. Max length is 255 characters. Valid characters: a-z, A-Z, 0-9, spaces, and _-./()#,@!+=&:[]!\$*

Description - required [Info](#)

launch-wizard-1 created 2025-11-11T05:36:49.724Z

Step 10: Click on edit button on the top right corner and select

Type: ssh

Source: anywhere

EC2 > **Instances** > Launch an instance

week-122

This security group will be added to all network interfaces. The name can't be edited after the security group is created. Max length is 255 characters. Valid characters: a-z, A-Z, 0-9, spaces, and _-./()#,@!+=&:[]!\$*

Description - required [Info](#)

launch-wizard-1 created 2025-11-11T05:36:49.724Z

Inbound Security Group Rules

▼ Security group rule 1 (TCP, 22, 0.0.0.0/0)

Type [Info](#)

ssh 

Protocol [Info](#)

TCP 

Port range [Info](#)

22 

Source type [Info](#)

Anywhere 

Source [Info](#)

 Add CIDR, prefix list or security group

Description - optional [Info](#)

e.g. SSH for admin desktop

Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only. 

[Add security group rule](#)

▼ Summary

Number of instances [Info](#)

1

Software Image (AMI)

Canonical, Ubuntu, 24.04, amd6...[read more](#)

ami-0ecb62995f68bb549

Virtual server type (instance type)

t3.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

[Cancel](#) [Launch instance](#) [Preview code](#)

Step 11: in configure storage give 8GB and give number of instances as 2 and click on launch instance

Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

[Add security group rule](#)

Configure storage info Advanced

1x 8 GiB gp3 Root volume, 3000 IOPS, Not encrypted

[Add new volume](#)

The selected AMI contains instance store volumes, however the instance does not allow any instance store volumes. None of the instance store volumes from the AMI will be accessible from the instance

Click refresh to view backup information
The tags that you assign determine whether the instance will be backed up by any Data Lifecycle Manager policies.

0 x File systems [Edit](#)

Advanced details Info

Summary

Number of instances Info
2
When launching more than 1 instance, consider EC2 Auto Scaling

Software image (AMI)
Canonical, Ubuntu, 24.04, amd64...[read more](#)
ami-0ecb62995f68bb549

Virtual server type (instance type)
t3.micro

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 8 GiB

[Cancel](#) [Launch instance](#) [Preview code](#)

Step 12: The launching of instance will start and successful message will be shown

Launching Instance
Creating security group rules 33%

Details

Please wait while we launch your instance.
Do not close your browser while this is loading.

Success
Successfully initiated launch of instances (i-0f868f6d463f89656, i-0a5aa6fe5d0039e34)

Launch log

Next Steps

What would you like to do next with these instances, for example "create alarm" or "create backup"

1 2 3 4 5 6

Create billing usage alerts

To manage costs and avoid surprise bills, set up email notifications for billing usage thresholds.

[Create billing alerts](#)

Connect to your instance

Once your instance is running, log into it from your local computer.

[Learn more](#)

Connect an RDS database

Configure the connection between an EC2 instance and a database to allow traffic flow between them.

[Connect an RDS database](#)

[Create a new RDS database](#)

[Learn more](#)

Create EBS snapshot policy

Create a policy that automates the creation, retention, and deletion of EBS snapshots

[Create EBS snapshot policy](#)

Manage detailed monitoring

Enable or disable detailed monitoring for the instance.

Create Load Balancer

Create an application, network gateway or classic Elastic

Create AWS budget

AWS Budgets allows you to create budgets, forecast

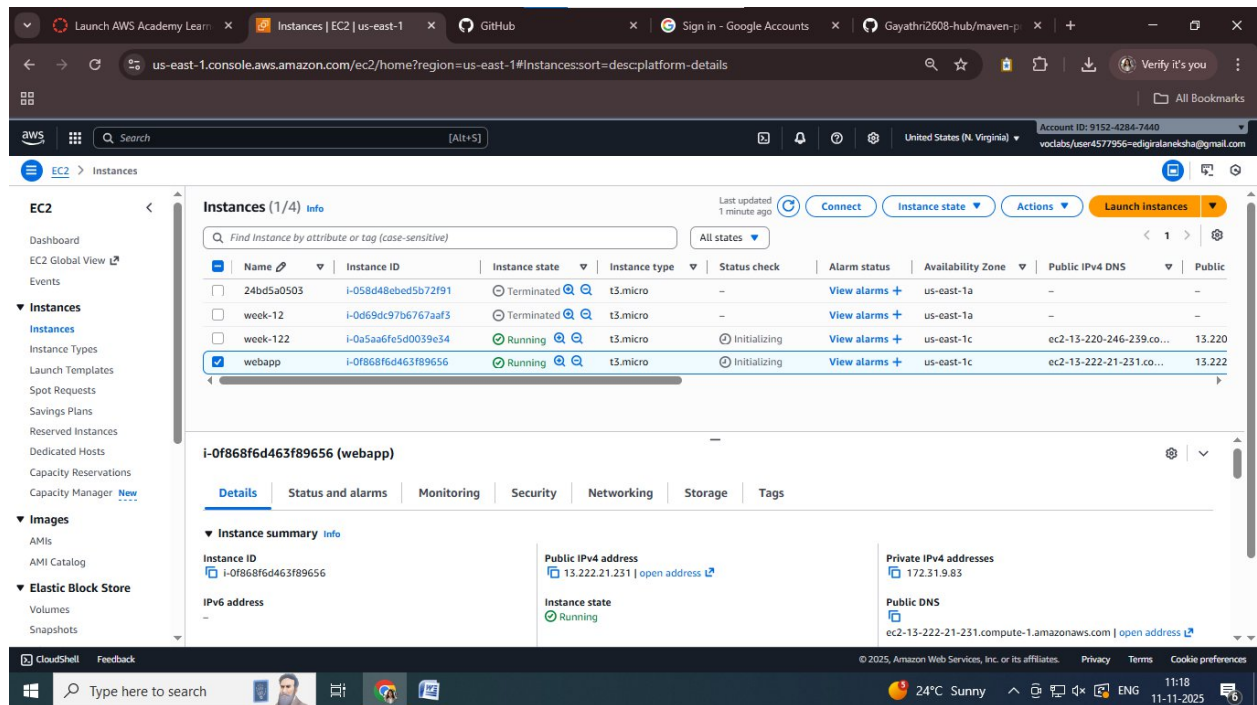
Manage CloudWatch alarms

Create or update Amazon CloudWatch alarms for the

CloudShell Feedback

© 2025, Amazon Web Services, Inc. or its affiliates. [Privacy](#) [Terms](#) [Cookie preferences](#)

Step 13: In the instances the created ones will be shown, you can also rename the instance , changed week-1222 to “webapp”



The screenshot shows the AWS Management Console for the us-east-1 region. The left sidebar contains navigation links for EC2, Images, and Elastic Block Store. The main content area displays a table of EC2 instances. The instance 'webapp' (ID: i-0f868f6d463f89656) is highlighted. Below the table, the details for this instance are shown, including its state (Running), type (t3.micro), and various addresses (Public IPv4, Private IPv4, Public DNS).

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public
24bd5a0503	i-058d48ebd5b72f91	Terminated	t3.micro	-	View alarms +	us-east-1a	-	-
week-12	i-0d69dc97b6767aaf3	Terminated	t3.micro	-	View alarms +	us-east-1a	-	-
week-122	i-0a5aa6fe5d0039e34	Running	t3.micro	Initializing	View alarms +	us-east-1c	ec2-13-220-246-239.co...	13.220
webapp	i-0f868f6d463f89656	Running	t3.micro	Initializing	View alarms +	us-east-1c	ec2-13-222-21-231.co...	13.222

i-0f868f6d463f89656 (webapp)

Details | Status and alarms | Monitoring | Security | Networking | Storage | Tags

Instance summary

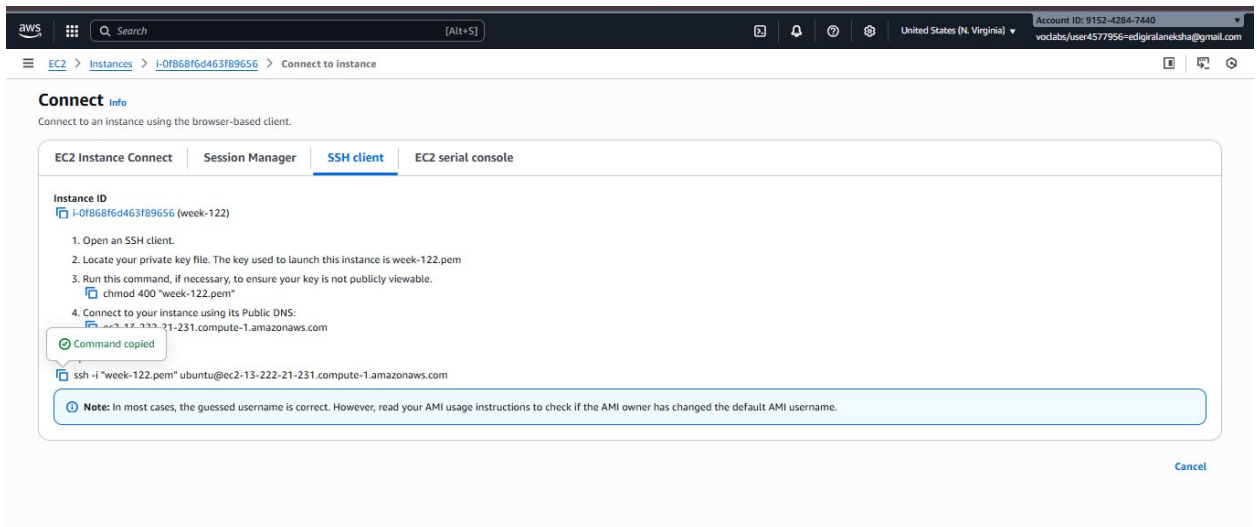
Instance ID: i-0f868f6d463f89656

Public IPv4 address: 13.222.21.231 | open address

Private IPv4 addresses: 172.31.9.83

Public DNS: ec2-13-222-21-231.compute-1.amazonaws.com | open address

Step 14: click on connect and select “SSH Client” and copy the ssh command



The screenshot shows the 'Connect' page in the AWS Management Console for the instance 'webapp'. The 'SSH client' tab is selected, and the SSH command is displayed. The command is: `ssh -i "week-122.pem" ubuntu@ec2-13-222-21-231.compute-1.amazonaws.com`. A note indicates that the guessed username is correct, but users should read their AMI usage instructions to check if the AMI owner has changed the default AMI username.

Connect

Connect to an instance using the browser-based client.

EC2 Instance Connect | Session Manager | **SSH client** | EC2 serial console

Instance ID: i-0f868f6d463f89656 (week-122)

1. Open an SSH client.
2. Locate your private key file. The key used to launch this instance is week-122.pem
3. Run this command, if necessary, to ensure your key is not publicly viewable.
`chmod 400 "week-122.pem"`
4. Connect to your instance using its Public DNS:
`ssh -i "week-122.pem" ubuntu@ec2-13-222-21-231.compute-1.amazonaws.com`

Note: In most cases, the guessed username is correct. However, read your AMI usage instructions to check if the AMI owner has changed the default AMI username.

Step 15: Navigate to the path where the file with .pem extension is present(week-122.pem) and paste the command

```
PS C:\Users\User\downloads> ssh -i "week-122.pem" ubuntu@ec2-13-222-21-231.compute-1.amazonaws.com
The authenticity of host 'ec2-13-222-21-231.compute-1.amazonaws.com (13.222.21.231)' can't be established.
ED25519 key fingerprint is SHA256:NEGegcHQjt8om/1AVLSqfmafMphv5Ad4A1Mwo8qEC0.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'ec2-13-222-21-231.compute-1.amazonaws.com' (ED25519) to the list of known hosts.
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-1015-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

System information as of Tue Nov 11 05:50:06 UTC 2025

System load:  0.08          Temperature:   -273.1 C
Usage of /:   25.9% of 6.71GB Processes:      118
Memory usage: 24%          Users logged in: 0
Swap usage:   0%           IPv4 address for ens5: 172.31.9.83

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@ip-172-31-9-83: $
```

Step 16: check the docker and git version

If they are not present, then go to administrative terminal using command

“sudo su”

Then update using the command “sudo apt-get update”

```

ubuntu@ip-172-31-9-83:~$ docker --version
Command 'docker' not found, but can be installed with:
sudo snap install docker          # version 28.4.0, or
sudo snap install docker          # version 28.1.1+1
sudo apt install docker.io        # version 28.2.2-0ubuntu1~24.04.1
sudo apt install podman-docker    # version 4.9.3+ds1-1ubuntu0.2
See 'snap info <snapname>' for additional versions.
ubuntu@ip-172-31-9-83:~$ git --version
git version 2.43.0
ubuntu@ip-172-31-9-83:~$ sudo su
root@ip-172-31-9-83:/home/ubuntu# sudo apt-get update

```

Step 17: use command “sudo apt-get install docker.io” to install docker

```

Get:50 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Components [212 B]
Get:51 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Packages [27.4 kB]
Get:52 http://security.ubuntu.com/ubuntu noble-security/multiverse Translation-en [5708 B]
Get:53 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Components [212 B]
Get:54 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 c-n-f Metadata [384 B]
Fetched 38.6 MB in 6s (6197 kB/s)
Reading package lists... Done
root@ip-172-31-9-83:/home/ubuntu# sudo apt-get install docker.io
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  bridge-utils containerd dns-root-data dnsmasq-base pigz runc ubuntu-fan
Suggested packages:
  ifupdown aufs-tools cgroupfs-mount | cgroup-lite debootstrap docker-buildx docker-compose-v2 docker-doc rinse
  zfs-fuse | zfsutils
The following NEW packages will be installed:
  bridge-utils containerd dns-root-data dnsmasq-base docker.io pigz runc ubuntu-fan
0 upgraded, 8 newly installed, 0 to remove and 10 not upgraded.
Need to get 76.0 MB of archives.
After this operation, 288 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 pigz amd64 2.8-1 [65.6 kB]
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 bridge-utils amd64 1.7.1-1ubuntu2 [33.9 kB]
Get:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 runc amd64 1.3.3-0ubuntu1~24.04.2 [8815 kB]
Get:4 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 containerd amd64 1.7.28-0ubuntu1~24.04.1 [38.4 MB]
Get:5 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 dns-root-data all 2024071801-ubuntu0.24.04.1 [5918 B]
Get:6 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 dnsmasq-base amd64 2.90-2ubuntu0.1 [376 kB]
Get:7 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 docker.io amd64 28.2.2-0ubuntu1~24.04.1 [28.3 MB]
Get:8 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 ubuntu-fan all 0.12.16+24.04.1 [34.2 kB]
Fetched 76.0 MB in 1s (81.3 MB/s)
Preconfiguring packages ...
Selecting previously unselected package pigz.
(Reading database ... 71735 files and directories currently installed.)
Preparing to unpack .../0-pigz_2.8-1_amd64.deb ...
Unpacking pigz (2.8-1) ...
Selecting previously unselected package bridge-utils.
Preparing to unpack .../1-bridge-utils_1.7.1-1ubuntu2_amd64.deb ...
Unpacking bridge-utils (1.7.1-1ubuntu2) ...
Selecting previously unselected package runc.
Preparing to unpack .../2-runc_1.3.3-0ubuntu1~24.04.2_amd64.deb ...
Unpacking runc (1.3.3-0ubuntu1~24.04.2) ...

```

Step 18: Clone the git repo that has maven project and change to that directory

```
No VM guests are running outdated hypervisor (qemu) binaries on this host.
root@ip-172-31-9-83:/home/ubuntu# git clone https://github.com/Gayathri2608-hub/maven-practice.git
Cloning into 'maven-practice'...
remote: Enumerating objects: 60, done.
remote: Counting objects: 100% (60/60), done.
remote: Compressing objects: 100% (40/40), done.
remote: Total 60 (delta 11), reused 34 (delta 2), pack-reused 0 (from 0)
Receiving objects: 100% (60/60), 13.39 KiB | 3.35 MiB/s, done.
Resolving deltas: 100% (11/11), done.
root@ip-172-31-9-83:/home/ubuntu# ls
maven-practice
root@ip-172-31-9-83:/home/ubuntu# cd maven-practice
root@ip-172-31-9-83:/home/ubuntu/maven-practice# ls
Dockerfile demo pom.xml readme src target
root@ip-172-31-9-83:/home/ubuntu/maven-practice#
```

Step 19: change to the project directory and check for Dockerfile, if not present create the Dockerfile – “nano Dockerfile” and then build the image

“sudo docker build -t image_name .” name of image:img1

```
root@ip-172-31-9-83:/home/ubuntu/maven-practice# ls
Dockerfile demo pom.xml readme src target
root@ip-172-31-9-83:/home/ubuntu/maven-practice# sudo docker build -t depl .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
Install the buildx component to build images with BuildKit:
https://docs.docker.com/go/buildx/

Sending build context to Docker daemon 120.8kB
Step 1/4 : FROM tomcat:9.0
9.0: Pulling from library/tomcat
4b3ffd8ccb52: Pulling fs layer
b48f960b380d: Pulling fs layer
58424d8c3e86: Pulling fs layer
4f4fb700ef54: Pulling fs layer
37b617836889: Pulling fs layer
891b6ad931b7: Pulling fs layer
ac0beccecf50: Pulling fs layer
4f4fb700ef54: Waiting
37b617836889: Waiting
891b6ad931b7: Waiting
ac0beccecf50: Waiting
b48f960b380d: Verifying Checksum
b48f960b380d: Download complete
4b3ffd8ccb52: Verifying Checksum
4b3ffd8ccb52: Download complete
4f4fb700ef54: Verifying Checksum
4f4fb700ef54: Download complete
37b617836889: Verifying Checksum
37b617836889: Download complete
891b6ad931b7: Verifying Checksum
891b6ad931b7: Download complete
ac0beccecf50: Verifying Checksum
ac0beccecf50: Download complete
58424d8c3e86: Verifying Checksum
58424d8c3e86: Download complete
4b3ffd8ccb52: Pull complete
```

Step 20: Run the image "sudo docker run -d -p 8081:8080 img1"

```
root@ip-172-31-9-83:/home/ubuntu/ar/maven-practice# sudo docker run -d -p 8081:8080 img1
c5fd91cf9a9b4f0625d4d2c0d896406e8da76929ed75a3f9ccc1699fbbb08535
root@ip-172-31-9-83:/home/ubuntu/ar/maven-practice#
```

Step 21: Check the images and the containers

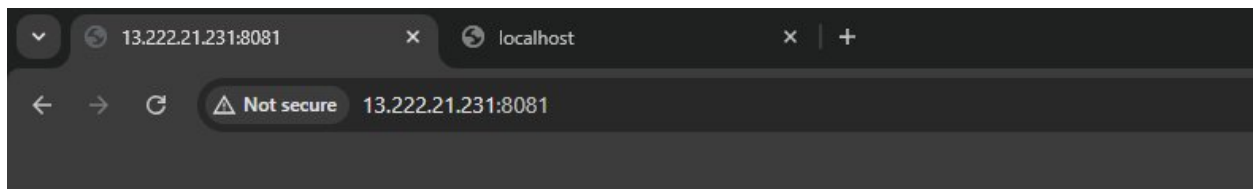
```
root@ip-172-31-9-83:/home/ubuntu/ar/maven-practice# sudo docker images
REPOSITORY TAG IMAGE ID CREATED SIZE
img1 latest a67a112ce8ac 2 minutes ago 413MB
dep1 latest 28efbe56e633 29 minutes ago 413MB
tomcat 9.0 2e4887a16e43 12 hours ago 413MB
root@ip-172-31-9-83:/home/ubuntu/ar/maven-practice# docker ps
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS
c5fd91cf9a9b img1 "catalina.sh run" About a minute ago Up About a minute 0.0.0.0:8081->8080/tcp, [::]:8081->8080/tcp
84e7f9ce5ec2 dep1 "catalina.sh run" 9 minutes ago Up 9 minutes 0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp
b62aadc8bb3b dep1 "catalina.sh run" 27 minutes ago Up 27 minutes 0.0.0.0:7070->8080/tcp, [::]:7070->8080/tcp
root@ip-172-31-9-83:/home/ubuntu/ar/maven-practice#
```

Step 22: Take the public IP address from the instances in AWS and open it in chrome along with the port number mapped.

Public IP- 13.222.21.231

Port used: 8081

Use: 13.222.21.231:8081, you will find your application that is deployed



Hello World! practice